NV3022B3 Datasheet

A-Si TFT LCD Single Chip Driver
80RGBx160 Resolution and 65,536 color

Version 0.6
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Introduction

NV3022B3 is a 65,536-color one-chip SOC driver for A-Si TFT liquid crystal display with resolution of 80RGBx160 dots, comprising a 240-channel source driver, a 160-channel gate driver, 25,600 bytes GRAM for graphic data of 80RGBx160 dots, and power supply circuit.

The NV3022B3 supports 9-/8-bit data bus interface and serial peripheral interfaces (SPI). The moving picture area can be specified in internal GRAM by window address function. The specified window area can be updated selectively, so that moving picture can be displayed simultaneously independent of still picture area.

NV3022B3 can operate with 1.65V I/O interface voltage, and an incorporated voltage follower circuit to generate voltage levels for driving an LCD. The NV3022B3 also supports a function to display in 8 colors idle mode, allowing for precise power control by software. So these features make the NV3022B3 an ideal LCD driver for medium or small size portable products such as digital cellular phones, smart phone, MP3 and PMP, on which long battery life is a major concern.

1. Features

- Display resolution options
  - 80(RGB) (H) X 160 (V)
- LCD Driver Output Circuits
  - 240 source outputs
  - 160 gate outputs
  - Common electrode output
- TFT-LCD driver with on-chip full display RAM: 25,600 bytes
- System Interfaces
  - 8-bits interface with 8080-series MCU
  - 8-bits interface with 6800-series MCU
  - 3-pin/4-pin serial interface
- Display color modes
  - Full color mode (idle mode off): 65,536-colors
  - Reduced color modes (idle mode on): 8-colors
- On Chip Functions
  - VCOM generator and adjustment
  - Timing generator
  - Oscillator
  - DC/DC converter
  - Line/frame inversion
  - Factory default value (Maker ID, Version ID, Module ID, etc) are stored on the display module
- Low–Power Consumption
  - Low operating power supplies
    - IOVCC = 1.65V ~ 3.6 V (interface I/O)
    - VCI = 2.5V ~ 3.6 V (analog)
- LCD Voltage Drive
  - Source/VCOM power supply voltage

- DDVDHAC – AGND = 4.5V ~ 5.2V
- VCL – AGND = -1.0V ~ -3.0V
- VCI － VCL ≦ 5.0V

➢ Gate driver output voltage

- VGH – AGND = 10V ~ 15V
- VGL – AGND = -7.5V ~ -13V
- VGH － VGL ≦ 28V

➢ VCOM driver output voltage

- VCOMH = 3.24V ~ 4.3425V
- VCOML = -0.405V ~ -1.5975V

◆ Operate Temperature Range: -30℃ to 85℃

2. Pin Descriptions

<table>
<tr><th>Pin Name</th><th>I/O</th><th>Descriptions</th></tr>
<tr><td colspan="3">Power Supply Pin</td></tr>
<tr><td>IOVCC</td><td>I</td><td>Power supply for interface logic circuits (1.65 ~ 3.6V).</td></tr>
<tr><td>VCI</td><td>I</td><td>Power supply for analog circuit.
Could connect to external power supply (VCI=2.5~3.6V).</td></tr>
<tr><td>AGND</td><td>I</td><td>System Ground for Analog System and Booster Circuit.</td></tr>
<tr><td>GND</td><td>I</td><td>System Ground for I/O System and Digital System.</td></tr>
<tr><td colspan="3">Interface Logic Pin</td></tr>
<tr><td>P68</td><td>I</td><td>8080/6800 MCU Interface Mode Selection
P68 = ‘1’ : Select 6800-MCU parallel interface
P68 = ‘0’ : Select 8080-MCU parallel interface
Note: If not used, please fix this pin at GND level.</td></tr>
<tr><td>IM2
IM1
IM0</td><td>I</td><td>MCU Parallel interface bus and Serial interface select :
<table>
<tr><th>IM2</th><th>IM1</th><th>IM0</th><th>SPI4W</th><th>Interface Type</th></tr>
<tr><td>1</td><td>X</td><td>X</td><td>X</td><td>MCU 8-bit Parallel</td></tr>
<tr><td>0</td><td>X</td><td>X</td><td>1</td><td>SPI 4 Wire Serial</td></tr>
<tr><td>0</td><td>X</td><td>X</td><td>0</td><td>SPI 3 Wire Serial(or 2 Data Line SPI)</td></tr>
</table></td></tr>
<tr><td>SPI4W</td><td>I</td><td>SPI4W=’0’, 3-line SPI or 2 Data Line SPI Enable.
SPI4W=’1’, 4-line SPI Enable.
If Not Used, Please fix this Pin at GND Level.</td></tr>
<tr><td>RESX</td><td>I</td><td>Chip Reset Pin (“Low Active”)
Note: Keep this pin without glitch</td></tr>
<tr><td>CSX</td><td>I</td><td>Chip Select Pin (“Low Active”)
Pull down when chip accessable</td></tr>
<tr><td>DCX</td><td>I</td><td>Reuse Pin according parallel and serial interface:
➢ MCU Interface: Distinguish Data (‘1’) or Command (‘0’).
➢ SPI Interface: Used as ‘SCL’ clock pin.
If not used, please connect this pin to GND.</td></tr>
<tr><td>RDX</td><td>I</td><td>➢ 8080-parallel interface: used as ‘Read’ enable.
➢ 6800-parallel interface:
1. when WRX=’0’ , RDX used as ‘write’ enable;
2. When WRX=’1’, RDX used as ‘Read’ enable.</td></tr>
<tr><td>WRX</td><td>I</td><td>➢ 8080-parallel interface, used as write enable.
➢ 6800-parallel interface, used to distinguish ‘write’ or ‘read’ operation.
➢ SPI 4-wire interface, used as D/CX.
➢ 2 Data Line SPI used as data input.</td></tr>
</table>

<table>
<tr><th>Pin Name</th><th>I/O</th><th>Descriptions</th></tr>
<tr><td>D[7:0]</td><td>I/O</td><td>➢ SPI interface : D[0] used as data bus;
➢ MCU interface : D[7:0] used as data bus</td></tr>
<tr><td>TE</td><td>O</td><td>Tearing effect output pin to synchronize MCU to frame writing</td></tr>
<tr><td>TEST12</td><td>O</td><td>TEST12 is used for SPI read data line</td></tr>
<tr><td colspan="3">Mode Selection Pin</td></tr>
<tr><td>GM2
GM1
GM0</td><td>I</td><td>
<table>
<tr><th>GM2</th><th>GM1</th><th>GM0</th><th>Description</th></tr>
<tr><td>0</td><td>0</td><td>0</td><td>80RGBX160(S1~240 and G2~161 output)</td></tr>
<tr><td>0</td><td>0</td><td>1</td><td>80RGBX160(S1~240 and G2~161 output)</td></tr>
<tr><td>0</td><td>1</td><td>0</td><td>80RGBX160(S1~240 and G2~161 output)</td></tr>
<tr><td>0</td><td>1</td><td>1</td><td>80RGBX160(S1~240 and G2~161 output)</td></tr>
<tr><td>1</td><td>0</td><td>0</td><td>80RGBX160(S1~240 and G2~161 output)</td></tr>
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<tr><td>1</td><td>1</td><td>1</td><td>80RGBX160(S1~240 and G2~161 output)</td></tr>
</table>
</td></tr>
<tr><td>VPP</td><td>I</td><td>When writing NVM, it needs external power supply voltage (7.5V).</td></tr>
<tr><td colspan="3">Driver Output pins</td></tr>
<tr><td>S1~S240</td><td>O</td><td>Source driver output pins.</td></tr>
<tr><td>G2~G161</td><td>O</td><td>Gate driver output pins.</td></tr>
<tr><td>VDD</td><td>O</td><td>Test pin for internal digital logic Power supply</td></tr>
<tr><td>GVDD</td><td>O</td><td>A power supply for Grayscale Voltage Generator</td></tr>
<tr><td>DDVDHAC</td><td>O</td><td>A power supply pin for source driver block that is generated from power block. Output of booster 1 circuit (output of 2-times output of VCI).</td></tr>
<tr><td>VCL</td><td>O</td><td>A power supply pin for generating VCOML</td></tr>
<tr><td>VGH</td><td>O</td><td>Power supply for Gate Driver</td></tr>
<tr><td>VGL</td><td>O</td><td>Negative power supply for Gate Driver</td></tr>
<tr><td>VCOMH</td><td>O</td><td>The high level of VCOM AC voltage.</td></tr>
<tr><td>VCOML</td><td>O</td><td>The low level of VCOM AC voltage.</td></tr>
<tr><td>VCOM</td><td>O</td><td>TFT display common electrode power supply. Alternates between voltage levels between VCOMH-VCOML.
Registers set the alternating cycle for operating or halting VCOM.</td></tr>
<tr><td>TEST_C<3:0></td><td>O</td><td>Digital function test outputs, it is not accessible to user. must be open.</td></tr>
<tr><td>TESTOP[8:1]</td><td>O</td><td>Analog function test outputs, it is not accessible to user. must be open.</td></tr>
<tr><td>SMX SMY</td><td></td><td>Dummy pin. if no used，please floating it.</td></tr>
<tr><td>TVBG、T1UA</td><td></td><td>Internal analog reference test pin，normal output is Hiz state,if no used，please floating it.</td></tr>
</table>

3. Bump Arrangement and Coordination

3.1. Bump Arrangement

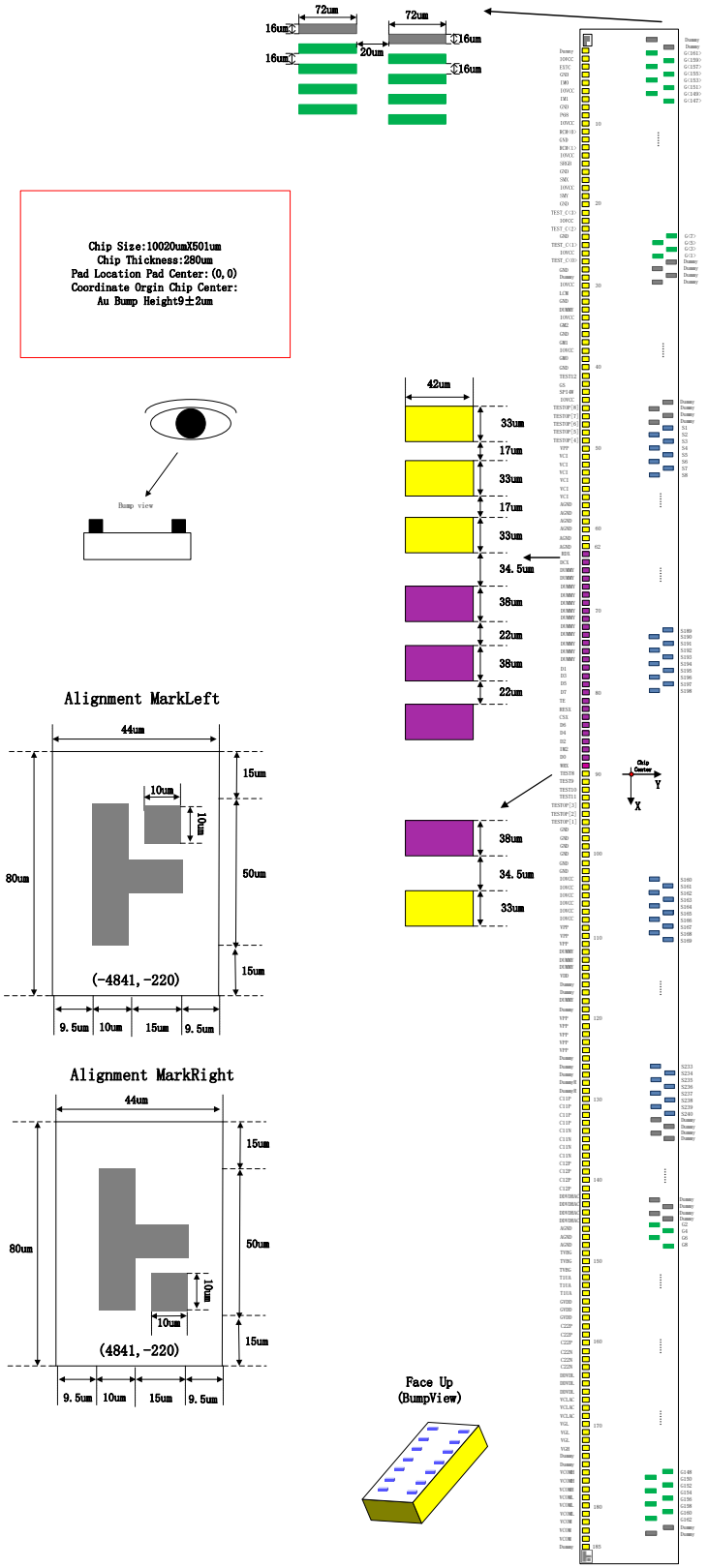


Figure 3-1 Pad Arrangement

3.2. Pin Connection

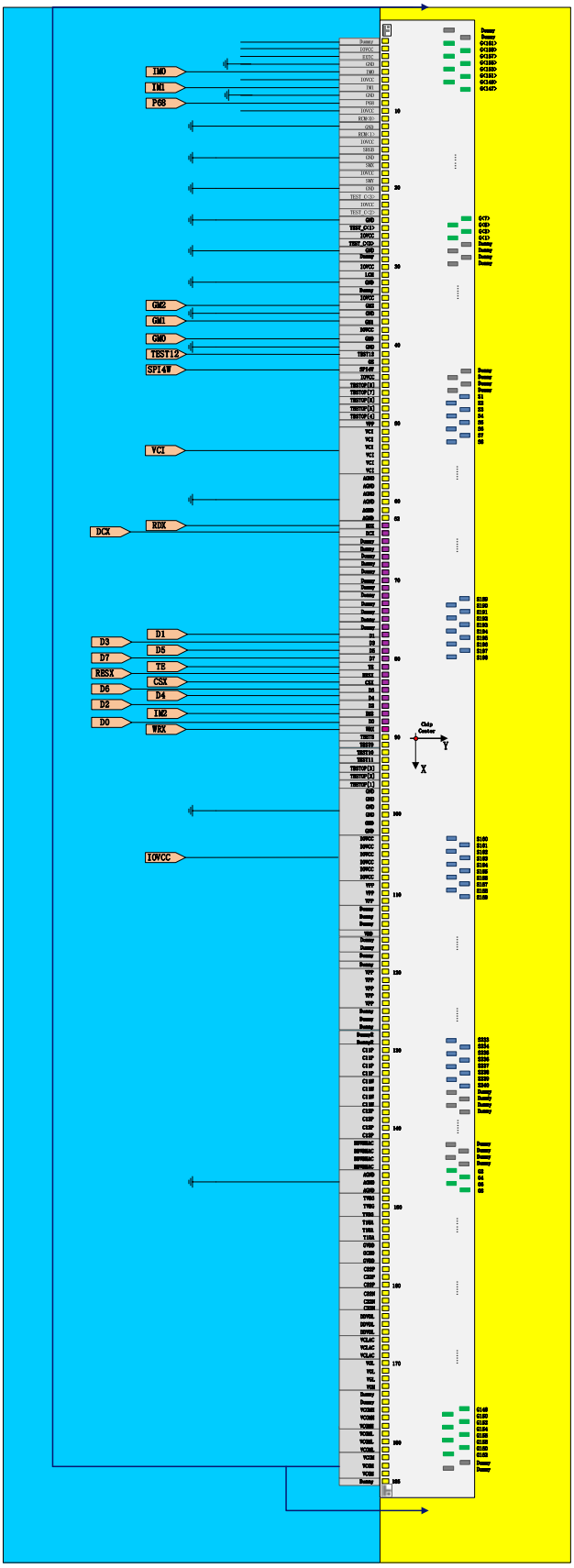


Figure 3-2 Pad Connection

3.3. Pad Coordination

No.	PIN Name	X	Y
1	DUMMY	-4750	-209.5
2	IOVCC	-4700	-209.5
3	EXTC	-4650	-209.5
4	GND	-4600	-209.5
5	IM0	-4550	-209.5
6	IOVCC	-4500	-209.5
7	IM1	-4450	-209.5
8	GND	-4400	-209.5
9	P68	-4350	-209.5
10	IOVCC	-4300	-209.5
11	RCM<0>	-4250	-209.5
12	GND	-4200	-209.5
13	RCM<1>	-4150	-209.5
14	IOVCC	-4100	-209.5
15	SRGB	-4050	-209.5
16	GND	-4000	-209.5
17	SMX	-3950	-209.5
18	IOVCC	-3900	-209.5
19	SMY	-3850	-209.5
20	GND	-3800	-209.5
21	TEST_C<3>	-3750	-209.5
22	IOVCC	-3700	-209.5
23	TEST_C<2>	-3650	-209.5
24	GND	-3600	-209.5
25	TEST_C<1>	-3550	-209.5
26	IOVCC	-3500	-209.5
27	TEST_C<0>	-3450	-209.5
28	GND	-3400	-209.5
29	DUMMY	-3350	-209.5
30	IOVCC	-3300	-209.5
31	LCM	-3250	-209.5
32	GND	-3200	-209.5
33	DUMMY	-3150	-209.5
34	IOVCC	-3100	-209.5
35	GM2	-3050	-209.5
36	GND	-3000	-209.5
37	GM1	-2950	-209.5
38	IOVCC	-2900	-209.5
39	GM0	-2850	-209.5
40	GND	-2800	-209.5
41	TEST12	-2750	-209.5
42	GS	-2700	-209.5
43	SPI4W	-2650	-209.5
44	IOVCC	-2600	-209.5
45	TESTOP<8>	-2550	-209.5
46	TESTOP<7>	-2500	-209.5
47	TESTOP<6>	-2450	-209.5
48	TESTOP<5>	-2400	-209.5
49	TESTOP<4>	-2350	-209.5
50	VPP	-2300	-209.5
51	VCI	-2250	-209.5
52	VCI	-2200	-209.5
53	VCI	-2150	-209.5
54	VCI	-2100	-209.5
55	VCI	-2050	-209.5
56	VCI	-2000	-209.5
57	AGND	-1950	-209.5
58	AGND	-1900	-209.5
59	AGND	-1850	-209.5
60	AGND	-1800	-209.5
61	AGND	-1750	-209.5
62	AGND	-1700	-209.5
63	RDX	-1630	-209.5
64	DCX	-1570	-209.5
65	DUMMY	-1510	-209.5
66	DUMMY	-1450	-209.5
67	DUMMY	-1390	-209.5
68	DUMMY	-1330	-209.5
69	DUMMY	-1270	-209.5
70	DUMMY	-1210	-209.5
71	DUMMY	-1150	-209.5
72	DUMMY	-1090	-209.5
73	DUMMY	-1030	-209.5
74	DUMMY	-970	-209.5
75	DUMMY	-910	-209.5
76	DUMMY	-850	-209.5
77	D1	-790	-209.5
78	D3	-730	-209.5
79	D5	-670	-209.5
80	D7	-610	-209.5
81	TE	-550	-209.5
82	RESX	-490	-209.5
83	CSX	-430	-209.5
84	D6	-370	-209.5
85	D4	-310	-209.5
86	D2	-250	-209.5
87	IM2	-190	-209.5
88	D0	-130	-209.5
89	WRX	-70	-209.5
90	TEST8	0	-209.5
91	TEST9	50	-209.5
92	TEST10	100	-209.5
93	TEST11	150	-209.5
94	TESTOP<3>	200	-209.5
95	TESTOP<2>	250	-209.5
96	TESTOP<1>	300	-209.5
97	GND	350	-209.5
98	GND	400	-209.5
99	GND	450	-209.5
100	GND	500	-209.5
101	GND	550	-209.5
102	GND	600	-209.5
103	IOVCC	650	-209.5
104	IOVCC	700	-209.5
105	IOVCC	750	-209.5
106	IOVCC	800	-209.5
107	IOVCC	850	-209.5
108	IOVCC	900	-209.5
109	VPP	950	-209.5
110	VPP	1000	-209.5
111	VPP	1050	-209.5
112	DUMMY	1100	-209.5
113	DUMMY	1150	-209.5
114	DUMMY	1200	-209.5
115	VDD	1250	-209.5
116	DUMMY	1300	-209.5
117	DUMMY	1350	-209.5
118	DUMMY	1400	-209.5
119	DUMMY	1450	-209.5
120	VPP	1500	-209.5
121	VPP	1550	-209.5
122	VPP	1600	-209.5
123	VPP	1650	-209.5
124	VPP	1700	-209.5
125	DUMMY	1750	-209.5
126	DUMMY	1800	-209.5
127	DUMMY	1850	-209.5
128	DummyR	1900	-209.5
129	DummyR	1950	-209.5
130	C11P	2000	-209.5
131	C11P	2050	-209.5
132	C11P	2100	-209.5
133	C11P	2150	-209.5
134	C11N	2200	-209.5
135	C11N	2250	-209.5
136	C11N	2300	-209.5
137	C11N	2350	-209.5

138	C12P	2400	-209.5
139	C12P	2450	-209.5
140	C12P	2500	-209.5
141	C12P	2550	-209.5
142	DDVDHAC	2600	-209.5
143	DDVDHAC	2650	-209.5
144	DDVDHAC	2700	-209.5
145	DDVDHAC	2750	-209.5
146	AGND	2800	-209.5
147	AGND	2850	-209.5
148	AGND	2900	-209.5
149	TVBG	2950	-209.5
150	TVBG	3000	-209.5
151	TVBG	3050	-209.5
152	T1UA	3100	-209.5
153	T1UA	3150	-209.5
154	T1UA	3200	-209.5
155	GVDD	3250	-209.5
156	GVDD	3300	-209.5
157	GVDD	3350	-209.5
158	C22P	3400	-209.5
159	C22P	3450	-209.5
160	C22P	3500	-209.5
161	C22N	3550	-209.5
162	C22N	3600	-209.5
163	C22N	3650	-209.5
164	DDVDL	3700	-209.5
165	DDVDL	3750	-209.5
166	DDVDL	3800	-209.5
167	VCLAC	3850	-209.5
168	VCLAC	3900	-209.5
169	VCLAC	3950	-209.5
170	VGL	4000	-209.5
171	VGL	4050	-209.5
172	VGL	4100	-209.5
173	VGH	4150	-209.5
174	DUMMY	4200	-209.5
175	DUMMY	4250	-209.5
176	VCOMH	4300	-209.5
177	VCOMH	4350	-209.5
178	VCOMH	4400	-209.5
179	VCOML	4450	-209.5
180	VCOML	4500	-209.5
181	VCOML	4550	-209.5
182	VCOM	4600	-209.5
183	VCOM	4650	-209.5
184	VCOM	4700	-209.5
185	DUMMY	4750	-209.5
186	DUMMY	4772	122.5
187	DUMMY	4756	214.5
188	G162	4740	122.5
189	G160	4724	214.5
190	G158	4708	122.5
191	G156	4692	214.5
192	G154	4676	122.5
193	G152	4660	214.5
194	G150	4644	122.5
195	G148	4628	214.5
196	G146	4612	122.5
197	G144	4596	214.5
198	G142	4580	122.5
199	G140	4564	214.5
200	G138	4548	122.5
201	G136	4532	214.5
202	G134	4516	122.5
203	G132	4500	214.5
204	G130	4484	122.5
205	G128	4468	214.5
206	G126	4452	122.5
207	G124	4436	214.5
208	G122	4420	122.5
209	G120	4404	214.5
210	G118	4388	122.5
211	G116	4372	214.5
212	G114	4356	122.5
213	G112	4340	214.5
214	G110	4324	122.5
215	G108	4308	214.5
216	G106	4292	122.5
217	G104	4276	214.5
218	G102	4260	122.5
219	G100	4244	214.5
220	G98	4228	122.5
221	G96	4212	214.5
222	G94	4196	122.5
223	G92	4180	214.5
224	G90	4164	122.5
225	G88	4148	214.5
226	G86	4132	122.5
227	G84	4116	214.5
228	G82	4100	122.5
229	G80	4084	214.5
230	G78	4068	122.5
231	G76	4052	214.5
232	G74	4036	122.5
233	G72	4020	214.5
234	G70	4004	122.5
235	G68	3988	214.5
236	G66	3972	122.5
237	G64	3956	214.5
238	G62	3940	122.5
239	G60	3924	214.5
240	G58	3908	122.5
241	G56	3892	214.5
242	G54	3876	122.5
243	G52	3860	214.5
244	G50	3844	122.5
245	G48	3828	214.5
246	G46	3812	122.5
247	G44	3796	214.5
248	G42	3780	122.5
249	G40	3764	214.5
250	G38	3748	122.5
251	G36	3732	214.5
252	G34	3716	122.5
253	G32	3700	214.5
254	G30	3684	122.5
255	G28	3668	214.5
256	G26	3652	122.5
257	G24	3636	214.5
258	G22	3620	122.5
259	G20	3604	214.5
260	G18	3588	122.5
261	G16	3572	214.5
262	G14	3556	122.5
263	G12	3540	214.5
264	G10	3524	122.5
265	G8	3508	214.5
266	G6	3492	122.5
267	G4	3476	214.5
268	G2	3460	122.5
269	DUMMY	3444	214.5
270	DUMMY	3428	122.5
271	DUMMY	3412	214.5
272	DUMMY	3396	122.5
273	DUMMY	3380	214.5
274	DUMMY	3364	122.5
275	DUMMY	3348	214.5
276	DUMMY	3332	122.5
277	DUMMY	3316	214.5
278	DUMMY	3300	122.5
279	DUMMY	3284	214.5
280	DUMMY	3268	122.5
281	DUMMY	3252	214.5
282	DUMMY	3236	122.5
283	DUMMY	3220	214.5
284	DUMMY	3204	122.5

285	DUMMY	3188	214.5
286	DUMMY	3172	122.5
287	DUMMY	3156	214.5
288	DUMMY	3140	122.5
289	DUMMY	3124	214.5
290	DUMMY	3108	122.5
291	DUMMY	3092	214.5
292	DUMMY	3076	122.5
293	DUMMY	3060	214.5
294	DUMMY	3044	122.5
295	DUMMY	3028	214.5
296	DUMMY	3012	122.5
297	DUMMY	2996	214.5
298	DUMMY	2980	122.5
299	DUMMY	2964	214.5
300	DUMMY	2948	122.5
301	DUMMY	2932	214.5
302	DUMMY	2916	122.5
303	DUMMY	2900	214.5
304	DUMMY	2884	122.5
305	DUMMY	2868	214.5
306	DUMMY	2852	122.5
307	DUMMY	2836	214.5
308	DUMMY	2820	122.5
309	DUMMY	2804	214.5
310	DUMMY	2788	122.5
311	DUMMY	2772	214.5
312	DUMMY	2756	122.5
313	DUMMY	2740	214.5
314	DUMMY	2724	122.5
315	DUMMY	2708	214.5
316	DUMMY	2692	122.5
317	DUMMY	2676	214.5
318	DUMMY	2660	122.5
319	DUMMY	2644	214.5
320	DUMMY	2628	122.5
321	DUMMY	2612	214.5
322	DUMMY	2596	122.5
323	DUMMY	2580	214.5
324	DUMMY	2564	122.5
325	DUMMY	2548	214.5
326	DUMMY	2532	122.5
327	DUMMY	2516	214.5
328	DUMMY	2500	122.5
329	DUMMY	2484	214.5
330	DUMMY	2468	122.5
331	DUMMY	2452	214.5
332	DUMMY	2436	122.5
333	DUMMY	2420	214.5
334	DUMMY	2404	122.5
335	DUMMY	2388	214.5
336	DUMMY	2372	122.5
337	DUMMY	2356	214.5
338	DUMMY	2340	122.5
339	DUMMY	2324	214.5
340	DUMMY	2308	122.5
341	DUMMY	2292	214.5
342	DUMMY	2276	122.5
343	DUMMY	2260	214.5
344	DUMMY	2244	122.5
345	DUMMY	2228	214.5
346	DUMMY	2212	122.5
347	DUMMY	2196	214.5
348	DUMMY	2180	122.5
349	DUMMY	2164	214.5
350	DUMMY	2148	122.5
351	S240	2132	214.5
352	S239	2116	122.5
353	S238	2100	214.5
354	S237	2084	122.5
355	S236	2068	214.5
356	S235	2052	122.5
357	S234	2036	214.5
358	S233	2020	122.5
359	S232	2004	214.5
360	S231	1988	122.5
361	S230	1972	214.5
362	S229	1956	122.5
363	S228	1940	214.5
364	S227	1924	122.5
365	S226	1908	214.5
366	S225	1892	122.5
367	S224	1876	214.5
368	S223	1860	122.5
369	S222	1844	214.5
370	S221	1828	122.5
371	S220	1812	214.5
372	S219	1796	122.5
373	S218	1780	214.5
374	S217	1764	122.5
375	S216	1748	214.5
376	S215	1732	122.5
377	S214	1716	214.5
378	S213	1700	122.5
379	S212	1684	214.5
380	S211	1668	122.5
381	S210	1652	214.5
382	S209	1636	122.5
383	S208	1620	214.5
384	S207	1604	122.5
385	S206	1588	214.5
386	S205	1572	122.5
387	S204	1556	214.5
388	S203	1540	122.5
389	S202	1524	214.5
390	S201	1508	122.5
391	S200	1492	214.5
392	S199	1476	122.5
393	S198	1460	214.5
394	S197	1444	122.5
395	S196	1428	214.5
396	S195	1412	122.5
397	S194	1396	214.5
398	S193	1380	122.5
399	S192	1364	214.5
400	S191	1348	122.5
401	S190	1332	214.5
402	S189	1316	122.5
403	S188	1300	214.5
404	S187	1284	122.5
405	S186	1268	214.5
406	S185	1252	122.5
407	S184	1236	214.5
408	S183	1220	122.5
409	S182	1204	214.5
410	S181	1188	122.5
411	S180	1172	214.5
412	S179	1156	122.5
413	S178	1140	214.5
414	S177	1124	122.5
415	S176	1108	214.5
416	S175	1092	122.5
417	S174	1076	214.5
418	S173	1060	122.5
419	S172	1044	214.5
420	S171	1028	122.5
421	S170	1012	214.5
422	S169	996	122.5
423	S168	980	214.5
424	S167	964	122.5
425	S166	948	214.5
426	S165	932	122.5
427	S164	916	214.5
428	S163	900	122.5
429	S162	884	214.5
430	S161	868	122.5
431	S160	852	214.5

432	S159	836	122.5
433	S158	820	214.5
434	S157	804	122.5
435	S156	788	214.5
436	S155	772	122.5
437	S154	756	214.5
438	S153	740	122.5
439	S152	724	214.5
440	S151	708	122.5
441	S150	692	214.5
442	S149	676	122.5
443	S148	660	214.5
444	S147	644	122.5
445	S146	628	214.5
446	S145	612	122.5
447	S144	596	214.5
448	S143	580	122.5
449	S142	564	214.5
450	S141	548	122.5
451	S140	532	214.5
452	S139	516	122.5
453	S138	500	214.5
454	S137	484	122.5
455	S136	468	214.5
456	S135	452	122.5
457	S134	436	214.5
458	S133	420	122.5
459	S132	404	214.5
460	S131	388	122.5
461	S130	372	214.5
462	S129	356	122.5
463	S128	340	214.5
464	S127	324	122.5
465	S126	308	214.5
466	S125	292	122.5
467	S124	276	214.5
468	S123	260	122.5
469	S122	244	214.5
470	S121	228	122.5
471	S120	-228	122.5
472	S119	-244	214.5
473	S118	-260	122.5
474	S117	-276	214.5
475	S116	-292	122.5
476	S115	-308	214.5
477	S114	-324	122.5
478	S113	-340	214.5
479	S112	-356	122.5
480	S111	-372	214.5
481	S110	-388	122.5
482	S109	-404	214.5
483	S108	-420	122.5
484	S107	-436	214.5
485	S106	-452	122.5
486	S105	-468	214.5
487	S104	-484	122.5
488	S103	-500	214.5
489	S102	-516	122.5
490	S101	-532	214.5
491	S100	-548	122.5
492	S99	-564	214.5
493	S98	-580	122.5
494	S97	-596	214.5
495	S96	-612	122.5
496	S95	-628	214.5
497	S94	-644	122.5
498	S93	-660	214.5
499	S92	-676	122.5
500	S91	-692	214.5
501	S90	-708	122.5
502	S89	-724	214.5
503	S88	-740	122.5
504	S87	-756	214.5
505	S86	-772	122.5
506	S85	-788	214.5
507	S84	-804	122.5
508	S83	-820	214.5
509	S82	-836	122.5
510	S81	-852	214.5
511	S80	-868	122.5
512	S79	-884	214.5
513	S78	-900	122.5
514	S77	-916	214.5
515	S76	-932	122.5
516	S75	-948	214.5
517	S74	-964	122.5
518	S73	-980	214.5
519	S72	-996	122.5
520	S71	-1012	214.5
521	S70	-1028	122.5
522	S69	-1044	214.5
523	S68	-1060	122.5
524	S67	-1076	214.5
525	S66	-1092	122.5
526	S65	-1108	214.5
527	S64	-1124	122.5
528	S63	-1140	214.5
529	S62	-1156	122.5
530	S61	-1172	214.5
531	S60	-1188	122.5
532	S59	-1204	214.5
533	S58	-1220	122.5
534	S57	-1236	214.5
535	S56	-1252	122.5
536	S55	-1268	214.5
537	S54	-1284	122.5
538	S53	-1300	214.5
539	S52	-1316	122.5
540	S51	-1332	214.5
541	S50	-1348	122.5
542	S49	-1364	214.5
543	S48	-1380	122.5
544	S47	-1396	214.5
545	S46	-1412	122.5
546	S45	-1428	214.5
547	S44	-1444	122.5
548	S43	-1460	214.5
549	S42	-1476	122.5
550	S41	-1492	214.5
551	S40	-1508	122.5
552	S39	-1524	214.5
553	S38	-1540	122.5
554	S37	-1556	214.5
555	S36	-1572	122.5
556	S35	-1588	214.5
557	S34	-1604	122.5
558	S33	-1620	214.5
559	S32	-1636	122.5
560	S31	-1652	214.5
561	S30	-1668	122.5
562	S29	-1684	214.5
563	S28	-1700	122.5
564	S27	-1716	214.5
565	S26	-1732	122.5
566	S25	-1748	214.5
567	S24	-1764	122.5
568	S23	-1780	214.5
569	S22	-1796	122.5
570	S21	-1812	214.5
571	S20	-1828	122.5
572	S19	-1844	214.5
573	S18	-1860	122.5
574	S17	-1876	214.5
575	S16	-1892	122.5
576	S15	-1908	214.5
577	S14	-1924	122.5
578	S13	-1940	214.5

579	S12	-1956	122.5
580	S11	-1972	214.5
581	S10	-1988	122.5
582	S9	-2004	214.5
583	S8	-2020	122.5
584	S7	-2036	214.5
585	S6	-2052	122.5
586	S5	-2068	214.5
587	S4	-2084	122.5
588	S3	-2100	214.5
589	S2	-2116	122.5
590	S1	-2132	214.5
591	DUMMY	-2148	122.5
592	DUMMY	-2164	214.5
593	DUMMY	-2180	122.5
594	DUMMY	-2196	214.5
595	DUMMY	-2212	122.5
596	DUMMY	-2228	214.5
597	DUMMY	-2244	122.5
598	DUMMY	-2260	214.5
599	DUMMY	-2276	122.5
600	DUMMY	-2292	214.5
601	DUMMY	-2308	122.5
602	DUMMY	-2324	214.5
603	DUMMY	-2340	122.5
604	DUMMY	-2356	214.5
605	DUMMY	-2372	122.5
606	DUMMY	-2388	214.5
607	DUMMY	-2404	122.5
608	DUMMY	-2420	214.5
609	DUMMY	-2436	122.5
610	DUMMY	-2452	214.5
611	DUMMY	-2468	122.5
612	DUMMY	-2484	214.5
613	DUMMY	-2500	122.5
614	DUMMY	-2516	214.5
615	DUMMY	-2532	122.5
616	DUMMY	-2548	214.5
617	DUMMY	-2564	122.5
618	DUMMY	-2580	214.5
619	DUMMY	-2596	122.5
620	DUMMY	-2612	214.5
621	DUMMY	-2628	122.5
622	DUMMY	-2644	214.5
623	DUMMY	-2660	122.5
624	DUMMY	-2676	214.5
625	DUMMY	-2692	122.5
626	DUMMY	-2708	214.5
627	DUMMY	-2724	122.5
628	DUMMY	-2740	214.5
629	DUMMY	-2756	122.5
630	DUMMY	-2772	214.5
631	DUMMY	-2788	122.5
632	DUMMY	-2804	214.5
633	DUMMY	-2820	122.5
634	DUMMY	-2836	214.5
635	DUMMY	-2852	122.5
636	DUMMY	-2868	214.5
637	DUMMY	-2884	122.5
638	DUMMY	-2900	214.5
639	DUMMY	-2916	122.5
640	DUMMY	-2932	214.5
641	DUMMY	-2948	122.5
642	DUMMY	-2964	214.5
643	DUMMY	-2980	122.5
644	DUMMY	-2996	214.5
645	DUMMY	-3012	122.5
646	DUMMY	-3028	214.5
647	DUMMY	-3044	122.5
648	DUMMY	-3060	214.5
649	DUMMY	-3076	122.5
650	DUMMY	-3092	214.5
651	DUMMY	-3108	122.5
652	DUMMY	-3124	214.5
653	DUMMY	-3140	122.5
654	DUMMY	-3156	214.5
655	DUMMY	-3172	122.5
656	DUMMY	-3188	214.5
657	DUMMY	-3204	122.5
658	DUMMY	-3220	214.5
659	DUMMY	-3236	122.5
660	DUMMY	-3252	214.5
661	DUMMY	-3268	122.5
662	DUMMY	-3284	214.5
663	DUMMY	-3300	122.5
664	DUMMY	-3316	214.5
665	DUMMY	-3332	122.5
666	DUMMY	-3348	214.5
667	DUMMY	-3364	122.5
668	DUMMY	-3380	214.5
669	DUMMY	-3396	122.5
670	DUMMY	-3412	214.5
671	DUMMY	-3428	122.5
672	DUMMY	-3444	214.5
673	G1	-3460	122.5
674	G3	-3476	214.5
675	G5	-3492	122.5
676	G7	-3508	214.5
677	G9	-3524	122.5
678	G11	-3540	214.5
679	G13	-3556	122.5
680	G15	-3572	214.5
681	G17	-3588	122.5
682	G19	-3604	214.5
683	G21	-3620	122.5
684	G23	-3636	214.5
685	G25	-3652	122.5
686	G27	-3668	214.5
687	G29	-3684	122.5
688	G31	-3700	214.5
689	G33	-3716	122.5
690	G35	-3732	214.5
691	G37	-3748	122.5
692	G39	-3764	214.5
693	G41	-3780	122.5
694	G43	-3796	214.5
695	G45	-3812	122.5
696	G47	-3828	214.5
697	G49	-3844	122.5
698	G51	-3860	214.5
699	G53	-3876	122.5
700	G55	-3892	214.5
701	G57	-3908	122.5
702	G59	-3924	214.5
703	G61	-3940	122.5
704	G63	-3956	214.5
705	G65	-3972	122.5
706	G67	-3988	214.5
707	G69	-4004	122.5
708	G71	-4020	214.5
709	G73	-4036	122.5
710	G75	-4052	214.5
711	G77	-4068	122.5
712	G79	-4084	214.5
713	G81	-4100	122.5
714	G83	-4116	214.5
715	G85	-4132	122.5
716	G87	-4148	214.5
717	G89	-4164	122.5
718	G91	-4180	214.5
719	G93	-4196	122.5
720	G95	-4212	214.5
721	G97	-4228	122.5
722	G99	-4244	214.5
723	G101	-4260	122.5
724	G103	-4276	214.5
725	G105	-4292	122.5

726	G107	-4308	214.5	736	G127	-4468	214.5	746	G147	-4628	214.5
727	G109	-4324	122.5	737	G129	-4484	122.5	747	G149	-4644	122.5
728	G111	-4340	214.5	738	G131	-4500	214.5	748	G151	-4660	214.5
729	G113	-4356	122.5	739	G133	-4516	122.5	749	G153	-4676	122.5
730	G115	-4372	214.5	740	G135	-4532	214.5	750	G155	-4692	214.5
731	G117	-4388	122.5	741	G137	-4548	122.5	751	G157	-4708	122.5
732	G119	-4404	214.5	742	G139	-4564	214.5	752	G159	-4724	214.5
733	G121	-4420	122.5	743	G141	-4580	122.5	753	G161	-4740	122.5
734	G123	-4436	214.5	744	G143	-4596	214.5	754	DUMMY	-4756	214.5
735	G125	-4452	122.5	745	G145	-4612	122.5	755	DUMMY	-4772	122.5

4. Interface Description

4.1. Interface Type Selection

The selection of a given interfaces are done by setting P68, IM2, IM1, and IM0 pins as show in below tables.

Table 4-1 Interface Type Selection

P68	IM2	IM1	IM0	Interface	Read Back Selection
X	0	X	X	Serial interface	Via the read instruction (8-bit read parameter)
0	1	X	X	8080 MCU 8-bit Parallel	RDX strobe (8-bit read data and 8-bit read parameter)
1	1	X	X	6800 MCU 8-bit Parallel	RDX strobe (8-bit read data and 8-bit read parameter)

4.2. Serial Interface

The Module uses a 3-wire 9-bit serial interface or 4-pins/8-bit bi-directional interface for communication between the micro controller and the LCD driver chip. The 3-pins serial use: CSX (chip enable), SCL (serial clock) and SDA (serial data input/output) and the 4-pins serial use: CSX (chip enable), D/CX (data/ command select), SCL (serial clock), and SDA (serial data input/output).

Table 4-2 Serial Interface Type Selection

IM2	SPI4W	Interface	Read back selection
0	0	3-Pins Serial Interface(or 2 Data Line SPI)	Via the read instruction (8-bit , 24-bit and 32-bit read parameter)
0	1	4-Pins Serial Interface	Via the read instruction (8-bit , 24-bit and 32-bit read parameter)

4.2.1 Command/Data Write

The write mode of the interface means the micro controller writes commands and data to the LCD driver. 3-pins serial data packet contains a control bit D/CX and a transmission byte, but in 4-pins serial case, data packet contains just transmission byte and control bit D/CX is transferred by the D/CX pin. If D/CX is “low”, the transmission byte is interpreted as a command byte. If D/CX is “high”, the

transmission byte is stored in the display data RAM (Memory write command), or command register as parameter.

Any instruction can be sent in any orders to the Driver. The MSB is transmitted first. The serial interface is initialized when CSX is high status. In this state, SCL clock pulse or SDA data have no effect. A falling edge on CSX enables the serial interface and indicated the start of data transmission.

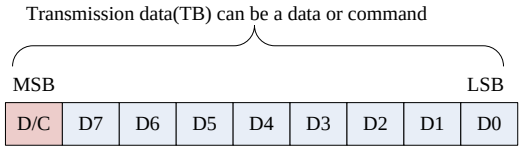


Figure 4-2-1-1 SPI 3-Wire Write Data Format

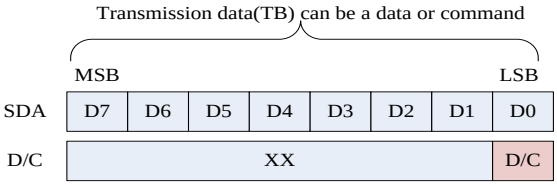


Figure 4-2-1-2 SPI 4-Wire Write Data Format

When CSX is “high”, SCL clock is ignored. At the falling edge of CSX, SCL can be high or low. SDA is sampled at the rising edge of SCL. D/CX indicates, whether the byte is command code (D/CX=’0’) or parameter/RAM data (D/CX=’1’). If CSX stays low after the last bit of command/data byte, the serial interface expects the D/CX bit (3-pin serial interface) or D7 (4-pins serial interface) of the next byte at the next rising edge of SCL.

4.2.2 Command/Data Read

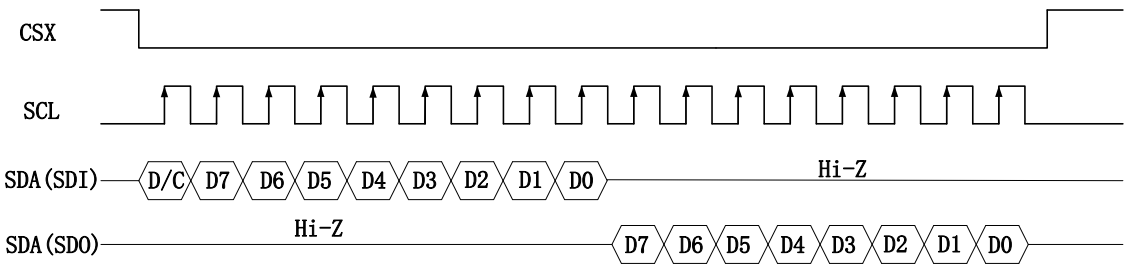


Figure 4-2-2-1 SPI 3-Wire 8-bit Read Operation
(For ‘DAH/DBH/DCH/0Ah/0Bh/0Ch/0Dh/0Eh/0Fh’ Command)

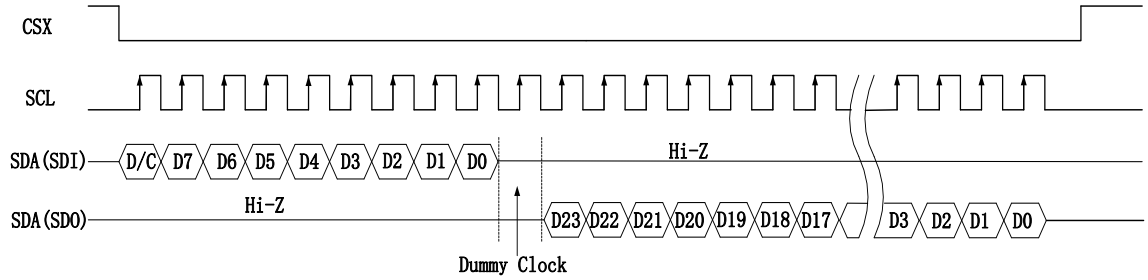


Figure 4-2-2-2 SPI 3-Wire 24-bit Read Operation
(For '04H' Command with one dummy clock)

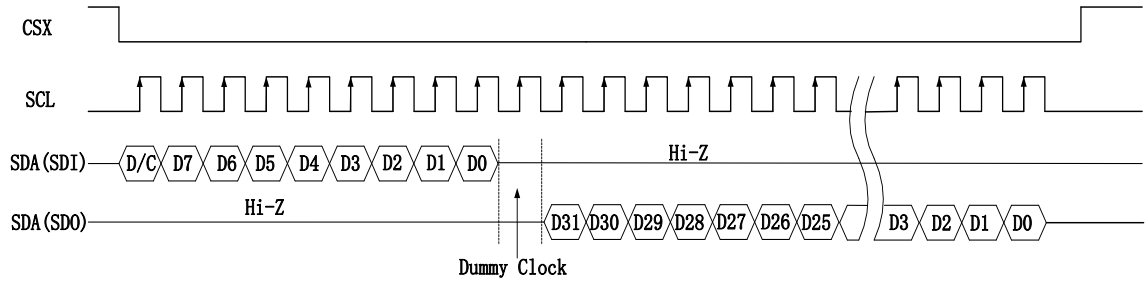


Figure 4-2-2-3 SPI 3-Wire 32-bit Read Operation
(For '09H' Command with one dummy clock)

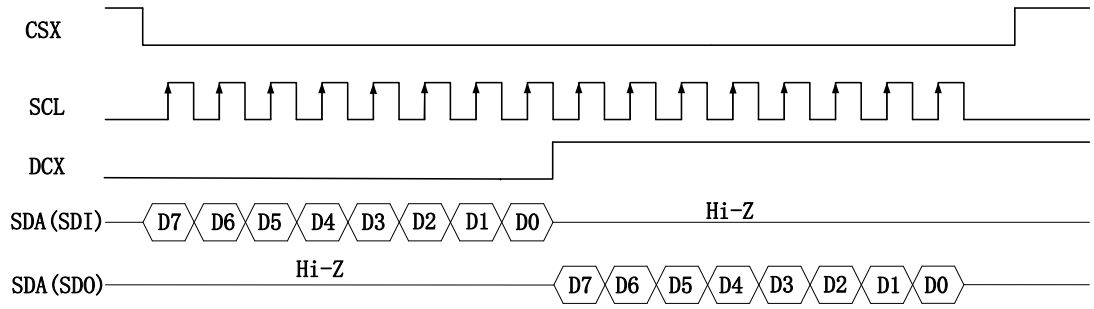


Figure 4-2-2-4 SPI 4-Wire 8-bit Read Operation
(For 'DAH/DBH/DCH/0Ah/0Bh/0Ch/0Dh/0Eh/0Fh' Command)

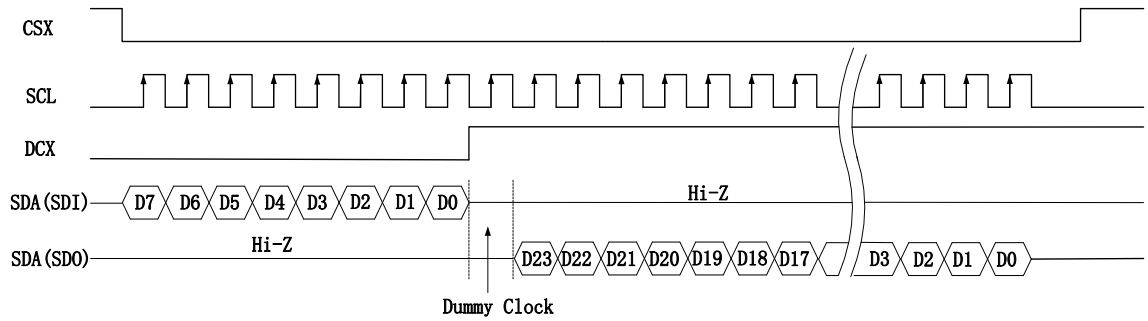


Figure 4-2-2-5 SPI 4-Wire 24-bit Read Operation
(For '04H' Command with one dummy clock)

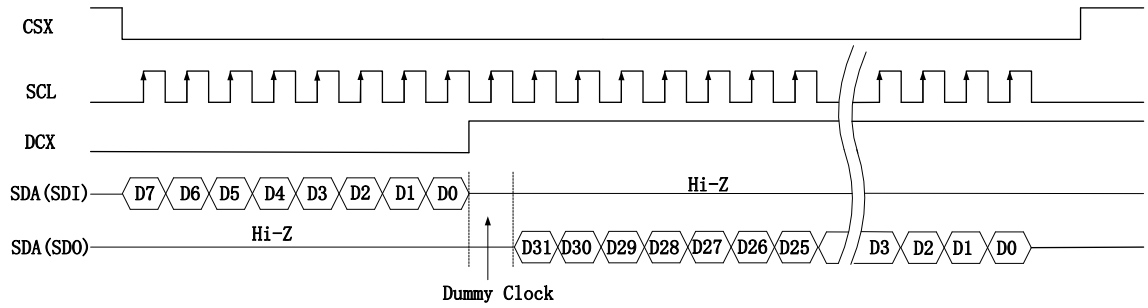


Figure 4-2-2-6 SPI 4-Wire 32-bit Read Operation
(For '09H' Command with one dummy clock)

4.3. 8080-Series Parallel Interface (P68='0')

The MCU uses an 11-wires 8-data parallel interface. The chip-select CSX (active low) enables and disables the parallel interface. RESX (active low) is an external reset signal. WRX is the parallel data write, RDX is the parallel data read and D[7:0] is parallel data.

The graphics controller chip reads the data at the rising edge of WRX signal. The D/CX is the data/command flag. When D/CX='1', D[7:0] bits are display RAM data or command parameters. When D/C='0', D[7:0] bits are commands.

The 8080-series bi-direction interface can be used for communication between the micro-controller and LCD driver chip. The selection of this interface is done when P68 pin is low state (VSS). Interface bus width can be selected with IM2, IM1 and IM0. The interface functions of 8080-series parallel interface are given in Table 4-3-1.

Table 4-3-1 8080 MCU Operation

P68	IM2	IM1	IM0	Interface	D/CX	RDX	WRX	Function
0	1	X	X	8-bit parallel	L	H	R	Write 8-bit command
					H	H	R	Write 8-bit display data or 8-bit parameter
					H	R	H	Read 8-bit display data or 8-bit parameter

Note: Reading operation applied for command code: DAh, DBh, DCh, 04h, 09h, 0Ah, 0Bh, 0Ch, 0Dh, 0Eh, 0Fh

4.3.1. Write Cycle/Sequence

The write cycle means that the host writes information (command or/and data) to the display via the interface. Each write cycle (WRX high-low-high sequence) consists of 3 control (D/CX, RDX, WRX) and data signals (D[7:0]). D/CX bit is a control signal, which tells if the data is a command or a data. The data signals are a command if the control signal is low (= '0') and vice versa it is data (= '1'). The write cycle is described in the following figure.

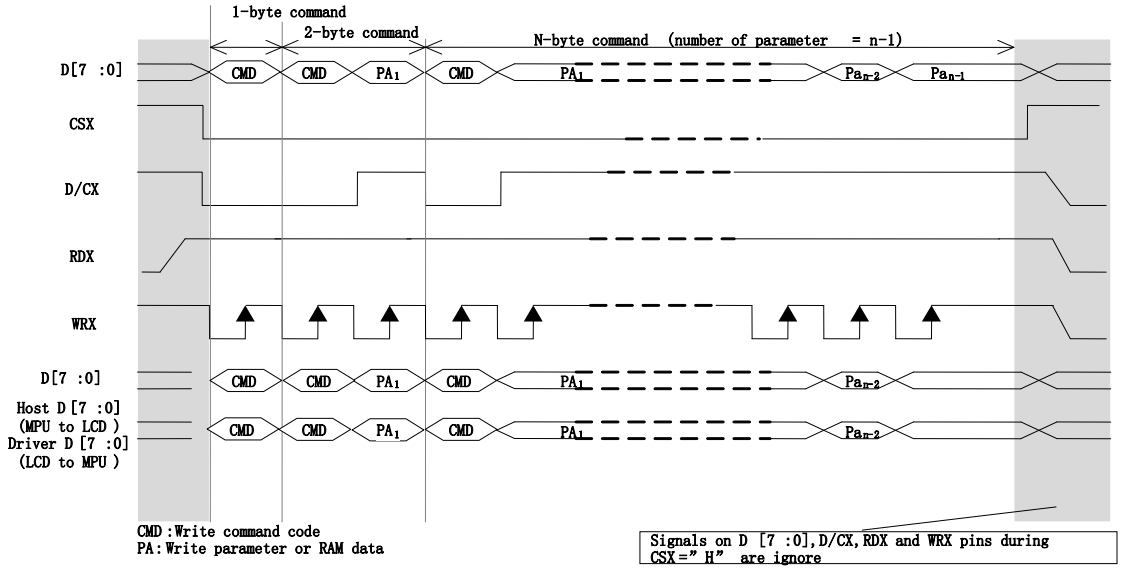


Figure 4-3-1 8080 MCU Write sequence

4.3.2. Read Cycle/Sequence

The read cycle (RDX high-low-high sequence) means that the host reads information from the display via interface. The display sends data (D[7:0]) to the host when there is a falling edge of RDX and the host reads data when there is a rising edge of RDX.

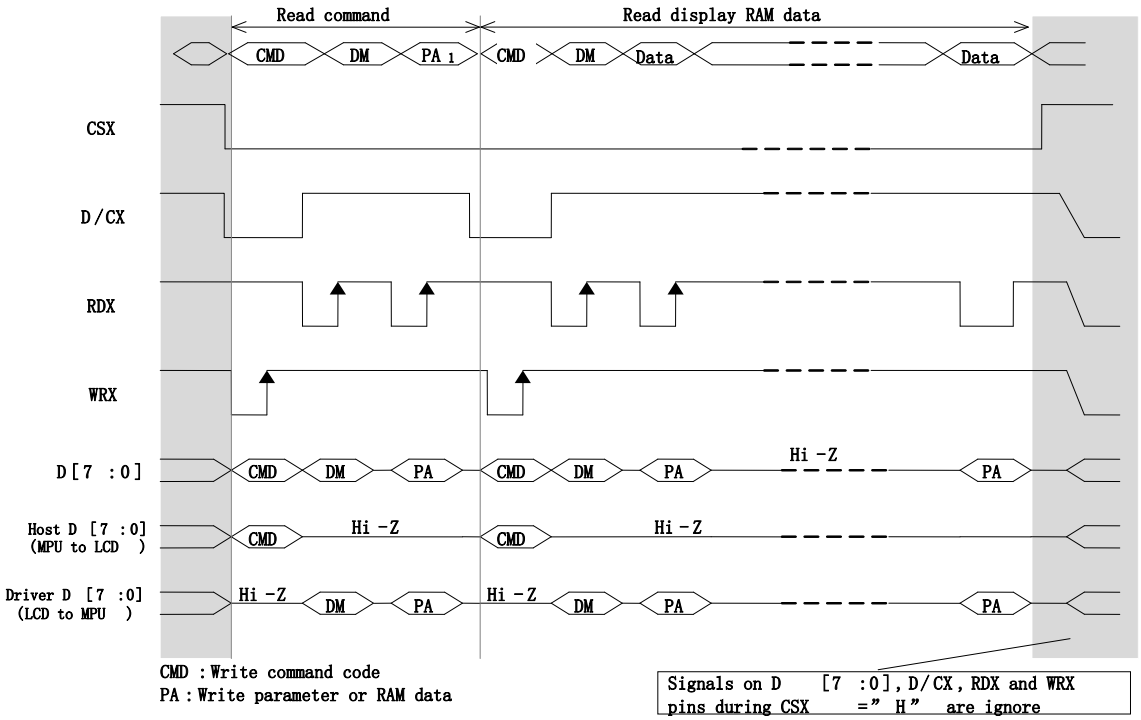


Figure 4-3-2 8080 Read sequence

4.4. 6800-Series Parallel Interface (P68='1')

The MCU uses a 11-wires 8-data parallel interface. The chip-select CSX (active low) enables and disables the parallel interface. RESX (active low) is an external reset signal. WRX is read/write flag, RDX is the parallel data read/write enable and D[7:0] is parallel data.

The Graphics Controller Chip reads the data at the falling edge of RDX signal when WRX = '1' and writes the data at the falling of the RDX signal when R/WX='0'. The D/CX is the data or command flag. When D/CX='1', D[7:0] bits are display RAM data or command parameters. When D/C='0', D[7:0] bits are commands.

The 6800-series bi-direction interface can be used for communication between the micro- controller and LCD driver chip. The selection of this interface is done when P68 pin is high state (IOVCC). Interface bus width can be selected with IM2, IM1 and IM0. The interface functions of 6800-series parallel interface are given in Table 4-4-1.

Table 4-4-1 6800 MCU Operation

P68	IM2	IM1	IM0	Interface	D/CX	WRX	RDX	Function
1	1	X	X	8-bit parallel	L	L	F	Write 8-bit command
					H	L	F	Write 8-bit display data or 8-bit parameter
					H	H	F	Read 8-bit display data or 8-bit parameter

Note: Reading operation applied for command code: DAh, DBh, DCh, 04h, 09h, 0Ah, 0Bh, 0Ch, 0Dh, 0Eh, 0Fh

4.4.1. Write Cycle Sequence

The write cycle means that the host writes information (command or/and data) to the display via the interface. Each write cycle consists of 3 control (D/CX, RDX, WRX) and data signals (D[7:0]). D/CX bit is a control signal, which tells if the data is a command or a data. The data signals are a command if the control signal is low (D/CX= ‘0’), vice versa, it is data (D/CX= ‘1’).WRX work as write or read flag, WRX=’0’ stands for write operation, WRX=’1’ means read operation. The detail write cycle is described in the Figure 4-4-1.

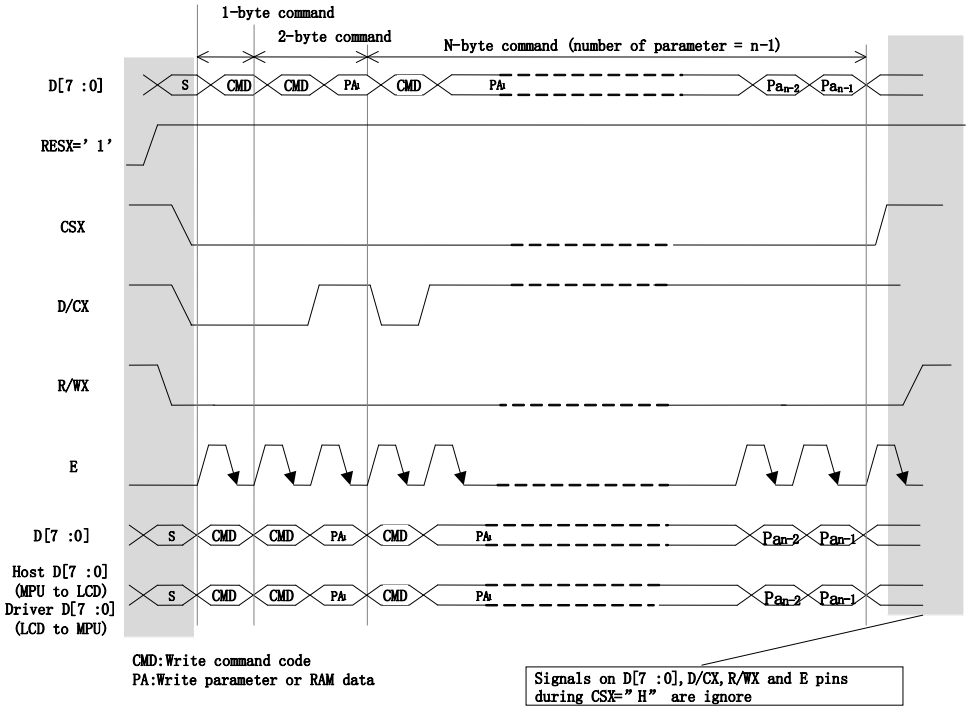


Figure 4-4-1 6800 Write Sequence

4.4.2. Read Cycle Sequence

The read cycle means that the host reads information (commend or/and data) to the display via the interface. Each read cycle consists of 3 control (D/CX, WRX, RDX) and data (D[7:0]). D/CX bit is control signal, which tells if the data is a command or a data. The data signals are the command if the control signal is low and vice versa it is data .WRX work as write or read flag,WRX=’0’ stands for write operation,WRX=’1’ means read operation.The detail read cycle is described in the figure Figure 4-4-2.

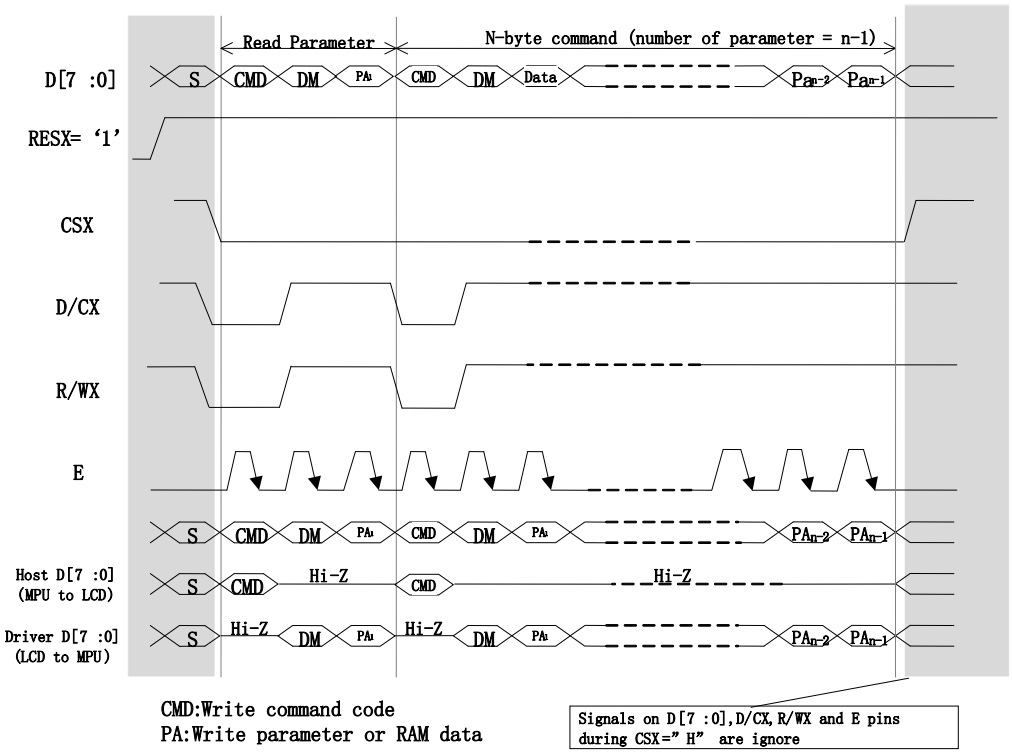


Figure 4-4-2 6800 Read Sequence

4.5. Serial Interface Recovery Function

If there is a break in data transmission by RESX pulse, while transferring a Command or Frame Memory Data or Multiple Parameter command Data, before Bit D0 of the byte has been completed, then Driver will reject the previous bits and should reset the interface that it will be ready to receive command data again when the chip select line (CSX) is next activated after RESX have been High state. See the following example.

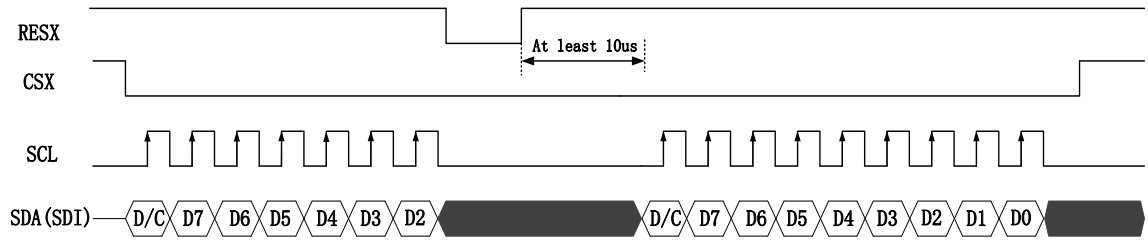


Figure 4-5-1 Serial interface recovery

If there is a break in data transmission by CSX pulse, while transferring a Command or Frame Memory Data or Multiple Parameter command Data, before Bit D0 of the byte has been completed, Then the Driver will reject the previous bits and should reset the interface such that it will be ready to receive the same byte re-transmitted when the chip select line (CSX) is next activated. See the following example.

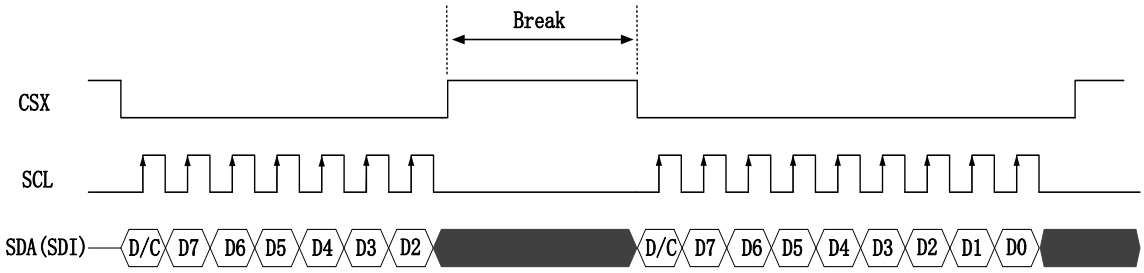


Figure 4-5-2 Serial interface recovery2

If 1, 2 or more parameter command is being sent and a break occurs while sending any parameter before the last one and if the host then sends a new command rather than re-transmitting the parameter that was interrupted, then the parameters that were successfully sent are stored and the parameter where the break occurred is rejected. The interface is ready to receive next byte as show below.

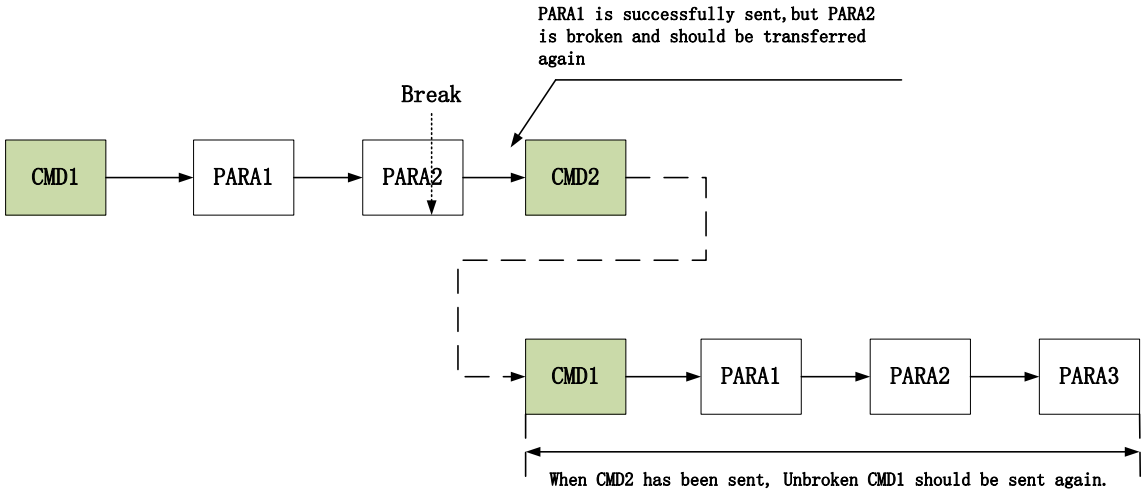


Figure 4-5-3 Serial interface recovery3

4.6. Display Data Transmission Pause

It will be possible when transferring a Command, Frame Memory Data or Multiple Parameter Data to invoke a pause in the data transmission. If the Chip Select Line is released after a whole byte of a Frame Memory Data or Multiple Parameter Data has been completed, then the Display Module will wait and continue the Frame Memory Data or Parameter Data Transmission from the point where it was paused. If the Chip Select Line is released after a whole byte of a command has been completed, then the Display Module will receive either the command's parameters (if appropriate) or a new command when the Chip Select Line is next enabled as shown below:

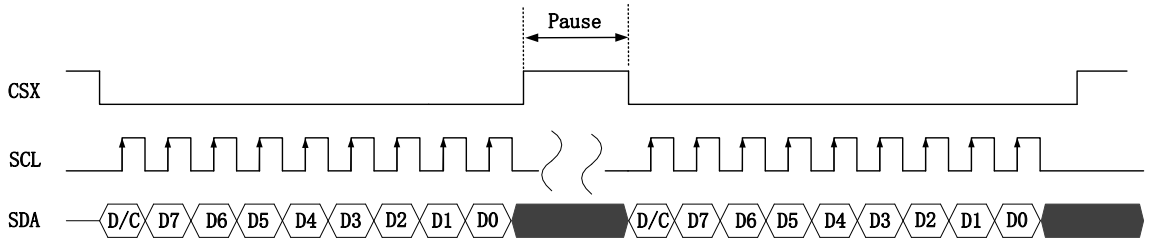


Figure 4-6 Serial interface recovery3

4.7. Display Data Transfer Mode

The data format is described for each interface. Data can be downloaded to the Frame Memory by 2 methods.

Method1:

The Image data is sent to the Frame Memory in successive Frame writes, each time the Frame Memory is filled, the Frame Memory pointer is reset to the start point and the next Frame is written.

Start Frame Write	Image Data Frame1	Image Data Frame2	Image Data Frame3

Method2:

After image data is sent and at the end of each Frame Memory download, a command is sent to stop Frame Memory Write. Then Start Memory Write command

is sent, and a new Frame is downloading.

Start Frame Write	Image Data Frame1	Any Command	Start Frame Write	Image Data Frame2	Any Command	. . . .	Any Command

4.8. Display Data Color Coding

4.8.1. Serial Interface

Different display data formats are available for three colors depth supported by the LCM listed below.

- 256 colors, RGB 3-3-2-bits input
- 4k colors, RGB 4-4-4-bits input
- 65K colors, RGB 5-6-5-bits input

Data format and Reconstruct ways are shown as follow figures:

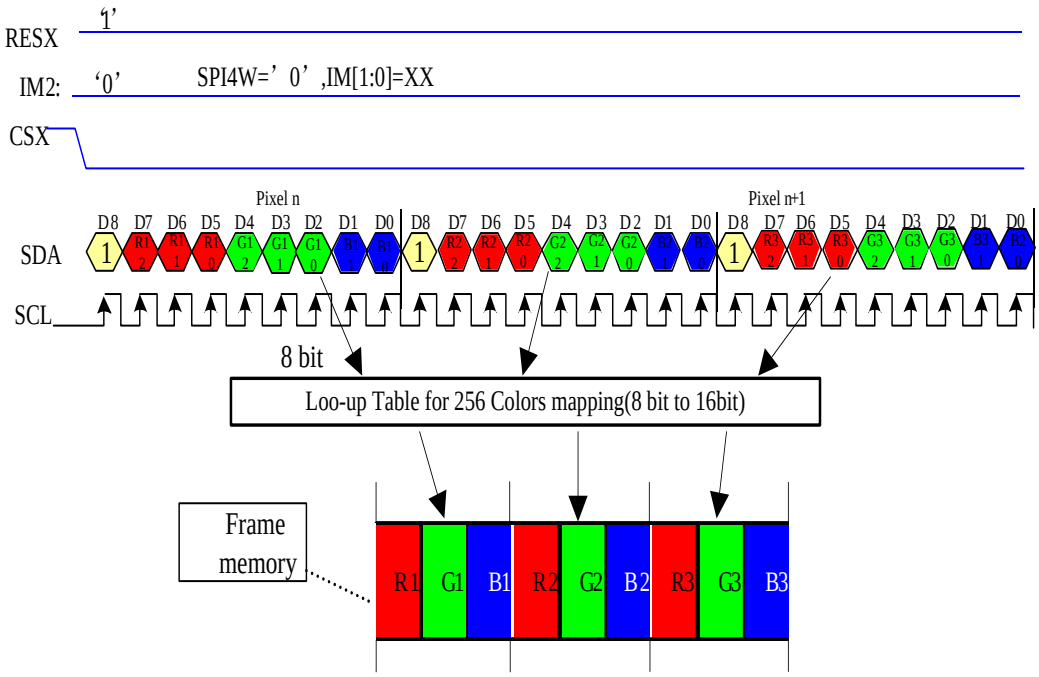


Figure 4-8-1-1 SPI 3-Wire 9-bit RGB3-3-2 Write Data Format

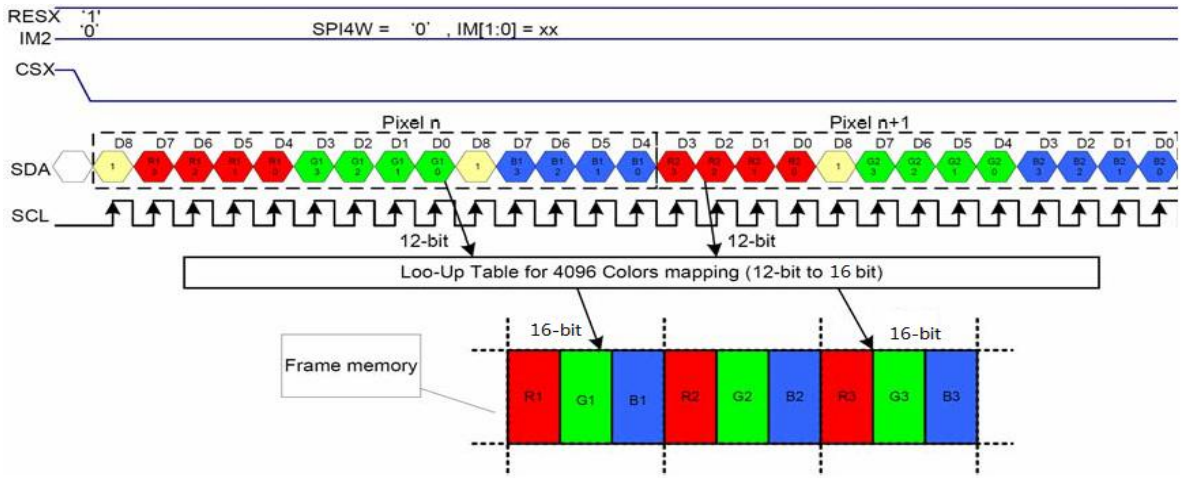


Figure 4-8-1-2 SPI 3-Wire 9-bit RGB4-4-4 Write Data Format

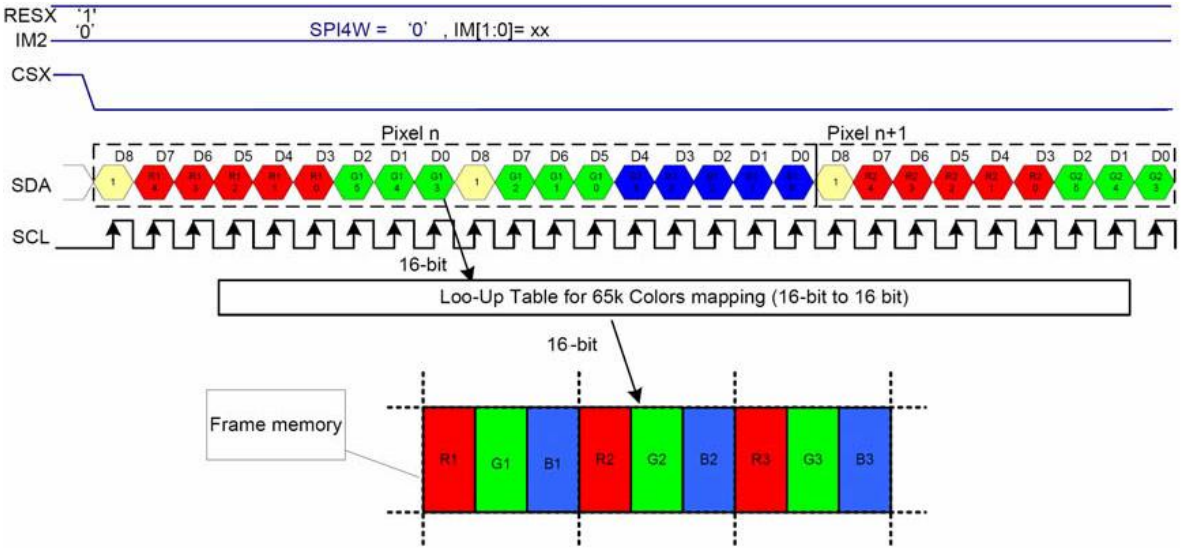


Figure 4-8-1-3 SPI 3-Wire 9-bit RGB5-6-5 Write Data Format

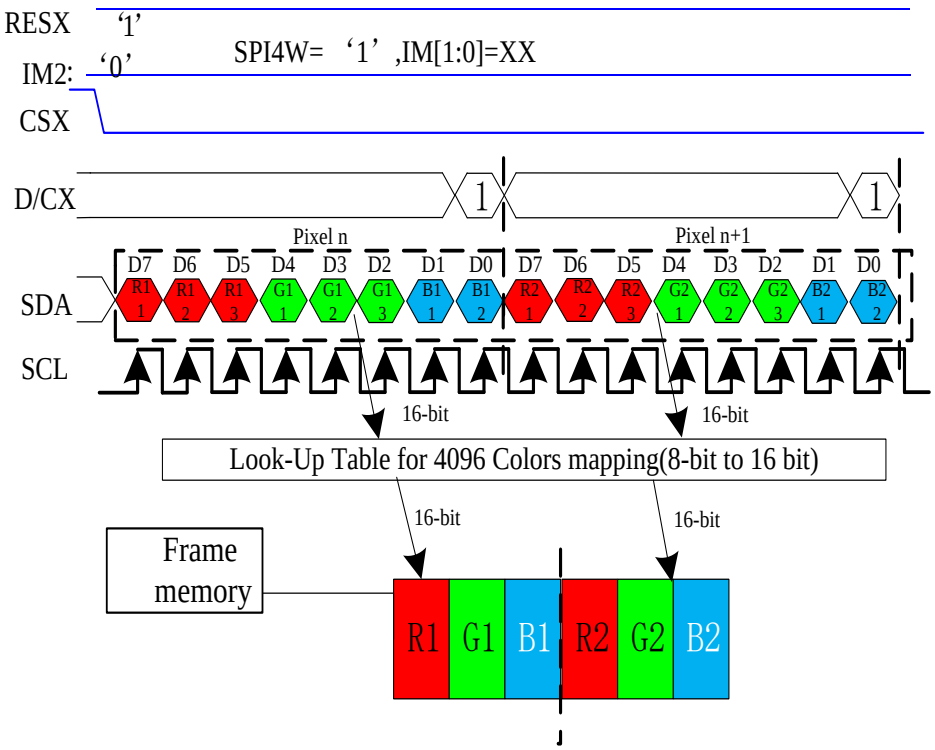


Figure 4-8-1-4 SPI 4-Wire 8-bit RGB3-3-2 Write Data Format

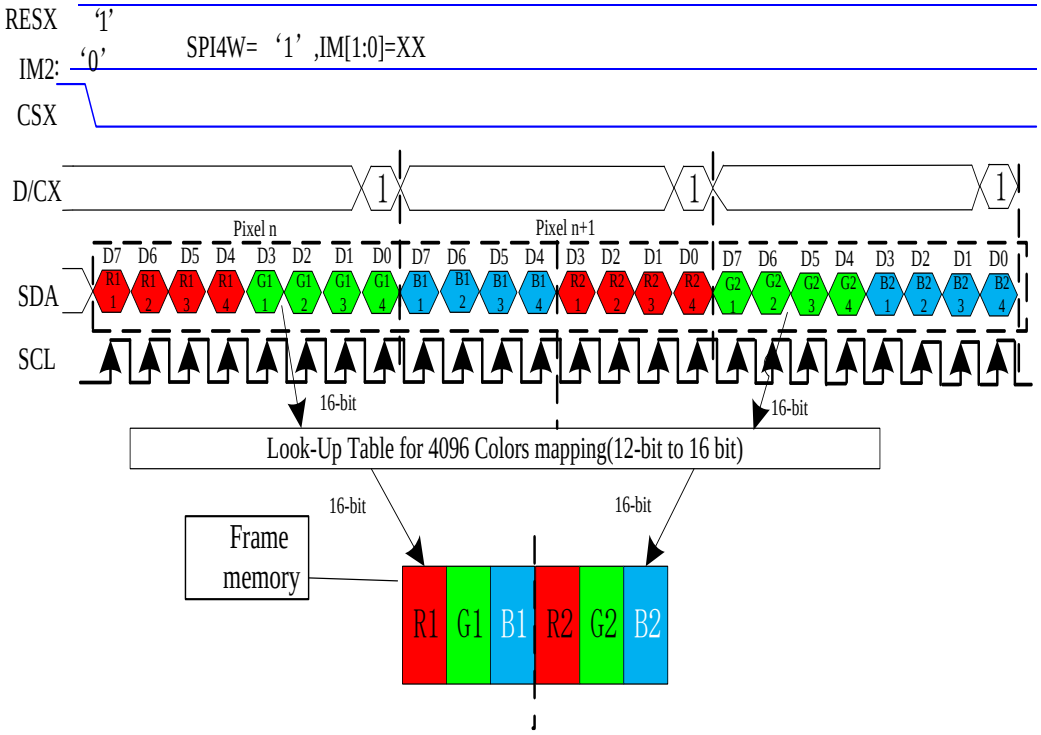


Figure 4-8-1-4 SPI 4-Wire 8-bit RGB4-4-4 Write Data Format

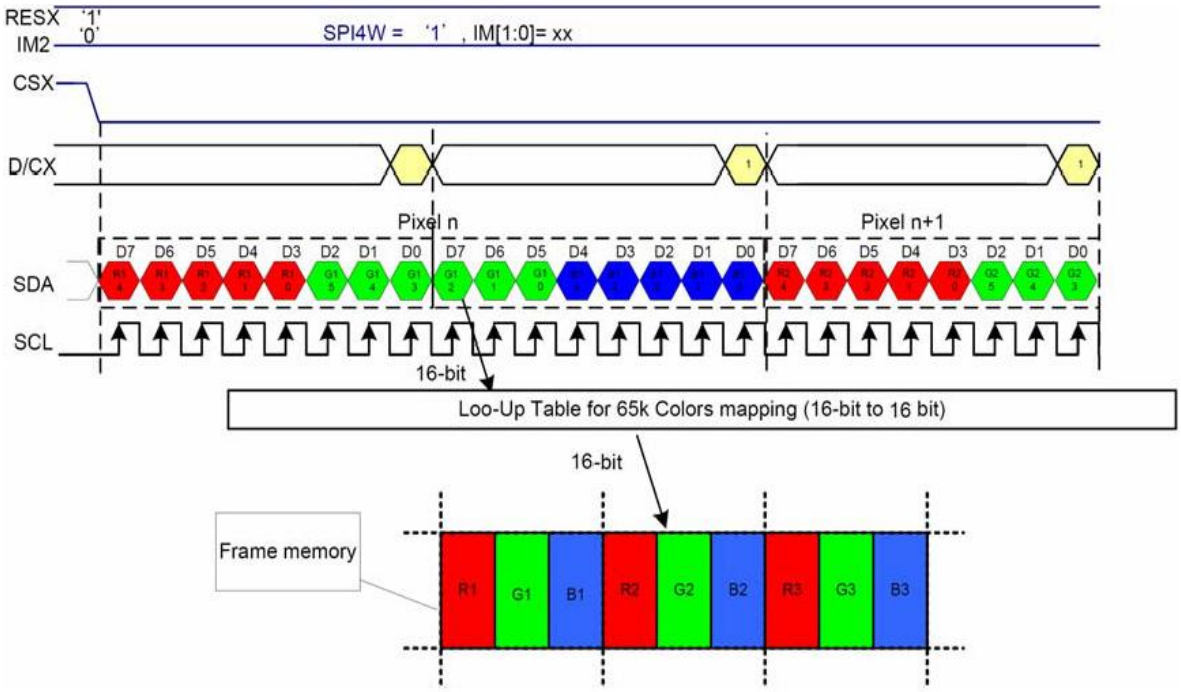


Figure 4-8-1-4 SPI 4-Wire 8-bit RGB5-6-5 Write Data Format

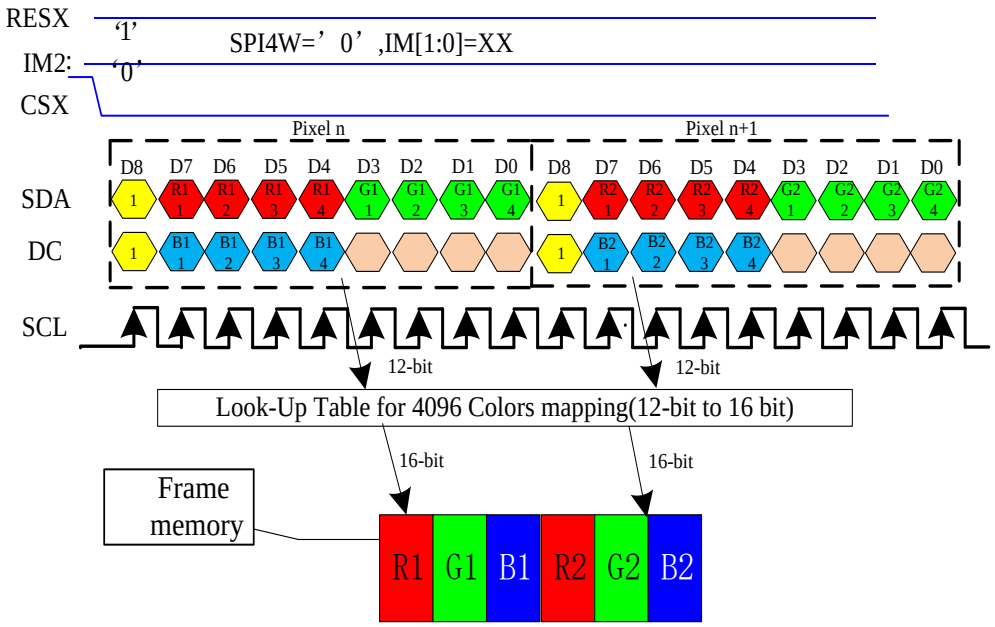


Figure 4-8-1-4 SPI_double 9-bit RGB4-4-4 Write Data Format

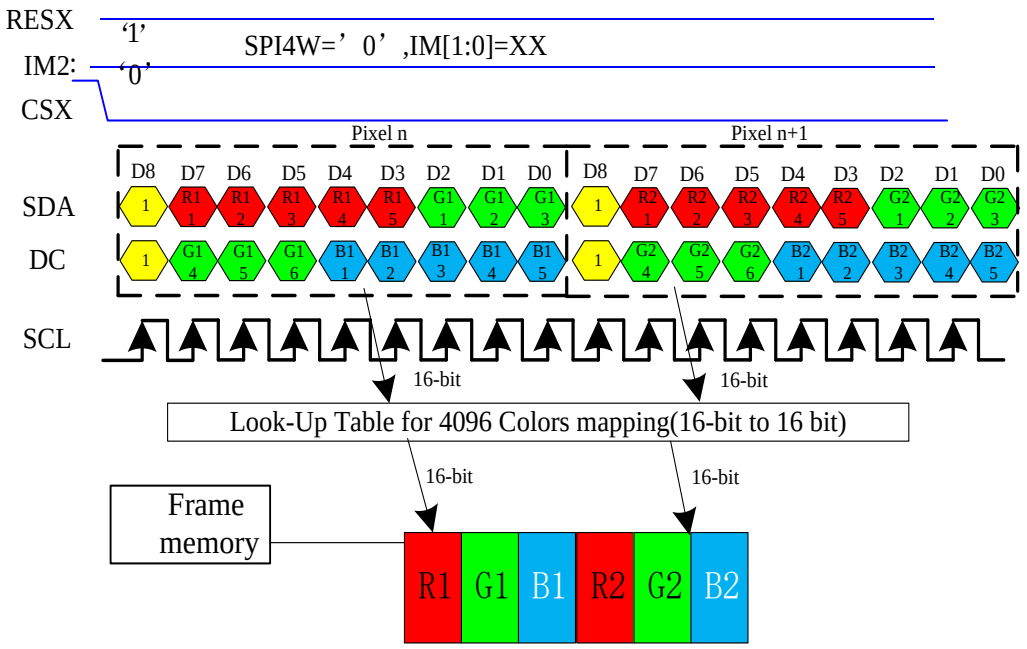


Figure 4-8-1-4 SPI_double 9-bit RGB5-6-5 Write Data Format

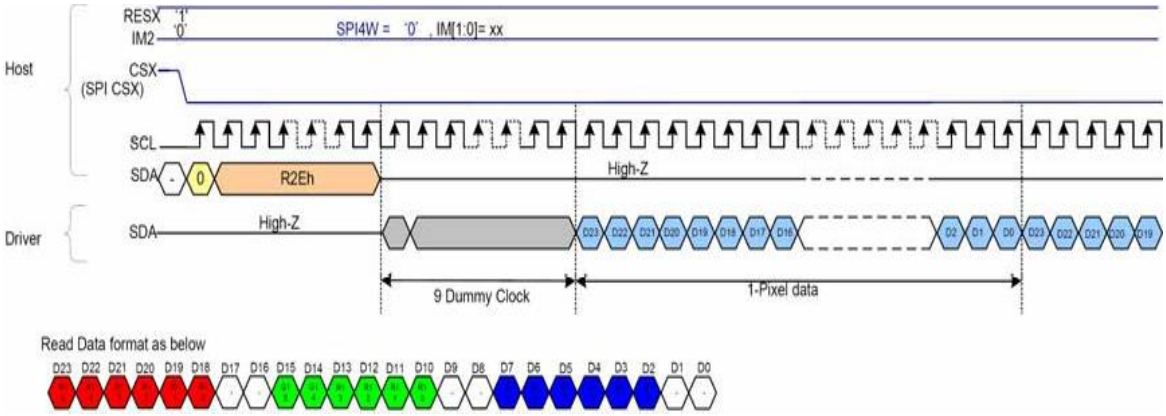


Figure 4-8-1-5 SPI 3-Wire Read Data Format

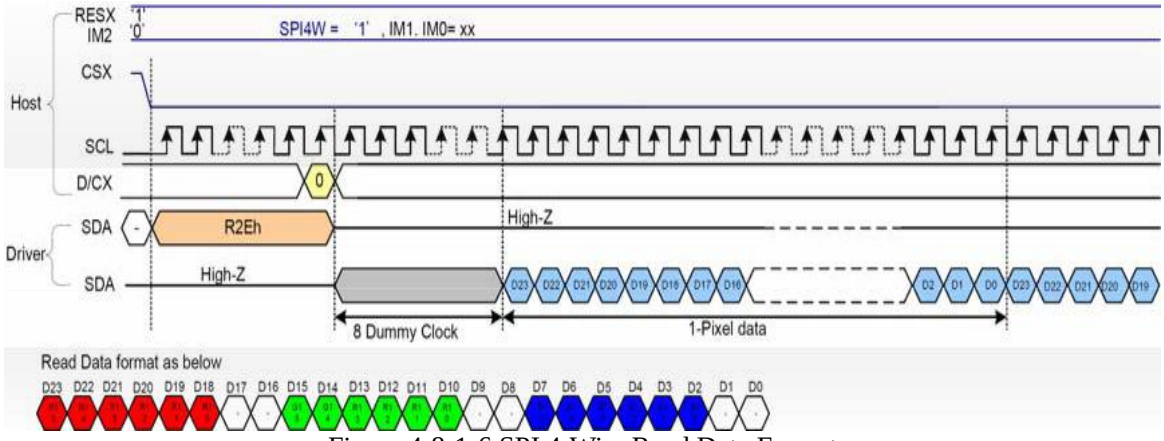


Figure 4-8-1-6 SPI 4-Wire Read Data Format

4.8.2 Parallel 8-bit Interface

Different display data formats are available for three colors depth supported by the LCM listed below.

- 4k colors, RGB 4-4-4-bits input
- 65K colors, RGB 5-6-5-bits input

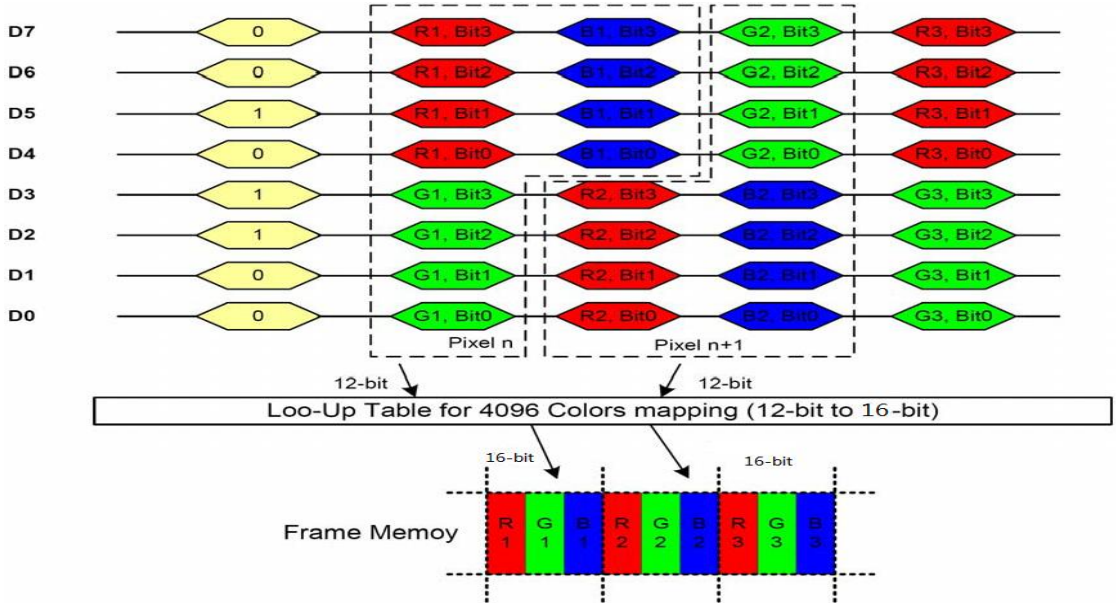


Figure 4-8-2-1 Two pixel (RGB4-4-4) in each three 8-bit write data packets

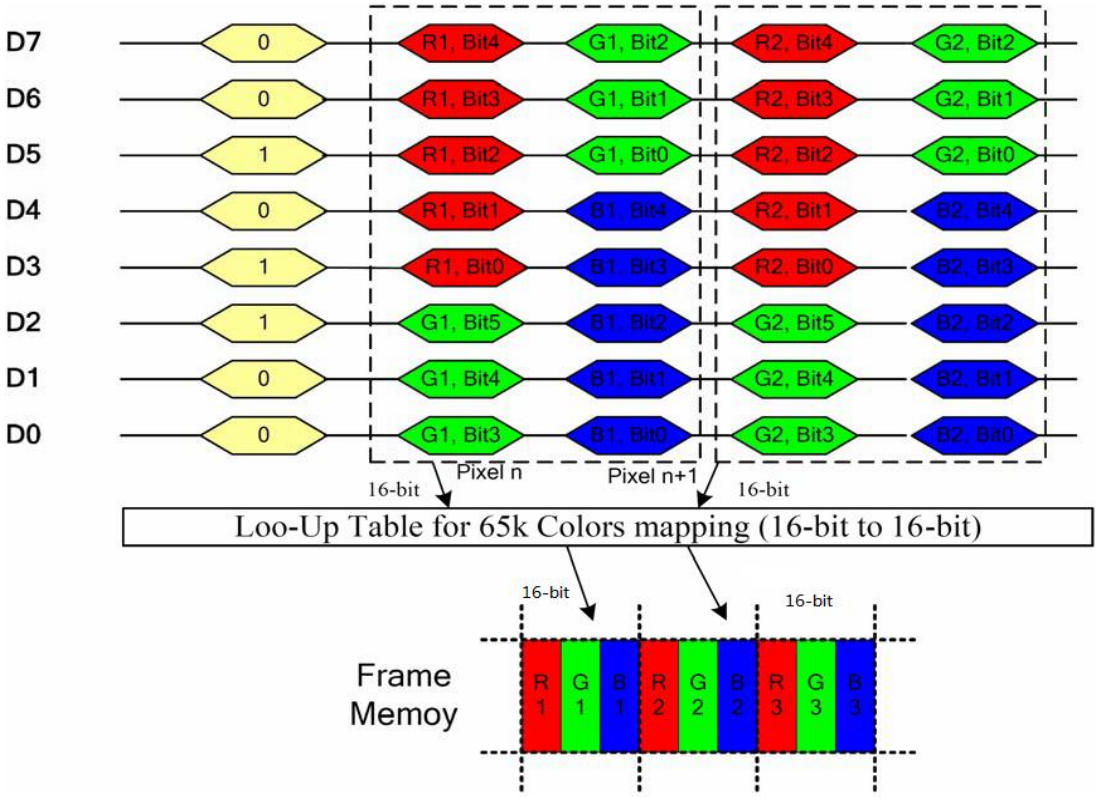
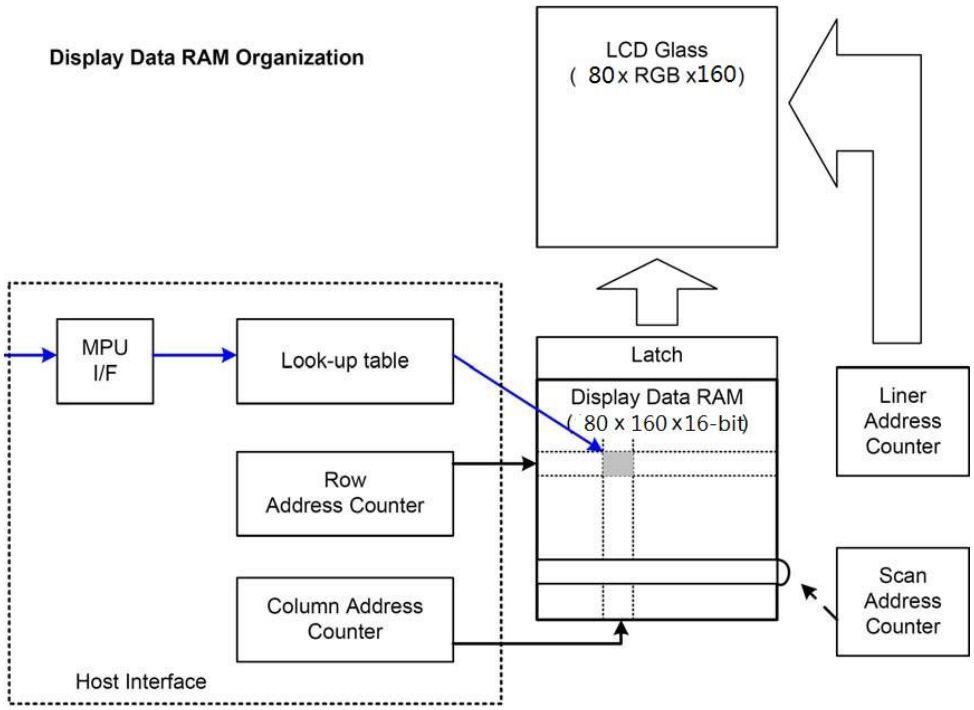


Figure 4-8-2-2 One pixel (RGB5-6-5) in each two 8-bit write data packets

5. Display Data with SRAM

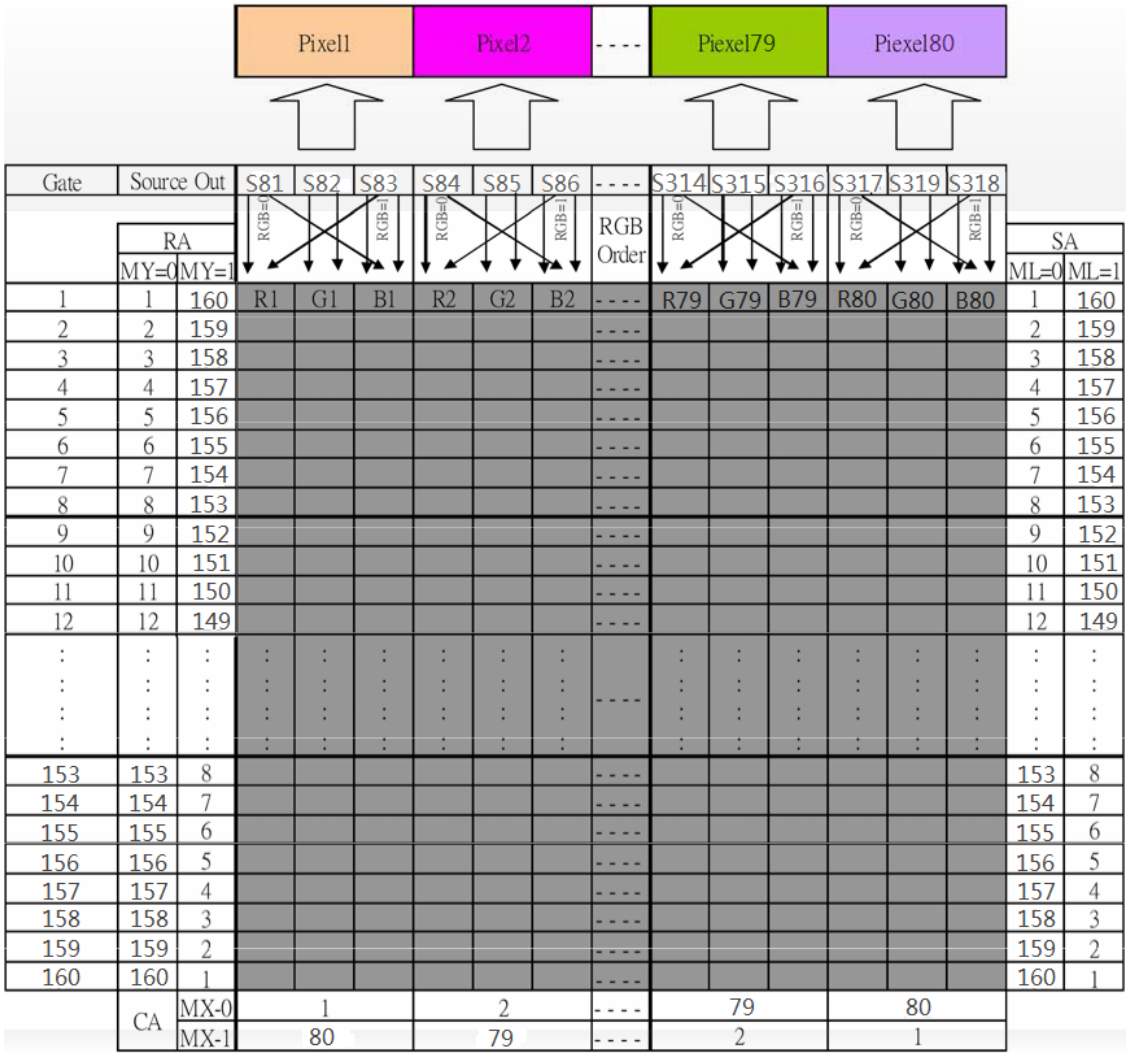
5.1. SRAM Organization

The display data SRAM stores display dots and consists of 204,800 bits (80x16x160 bits). There is no restriction on access to the RAM even when the display data on the same address is loaded to “Source”. There will be no abnormal visible effect on the display when there is a simultaneous Panel Read and Interface Read or Write to the same location of the Frame Memory.



5.2. Memory To Display Mapping

Display data are writen in different direction according to different MX/MY settings, and scan read from SRAM also controlled by ML configuration. The detail as below figure shows:



Gate	Source Out		S81	S82	S83	S84	S85	S86	- - - -	S314	S315	S316	S317	S319	S318		
	RA		RGB=0		RGB=1	RGB=0		RGB=1	RGB Order	RGB=0		RGB=1	RGB=0		RGB=1	SA	
	MY=0	MY=1														ML=0	ML=1
1	1	160	R1	G1	B1	R2	G2	B2	- - - -	R79	G79	B79	R80	G80	B80	1	160
2	2	159							- - - -							2	159
3	3	158							- - - -							3	158
4	4	157							- - - -							4	157
5	5	156							- - - -							5	156
6	6	155							- - - -							6	155
7	7	154							- - - -							7	154
8	8	153							- - - -							8	153
9	9	152							- - - -							9	152
10	10	151							- - - -							10	151
11	11	150							- - - -							11	150
12	12	149							- - - -							12	149
:	:	:	:	:	:	:	:	:	- - - -	:	:	:	:	:	:	:	:
:	:	:	:	:	:	:	:	:		:	:	:	:	:	:	:	:
:	:	:	:	:	:	:	:	:		:	:	:	:	:	:	:	:
:	:	:	:	:	:	:	:	:		:	:	:	:	:	:	:	:
153	153	8							- - - -							153	8
154	154	7							- - - -							154	7
155	155	6							- - - -							155	6
156	156	5							- - - -							156	5
157	157	4							- - - -							157	4
158	158	3							- - - -							158	3
159	159	2							- - - -							159	2
160	160	1							- - - -							160	1
	CA	MX-0	1			2			- - - -	79			80				
		MX-1	80			79			- - - -	2			1				

Figure 5-2 Display Data Mapping in SRAM

Note:
RA: Row Address
CA: Column Address
SA: Scan Address
MX: Mirror X-axis (Column address direction parameter), configured by MADCTR command
MY: Mirror Y-axis (Row address direction parameter), configured by MADCTR command
ML: Scan direction parameter, configured by MADCTR command
RGB: Red, Blue subpixel position change, configured by MADCTR command

5.3. MCU To SRAM Access Direction

The address counter set the addresses of the display data RAM for writing and reading. Data is written pixel-wise into the RAM matrix of DRIVER. The data for one pixel or two pixels is collected (RGB 5-6-5-bit), according to the data formats. As soon as this pixel-data information is complete the “Write access” is activated on the RAM. The address ranges are X=0 to X=79 (4Fh) and Y=0 to Y=159 (9Fh). Addresses outside these ranges are not allowed. Before writing to the RAM a

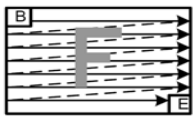
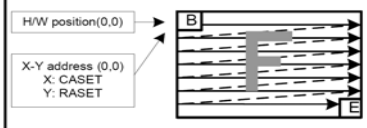
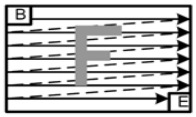
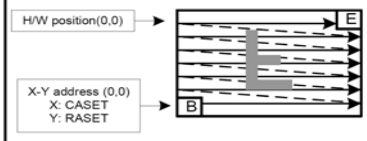
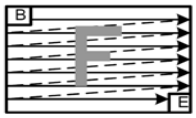
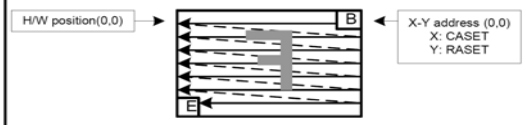
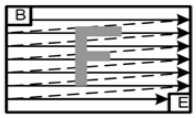
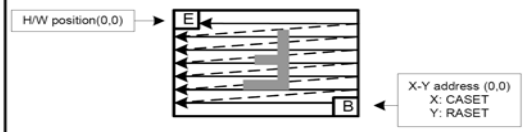
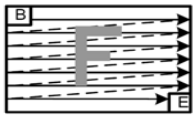
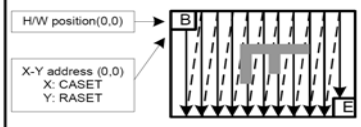
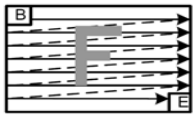
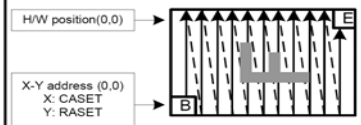
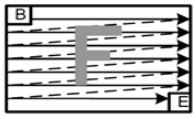
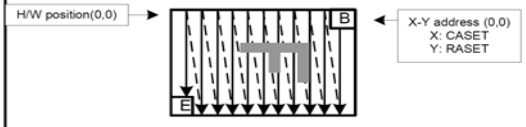
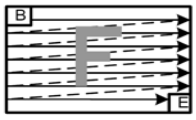
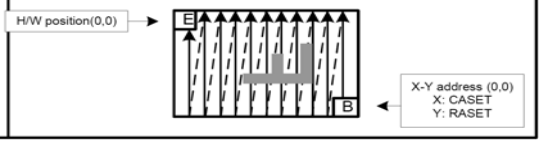
window must be defined into which will be written. The window is programmable via the command register XS, YS designating the start address and XE, YE designating the end address.

For example, the whole display contents will be written, the window is defined by the following values: XS=0(0h) YS=0(0h) and XE=79(4Fh), YE=159(9Fh) In vertical addressing mode (MV=1), the Y-address increments after each byte, after the last Y-address (Y=YE), Y wraps around to YS and X increments to address the next column. In horizontal addressing mode (V=0), the X-address increments after each byte, after the last X-address(X=XE), X wraps around to XS and Y increments to address the next row. After every last address (X=XE and Y=YE) the address pointers wrap around to address (X=XS and Y=YS).

For flexibility in handling a wide variety of display architectures, the commands “CASET, RASET” and “MADCTR”, define flags MX and MY, which allows mirroring of the X-address and Y-address. All combinations of flags are allowed. Below table shows the available combinations of writing to the display RAM. When MX, MY and MV will be changed the data bust be rewritten to the display RAM. For each image orientation, the controls for the column and page counters apply as below:

Condition	Column Counter	Row Counter
When RAMWR/RAMRD command is accepted	Return to “Start Column (XS)”	Return to “Start Row (YS)”
Complete Pixel Read/Write action	Increment by 1	No change
The Column counter value is larger than “End Column (XE)”	Return to “Start Column (XS)”	Increment by 1
The Column counter value is larger than “End Column (XE)” and the Row counter value is larger than “End Row (YE)”	Return to “Start Column (XS)”	Return to “Start Row (YS)”

Table 5-3 Frame Data Write Direction According to the MADCTR parameters (sys_mv, sys_mx and sys_my)

Display Data Direction	MADCTR Parameter			Image in the Memory (MPU)	Image in the Driver (DDRAM)
	MV	MX	MY		
Normal	0	0	0		
Y-Mirror	0	0	1		
X-Mirror	0	1	0		
X-Mirror Y-Mirror	0	1	1		
X-Y Exchange	1	0	0		
X-Y Exchange Y-Mirror	1	0	1		
XY Exchange	1	1	0		
XY Exchange	1	1	1		

6. Tearing Effect

The Tearing Effect output line supplies to the MCU a Panel synchronization signal. This signal can be enabled or disabled by the Tearing Effect Line Off & On commands. The mode of the Tearing Effect Signal is defined by the Parameter of the Tearing Effect Line On command. The signal can be used by the MCU to synchronize Frame Memory Writing when displaying video images.

6.1. Tearing Effect Line Modes

Mode 1, the Tearing Effect Output signal consists of V-Sync information only:



Tvdh = The LCD display is not updated from the Frame Memory.

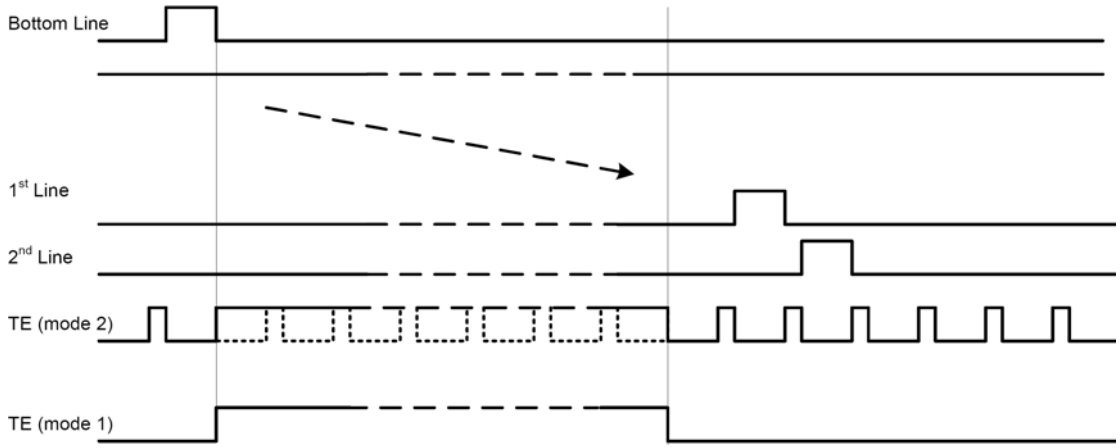
Tvdl = The LCD display is updated from the Frame Memory (except Invisible Line – see below).

Mode 2, the Tearing Effect Output signal consists of V-Sync and H-Sync information, There are one V-sync and 160 H-sync pulses per field:



Thdh = The LCD display is not updated from the Frame Memory.

Thdl = The LCD display is updated from the Frame Memory (except Invisible Line – see above).



Note: During Sleep In Mode, the Tearing Effect Output Pin is active Low.

6.2. Tearing Effect Line Timing

The Tearing Effect signal is described below:

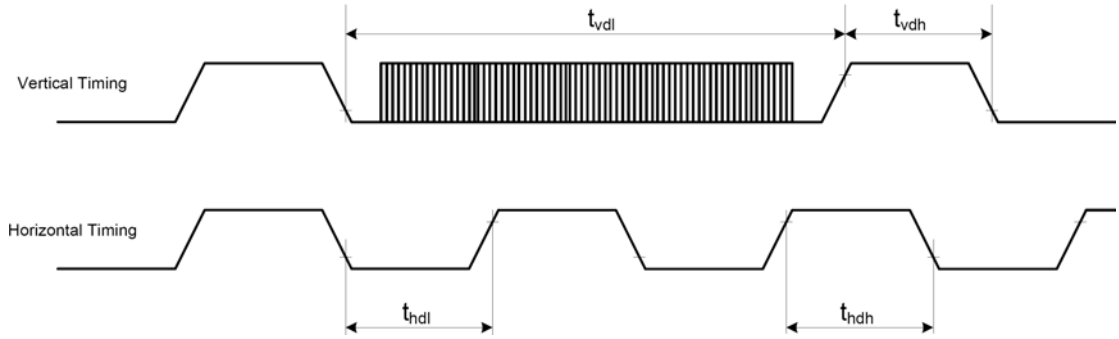


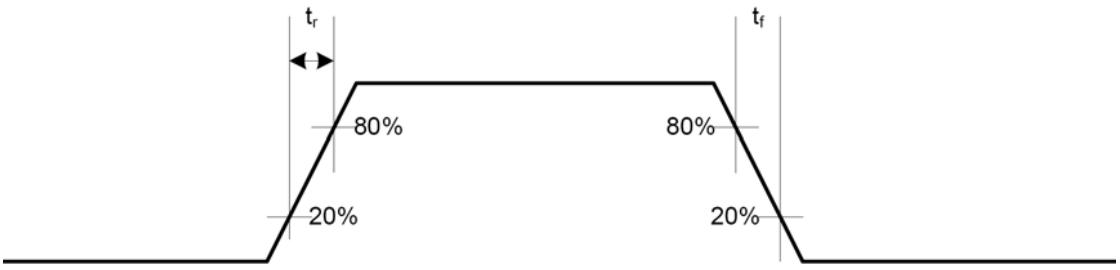
Table 6-2 AC characteristics of Tearing Effect Signal Idle Mode Off/On (Frame Rate = 58.9Hz)

Symbol	Parameter	MIN	MAX	Unit	Descritpion
tvdl	Vertical Timing Low Duration	13	-	ms	
tvdh	Vertical Timing High Duration	1000	-	µs	
thdl	Horizontal Timing Low Duration	33	-	µs	
thdh	Horizontal Timing High Duration	25	500	µs	

Notes:

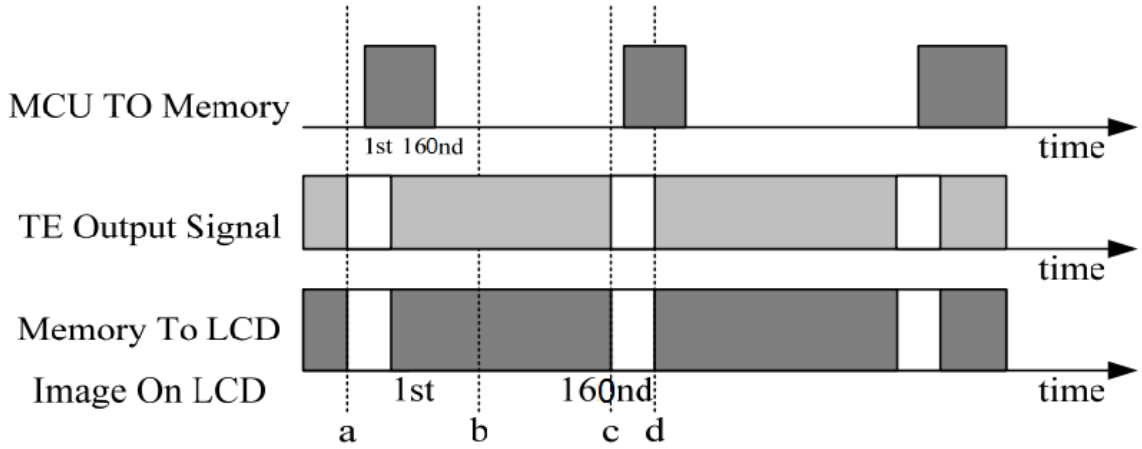
The timings in Table 7-2 apply when MADCTR B4=0

The signal's rise and fall times (tf, tr) are stipulated to be equal to or less than 15ns.



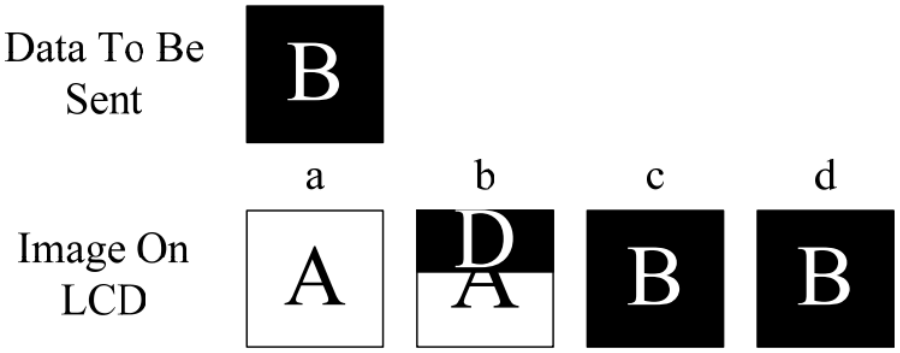
The Tearing Effect Output Line is fed back to the MCU and should be used as shown below to avoid Tearing Effect:

6.2.1. Example 1 MCU Write is Faster than Panel Read

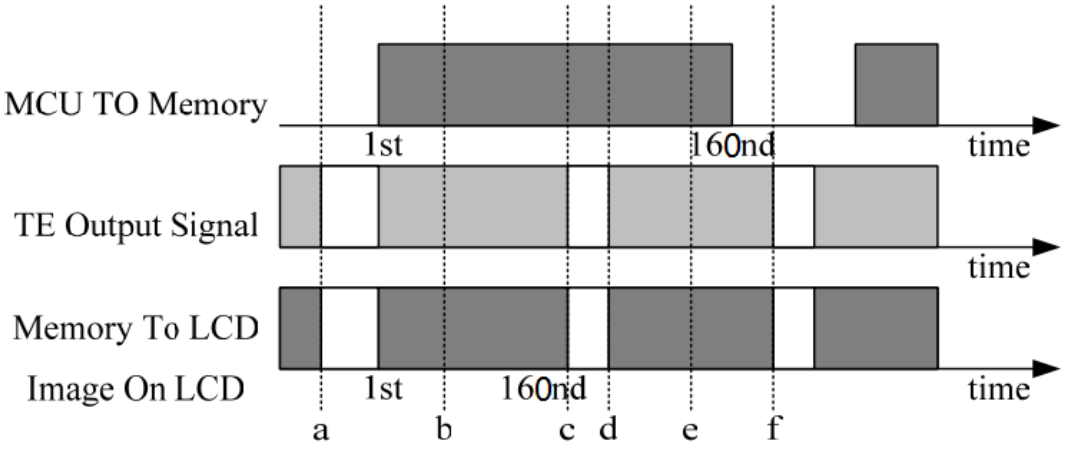


The MCU to Frame Memory write begins just after Panel Read has commenced i.e. after one horizontal sync pulse of the Tearing Effect Output Line. This allows time for the image to download behind the Panel Read pointer and finishing download during the subsequent Frame before the Read Pointer “catches” the MCU to Frame memory write position.

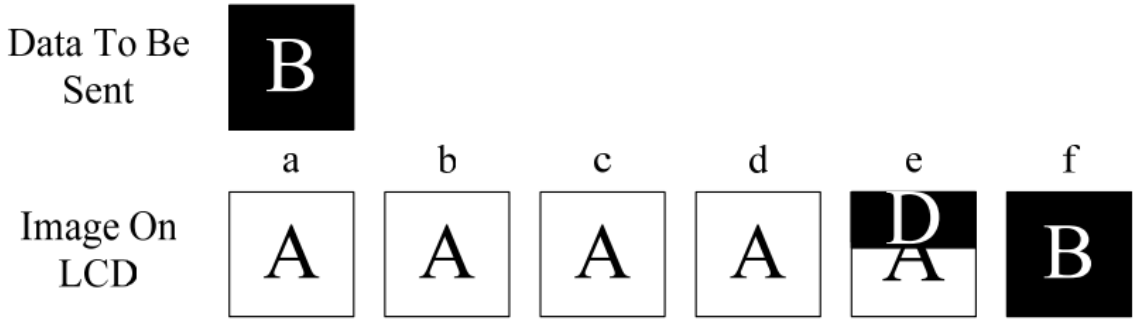
Frame refresh has a complete new image:



6.2.2. Example 2 MCU Write is slower than Panel Read



The MCU to Frame Memory write begins just after Panel Read has commenced i.e. after one horizontal sync pulse of the Tearing Effect Output Line. This allows time for the image to download behind the Panel Read pointer and finishing download during the subsequent Frame before the Read Pointer “catches” the MCU to Frame memory write position.



7. Power ON/OFF Sequence

IOVCC and VCI can be applied in any order.

VCI and IOVCC can be powered down in any order.

During power off, if LCD is in the Sleep Out mode, VCI and IOVCC must be powered down minimum 120 msec after RESX has been released.

During power off, if LCD is in the Sleep In Mode, IOVCC or VCI can be powered down minimum 0 msec after RESX has been released.

CSX can be applied at any timing or can be permanently grounded. RESX has priority over CSX.

Notes:

There will be no damage to the display module if the power sequences are not met.

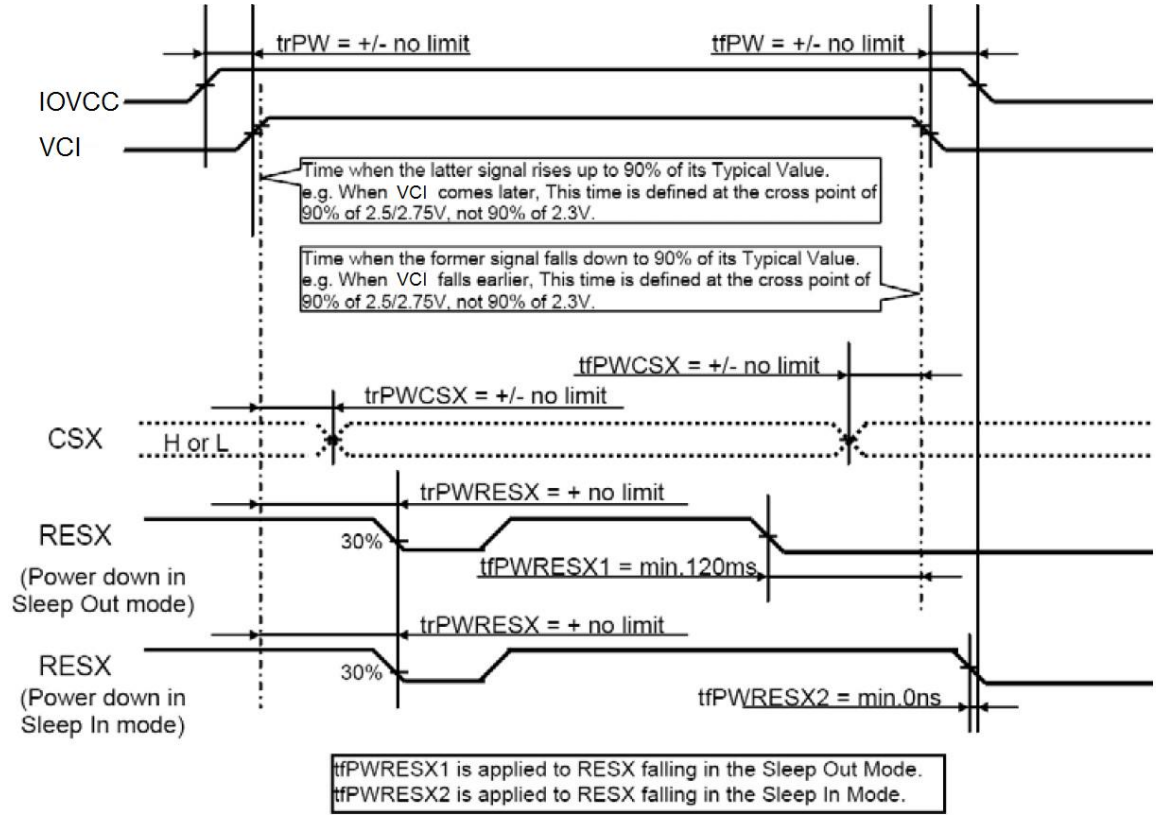
There will be no abnormal visible effects on the display panel during the Power On/Off Sequences.

There will be no abnormal visible effects on the display between end of Power On Sequence and before receiving Sleep Out command. Also between receiving Sleep In command and Power Off Sequence.

If RESX line is not held stable by host during Power On Sequence as defined in Sections 8.1 and 8.2, then it will be necessary to apply a Hardware Reset (RESX) after Host Power On Sequence is complete to ensure correct operation. Otherwise function is not guaranteed.

7.1. Case 1 - RESX line is held high or Unstable by Host at Power On

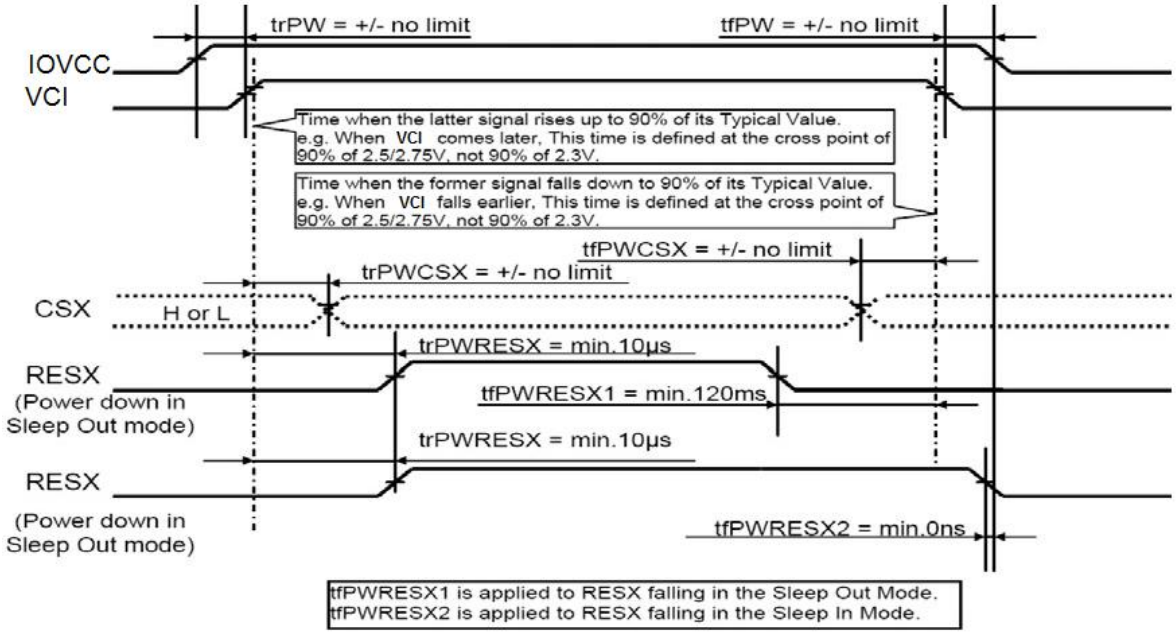
If RESX line is held high or unstable by the host during Power On, then a Hardware Reset must be applied after both VCI and IOVCC have been applied – otherwise correct functionality is not guaranteed. There is no timing restriction upon this hardware reset.



Note: Unless otherwise specified, timings herein show cross point at 50% of signal/power level.

7.2. Case 2 - RESX line is held low by Host at Power On

If RESX line is held Low (and stable) by the host during Power On, then the RESX must be held low for minimum 10 µsec after both VCI and IOVCC have been applied.



Note: Unless otherwise specified, timings herein show cross point at 50% of signal/power level.

7.3. Uncontrolled Power Off

The uncontrolled power off means a situation when e.g. there is removed a battery without the controlled power off sequence. The display module must meet following requirements:

There cannot be any damages for the display module or the display module cannot cause any damages for the host or lines of the interface.

There cannot be any abnormal visible effects (= Display must be blank) within 1 second on the display and remains blank until “Power On Sequence” powers it up.

8. Power Level Definition

8.1. Power Levels

6 level modes are defined they are in order of Maximum Power consumption to Minimum Power Consumption:

1. Normal Mode On (full display), Idle Mode Off, Sleep Out. In this mode, the display is able to show maximum 65536 colors.

2. Partial Mode On, Idle Mode Off, Sleep Out. In this mode part of the display is used with maximum 65536 colors.

3. Normal Mode On (full display), Idle Mode On, Sleep Out. In this mode, the full display area is used but with 8 colors.

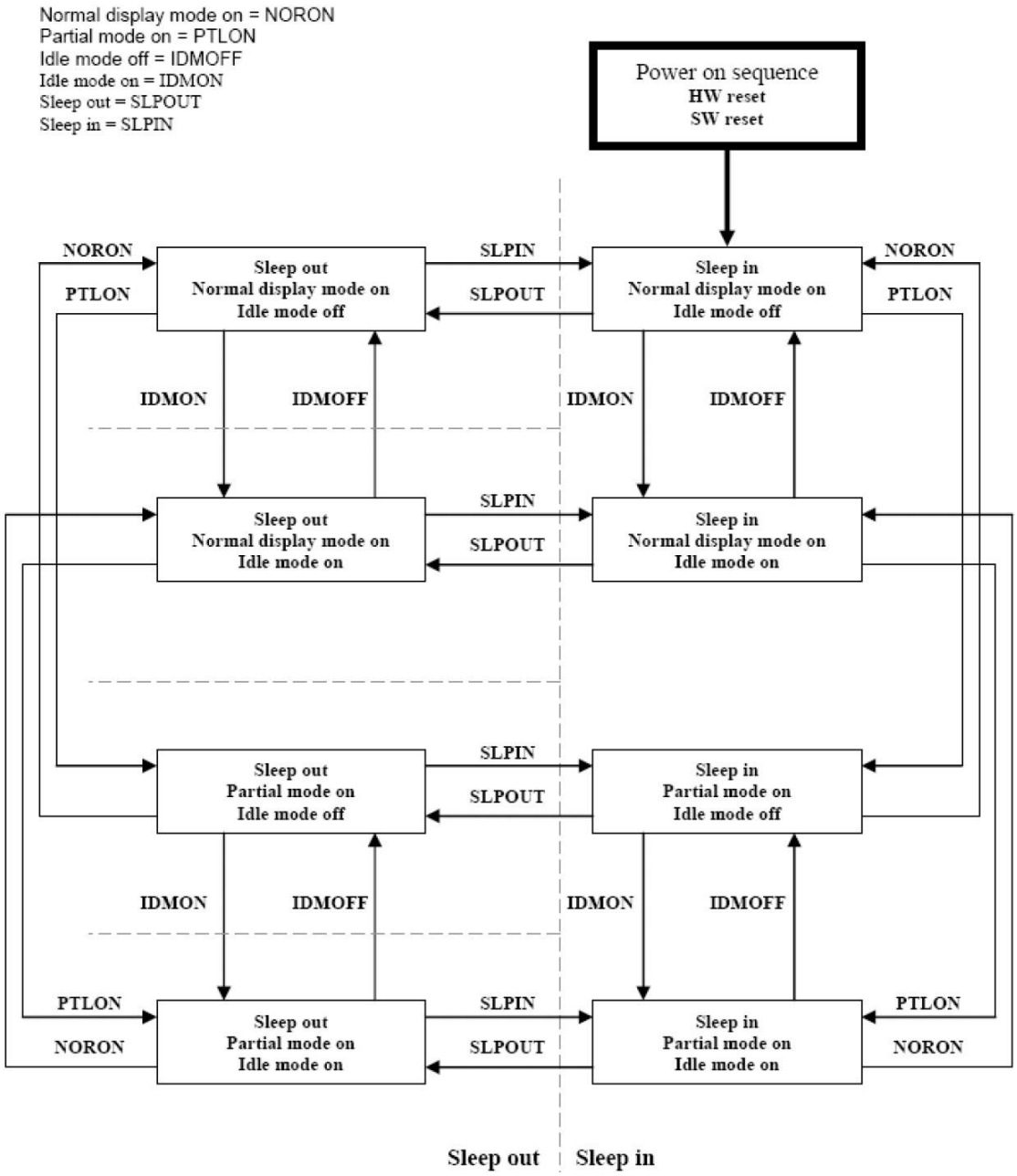
4. Partial Mode On, Idle Mode On, Sleep Out. In this mode, part of the display is used but with 8 colors.

5. Sleep In Mode. In this mode, the DC/DC converter, internal oscillator and panel driver circuit are stopped. Only the MCU interface and memory works with IOVCC power supply. Contents of the memory are safe.

6. Power Off Mode. In this mode, both VCI and IOVCC are removed.

Note: Transition between modes 1-5 is controllable by MCU commands. Mode 6 is entered only when both Power supplies are removed. If RESX line is held high or unstable by the host during Power On, then a Hardware Reset must be applied after both VCI and IOVCC have been applied – otherwise correct functionality is not guaranteed. There is no timing restriction upon

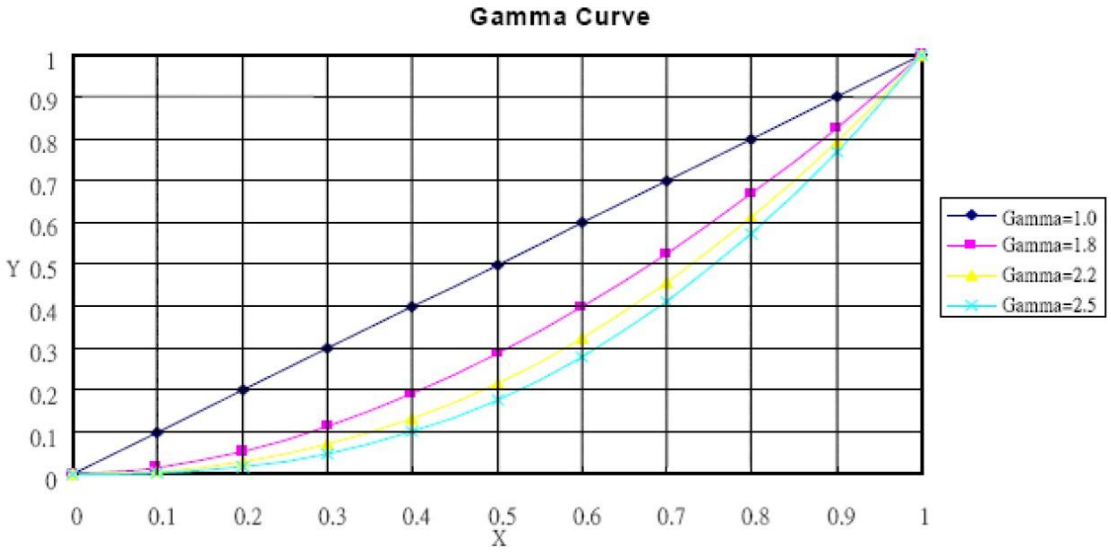
8.2. Power Flow Chart



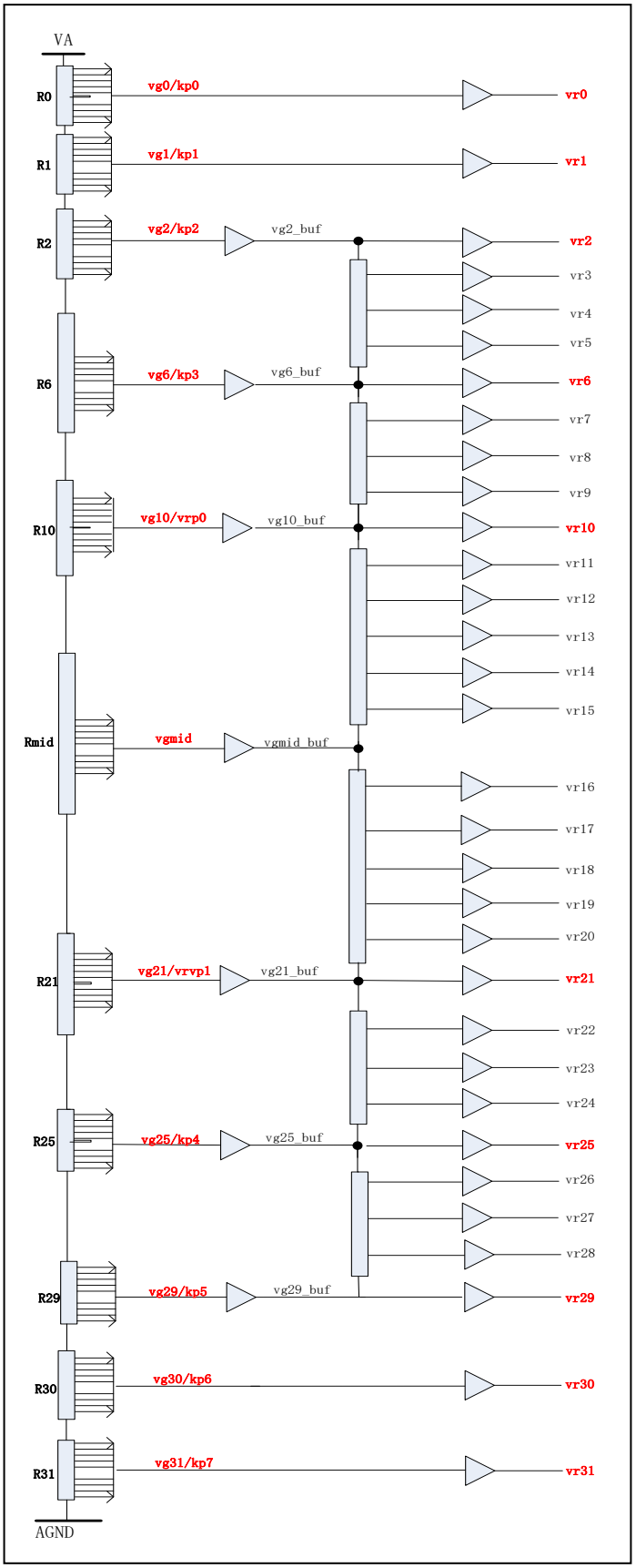
Note 1: There is not any abnormal visual effect when there is changing from one power mode to another power mode.
Note 2: There is not any limitation, which is not specified by Nokia, when there is changing from one power mode to another power mode.
Note 3: It is recommended that it should be enter Sleep in before power off.

9. Gamma Curve

9.1. Gamma curve according to the Gamma 1.0/1.8/2.2/2.5



9.2. Gamma Structure



10. Reset Function

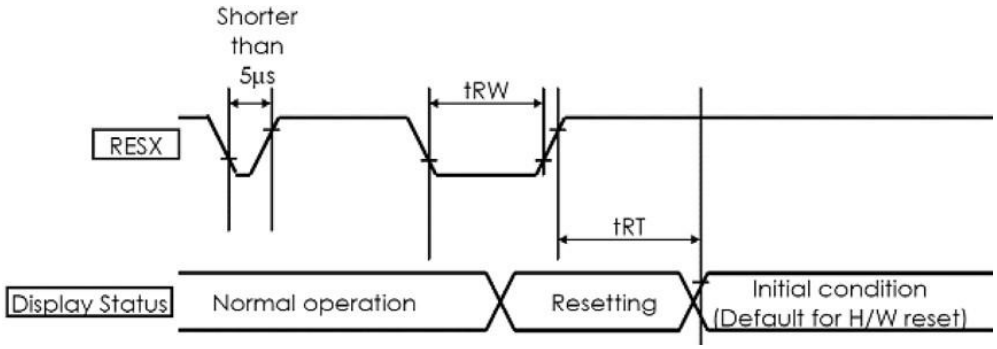
10.1. State of Register After Different Reset

The registers for resolution 80(RGB) x160 are initialized are listed below:

Item	After Power On	After Hardware Reset
Frame memory	Random	No Change
Sleep In/Out	In	In
Display On/Off	Off	Off
Display mode(normal/partial)	Normal	Normal
Display Inversion On/Off	Off	Off
Display Idle Mode On/Off	Off	Off
Column:Start Address(XS)	0000h	0000h
Column:End Address(XE)	007Fh	007Fh
Row:Start Address(YS)	0000h	0000h
Row:End Address(YE)	009Fh	009Fh
Gamma Setting	GC0	GC0
Color Set	64K	64K
Partial:Start Address(PSL)	0000h	0000h
Partial:End Address(PEL)	009Fh	009Fh
Scroll:Vertical scrolling	Off	Off
Scroll:Top Fixed Area(TFA)	0000h	0000h
Scroll:Scroll Area(VSA)	00A0h	00A0h
Scroll:Bottom Fixed Area(BFA)	0000h	0000h
Scroll Start Address (SSA)	0000h	0000h
Tearing:On/Off	Off	Off
Tearing Effect Mode*3	0(Mode1)	0(Mode1)
Memory Data Access Control (MY/MX/MV/ML/MH/RGB)	0/0/0/0/0/0	0/0/0/0/0/0
Interface Pixel Color Format	6(16-Bit/Pixel)	6(16-Bit/Pixel)
RDDPM	08h	08h
RDDMADCTR	0	0
RDDCOLMOD	6(16-Bit/Pixel)	6(16-Bit/Pixel)
RDDIM	00h	00h
RDDSM	00h	00h
RDDSER	00h	00h
ID1	33h	33h
ID2	30h	30h
ID3	23h	23h

Notes:
1. There will be no abnormal visible effects on the display when S/W or H/W Reset are applied.
2. After Powered-On Reset finishes within 10μs after both VCI & IOVCC are applied.
3. Mode 1 means Tearing Effect Output Line consists of V-Blanking Information only.

10.2. Reset Timing



(GND=0V, IOVCC=1.65V to 3.6V, VCI=2.5V to 3.6V, Ta = -30 to 85°C)

Symbol	Parameter	Related Pins	MIN	TYP	MAX	Note	Unit
tRW	Valid Reset low pulse width	RESX	10	-	-	-	us
tRT	Valid Reset Complete width	RESX	-	-	5	When reset applied during Sleep in mode	ms
		RESX	-	-	120	When reset applied during Sleep out mode	ms

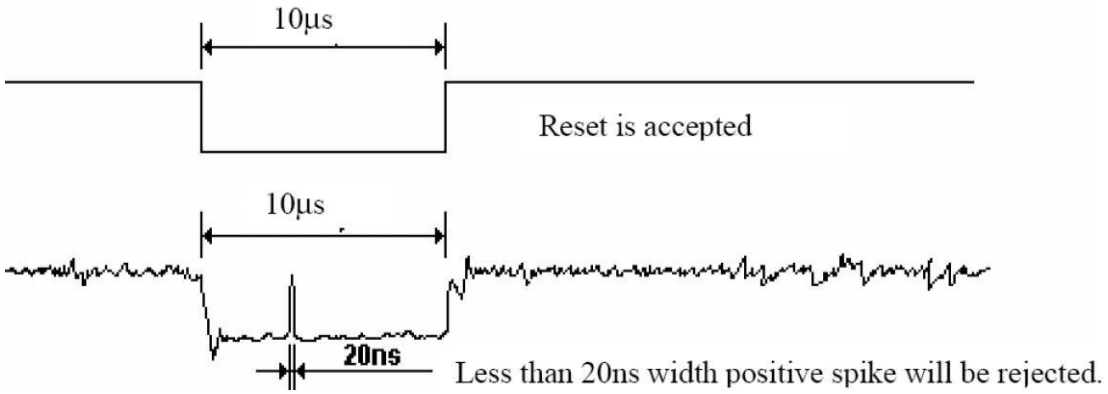
Note:

1> Spike due to an electrostatic discharge on RESX line does not cause system reset according to the table below.

RESX Pulse	Action
Shorten than 5us	Reset Rejected
Longer than 10us	Reset
Between 5us and 10us	Reset starts(It depends on voltage and temperature condition)

2> During the resetting period, the display will be blanked (The display is entering blanking sequence, which maximum time is 120 ms, when Reset Starts in Sleep Out –mode. The display remains the blank state in Sleep In mode) and then return to Default condition for Hardware Reset.

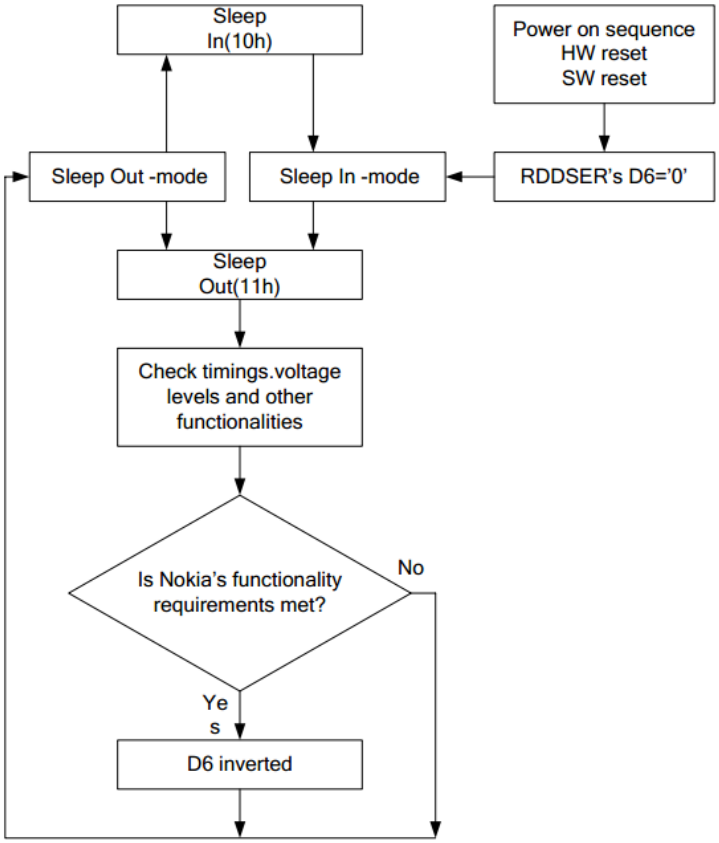
3> Spike Rejection also applies during a valid reset pulse as shown below:



4> It is necessary to wait 5msec after releasing RESX before sending commands. Also Sleep Out command cannot be sent for 120msec.

11. Sleep Out and Self-Diagnostic Functions of Display

Sleep Out-command (11h) is a trigger for an internal function of the display module, which indicates, if the display module is still running and meets functionality requirements. The internal function (= the display controller) is comparing, if the display module is still meeting functionality requirements (only Booster voltage level). If functionality requirement is met, there is inverted a bit, which defined in command ("0Fh", The used bit of this command is D6). If functionality requirement is not same, this bit (D6) is not inverted. The flow chart for this internal function is following:



Note: There is needed 120msec after Sleep Out command, when there is changing from Sleep In mode to Sleep Out mode, before there is possible to check if Nokia's functionality requirements are met and a value of RDDSDR's D6 is valid. Otherwise, there is 5msec delay for D6's value, when Sleep Out command is sent in Sleep Out mode.

12. Command

12.1. System Function Command List and Description

Addr	Name	R/W	D7	D6	D5	D4	D3	D2	D1	D0
0x00	NOP	W	0	0	0	0	0	0	0	0
0x04	RDDID	R	sys_id1[7:0]							
			sys_id2[7:0]							
			sys_id3[7:0]							
0x09	RDDST	R	boost_on	my	mx	mv	ml	rgb		
			pixel_fmt[3:0]				idle_on	ptl_on	slpout	normal_on
			scroll_on	1'b0	invon	0	0	dispon	te_on	0
			0	0	telom	rgb_on				
0x0A	RDDPM	R	boost_on	idle_on	ptl_on	slpout	normal_on	dispon	0	0
0x0B	RDD_MADCTR	R	my	mx	mv	ml	rgb	0	0	0
0x0C	RDD_COLMOD	R						ifpf[2:0]		
0x0D	RDDIM	R	scroll_on	0	invon	0	0	0	0	0
0x0E	RDDSM	R	te_on	telom	0	0	0	0	0	0
0x0F	RDDSDR	R	4{self_diag}				fifo_intf2osc_empty	fifo_intf2osc_full	fifo_osc2tcon_empty	fifo_osc2tcon_full
0x10	SLPIN	W								
0x11	SLPOUT	W								
0x12	PTLON	W								
0x13	NORON	W								
0x20	INVOFF	W								
0x21	INVON	W								
0x28	DISPOFF	W								
0x29	DISPON	W								
0x2A	CASE	W	col_st_set_pre[7:0]							

Addr	Name	R/W	D7	D6	D5	D4	D3	D2	D1	D0
	T		col_ed_set_pre[7:0]							
0x2B	RASET	W	row_st_set_pre[7:0]							
			row_ed_set_pre[7:0]							
0x2C	RAMWR	W								
0x2E	RAMRD	W								
0x30	PTLAR	W	ptl_st_set[7:0]							
			ptl_ed_set[7:0]							
0x31	PTLDISP	W								ddl
0x33	SCRLAR	W	tfa[7:0]							
			vsa_set_pre[7:0]							
0x34	TEOFF	W								
0x35	TEON	W								telom
0x36	MADCTR	W	sys_my	sys_mx	sys_mv	sys_ml	sys_rgb			
0x37	VSCSAD	W	ssa[7:0]							
0x38	IDMOFF	W								
0x39	IDMON	W								
0x3A	COLMOD	W						ifpf[2:0]		
0xDA	RDID1	R	SYS_ID3[7:0]							
0xDB	RDID2	R	sys_id2[7:0]							
0xDC	RDID3	R	sys_id3[7:0]							

12.1.1. NOP (00h)

<table>
<tr><th>00H</th><th colspan="11">No Operation</th></tr>
<tr><td></td><td>D/CX</td><td>RDX</td><td>WRX</td><td>D7</td><td>D6</td><td>D5</td><td>D4</td><td>D3</td><td>D2</td><td>D1</td><td>D0</td><td>HEX</td></tr>
<tr><td>Command</td><td>L</td><td>H</td><td>W</td><td>0</td><td>0</td><td>0</td><td>0</td><td>0</td><td>0</td><td>0</td><td>0</td><td>00H</td></tr>
<tr><td>Parameter</td><td colspan="12">No Parameter</td></tr>
<tr><td>Description</td><td colspan="12">This command is an empty command. It does not have any effect on the display module. However it can be used to terminate Frame Memory White or Read as described in RAMWR (Memory White) and RAMRD (Memory Read) Commands.
X=Don’t care.</td></tr>
<tr><td>Restriction</td><td colspan="12">None</td></tr>
<tr><td>Register Availability</td><td colspan="12"><table><tr><th>Status</th><th>Availability</th></tr><tr><td>Normal Mode on,Idle Mode Off,Sleep Out</td><td>Yes</td></tr><tr><td>Normal Mode on,Idle Mode On,Sleep Out</td><td>Yes</td></tr><tr><td>Partial Mode on,Idle Mode Off,Sleep Out</td><td>Yes</td></tr><tr><td>Partial Mode on,Idle Mode On,Sleep Out</td><td>Yes</td></tr><tr><td>Sleep In</td><td>Yes</td></tr></table></td></tr>
<tr><td>Default</td><td colspan="12"><table><tr><th>Status</th><th>Default Value</th></tr><tr><td>Power On Sequence</td><td>N/A</td></tr><tr><td>HW Reset</td><td>N/A</td></tr></table></td></tr>
</table>

12.1.2. RDDID (04h)

<table>
<tr><td>04H</td><td colspan="12">Read Display ID</td></tr>
<tr><td></td><td>D/CX</td><td>RDX</td><td>WRX</td><td>D7</td><td>D6</td><td>D5</td><td>D4</td><td>D3</td><td>D2</td><td>D1</td><td>D0</td><td>HEX</td></tr>
<tr><td>Command</td><td>L</td><td>R</td><td>H</td><td>0</td><td>0</td><td>0</td><td>0</td><td>0</td><td>1</td><td>0</td><td>0</td><td>04H</td></tr>
<tr><td>Parameter1</td><td>H</td><td>R</td><td>H</td><td>X</td><td>X</td><td>X</td><td>X</td><td>X</td><td>X</td><td>X</td><td>X</td><td>X</td></tr>
<tr><td>Parameter2</td><td>H</td><td>R</td><td>H</td><td colspan="8">SYS_ID1[7:0]</td><td>33H</td></tr>
<tr><td>Parameter3</td><td>H</td><td>R</td><td>H</td><td colspan="8">SYS_ID2[7:0]</td><td>30H</td></tr>
<tr><td>Parameter4</td><td>H</td><td>R</td><td>H</td><td colspan="8">SYS_ID3[7:0]</td><td>23H</td></tr>
<tr><td>Description</td><td colspan="12">This read byte returns 24-bit display identification.
The 1st Parameter is dummy read.
The 2nd Parameter (SYS_ID1): LCD module’s manufacture ID.
The 3rd Parameter (SYS_ID2): LCD module/driver version ID.
The 4th Parameter (SYS_ID3): LCD module/driver version ID.
Note: Commands RDDID1/2/3 (DAh, DBh, DCh) read data correspond to the parameters 1, 2, 3 of command 04h, respectively.</td></tr>
<tr><td>Restriction</td><td colspan="12">None</td></tr>
<tr><td>Register Availability</td><td colspan="12"><table><tr><th>Status</th><th>Availability</th></tr><tr><td>Normal Mode on,Idle Mode Off,Sleep Out</td><td>Yes</td></tr><tr><td>Normal Mode on,Idle Mode On,Sleep Out</td><td>Yes</td></tr><tr><td>Partial Mode on,Idle Mode Off,Sleep Out</td><td>Yes</td></tr><tr><td>Partial Mode on,Idle Mode On,Sleep Out</td><td>Yes</td></tr><tr><td>Sleep In</td><td>Yes</td></tr></table></td></tr>
<tr><td>Default</td><td colspan="12"><table><tr><th rowspan="2">Status</th><th colspan="3">Default Value</th></tr><tr><th>ID1</th><th>ID2</th><th>ID3</th></tr><tr><td>Power On Sequence</td><td>8’h33</td><td>8’h30</td><td>8’h23</td></tr><tr><td>HW Reset</td><td>8’h33</td><td>8’h30</td><td>8’h23</td></tr></table></td></tr>
</table>

12.1.3. RDDST (09h)

09H	Read Display Status											
	D/CX	**RDX**	**WRX**	D7	D6	D5	D4	D3	D2	D1	D0	HEX
Command	L	R	H	0	0	0	0	1	0	0	1	09H
Parameter1	H	R	H	X	X	X	X	X	X	X	X	X
Parameter2	H	R	H	BOOST_ON	MY	MX	MV	ML	RGB			04H
Parameter3	H	R	H	PIXEL_FMT[3:0]				IDLE_ON	PTL_ON	SLPOUT	NORMAL_ON	71H
Parameter4	H	R	H	SCROLL_ON	0	INVON	0	0	DISPON	TE_ON	0	00H
Parameter5	H	R	H	0	0	TELOM	RGB_ON					00H

Description

Bit	Description	Value
BOOST_ON	Booster Voltage Status	“1”=Booster on,“0”=Booster off
MY	Row Address Order	“1”=Decrement. (Bottom to Top), when MADCTR (36h) D7=“1”
		“0”=Increment.(Top to Bottom),when MADCTR (36h) D7=“0”
MX	Column Adress Order	“1”=Decrement.(Right to Left),when MADCTR(36h)D6=“1”
		“0”=Increment.(Left to Right),when MADCTR(36h)D6=“0”
MV	Row/Column Exchange	“1”=Row/column exchange. when MADCTR (36h) D5=“1”
		“0”=Normal(MV=0).when MADCTR (36h) D5=“0”
ML	Vertical refresh Order	“1”=Decrement.(LCD refresh Bottom to Top, when MADCTR (36h) D4=“1”
		“0”=Increment.(LCD refresh Top to Bottom), when MADCTR(36h)D4=“0”
RGB	RGB/BGR Order	“1”=BGR.when MADCTR(36h)D3=“1”
		“0”=RGB.when MADCTR(36h)D3=“0”
PIXEL_FMT[3:0]	Interface Color Pixel Format Definition	“0011”=12-bit/pixel
		“0101”=16-bit/pixel
		“0111”=8-bit/pixel
		The other=not defined
IDLE_ON	Idle Mode On/Off	“1”=On,“0”=Off
PTL_ON	Partial Mode On/Off	“1”=On,“0”=Off
SLPOUT	Sleep In/Out	“1”=On,“0”=Off
NORMAL_ON	Display Normal Mode On/Off	“1”=Normal Display, “0”=Normal Display Off
SCROLL_ON	Vertical Scrolling Status	“1”=Scroll On,“0”=Scroll Off
INVON	Inversion Status	“1”=On,“0”=Off
DISPON	Display On/Off	“1”=On,“0”=Off

<table>
<tr><td></td><td>
<table>
<tr><td>TE_ON</td><td>Tearing effect line on/off</td><td>“1”=On,“0”=Off</td></tr>
<tr><td>TELOM</td><td>Tearing effect line mode</td><td>“0”=mode1,“1”=mode2,</td></tr>
<tr><td>RGB_ON</td><td>RGB interfacee flag</td><td>“0x1A”=RGB ON,”0x00”=RGB OFF</td></tr>
</table>
</td></tr>
<tr><td>Restriction</td><td>None</td></tr>
<tr><td>Register Availability</td><td>
<table>
<tr><th>Status</th><th>Availability</th></tr>
<tr><td>Normal Mode on,Idle Mode Off,Sleep Out</td><td>Yes</td></tr>
<tr><td>Normal Mode on,Idle Mode On,Sleep Out</td><td>Yes</td></tr>
<tr><td>Partial Mode on,Idle Mode Off,Sleep Out</td><td>Yes</td></tr>
<tr><td>Partial Mode on,Idle Mode On,Sleep Out</td><td>Yes</td></tr>
<tr><td>Sleep In</td><td>Yes</td></tr>
</table>
</td></tr>
<tr><td>Default</td><td>
<table>
<tr><th rowspan="2">Status</th><th colspan="4">Default Value(ST31 to ST0)</th></tr>
<tr><th>ST[31-24]</th><th>ST[23-16]</th><th>ST[15-8]</th><th>ST[7-0]</th></tr>
<tr><td>Power on Sequence</td><td>0000-0100</td><td>0111-0001</td><td>0000-0000</td><td>0000-0000</td></tr>
<tr><td>HW Reset</td><td>0000-0100</td><td>0111-0001</td><td>0000-0000</td><td>0000-0000</td></tr>
</table>
</td></tr>
</table>

12.1.4. RDDPM (0Ah)

<table>
<tr><th>0AH</th><th colspan="11">Read Display Power Mode</th></tr>
<tr><td></td><td>D/CX</td><td>RDX</td><td>WRX</td><td>D7</td><td>D6</td><td>D5</td><td>D4</td><td>D3</td><td>D2</td><td>D1</td><td>D0</td><td>HEX</td></tr>
<tr><td>Command</td><td>L</td><td>R</td><td>H</td><td>0</td><td>0</td><td>0</td><td>0</td><td>1</td><td>0</td><td>1</td><td>0</td><td>0AH</td></tr>
<tr><td>Parameter1</td><td>H</td><td>R</td><td>H</td><td>X</td><td>X</td><td>X</td><td>X</td><td>X</td><td>X</td><td>X</td><td>X</td><td>X</td></tr>
<tr><td>Parameter2</td><td>H</td><td>R</td><td>H</td><td>BOOST_ON</td><td>IDLE_ON</td><td>PTL_ON</td><td>SLPOUT</td><td>NORMAL_ON</td><td>DISPON</td><td>0</td><td>0</td><td>08H</td></tr>
<tr><td>Description</td><td colspan="11">
<table>
<tr><th>Bit</th><th>Description</th><th>Value</th></tr>
<tr><td>BOOST_ON</td><td>Booster Voltage Status</td><td>“1”=Booster On
“0”Booster Off</td></tr>
<tr><td>IDLE_ON</td><td>Idle mode On/Off</td><td>“1”=Idle mode On
“0”=Idle mode Off</td></tr>
<tr><td>PTL_ON</td><td>Partial Mode On/Off</td><td>“1”=Partial Mode On
“0”=Partial Mode Off</td></tr>
<tr><td>SLPOUT</td><td>Sleep In/Off</td><td>“1”=Sleep Out
“0”=Sleep In</td></tr>
<tr><td>NORMAL_ON</td><td>Display Normal Mode On/Off</td><td>“1”=Normal Display
“0”=Partial Display</td></tr>
<tr><td>DISPON</td><td>Display On/Off</td><td>“1”=Display On
“0”=Display Off</td></tr>
</table>
</td></tr>
<tr><td>Restriction</td><td colspan="11">None</td></tr>
<tr><td>Register Availability</td><td colspan="11">
<table>
<tr><th>Status</th><th>Availability</th></tr>
<tr><td>Normal Mode on,Idle Mode Off,Sleep Out</td><td>Yes</td></tr>
<tr><td>Normal Mode on,Idle Mode On,Sleep Out</td><td>Yes</td></tr>
<tr><td>Partial Mode on,Idle Mode Off,Sleep Out</td><td>Yes</td></tr>
<tr><td>Partial Mode on,Idle Mode On,Sleep Out</td><td>Yes</td></tr>
<tr><td>Sleep In</td><td>Yes</td></tr>
</table>
</td></tr>
<tr><td>Default</td><td colspan="11">
<table>
<tr><th>Status</th><th>Default Value(D7 to D0)</th></tr>
<tr><td>Power On Sequence</td><td>0000_1000(8’h08)</td></tr>
<tr><td>HW Reset</td><td>0000_1000(8’h08)</td></tr>
</table>
</td></tr>
</table>

12.1.5. RDD_MADCTR (0Bh)

0BH	Read Display MADCTR											
	D/CX	RDX	WRX	D7	D6	D5	D4	D3	D2	D1	D0	HEX
Command	L	R	H	0	0	0	0	1	0	1	1	0BH
Parameter1	H	R	H	X	X	X	X	X	X	X	X	X
Parameter2	H	R	H	MY	MX	MV	ML	RGB	0	0	0	08H

Description

Bit	Description	Value
MY	Page Address Order	“1”=Decrement,“0”=Increment
MX	Column Adress Order	“1”=Decrement,“0”=Increment
MV	Page/Column Order	“1”=Row/column exchange (MV=1) “0”=Normal(MV=0)
ML	Line Address Order	“1”=LCD Refresh Botton to top “0”=LCD Refresh top to Botton
RGB	RGB/BGR Order	“1”=BGR, “0”=RGB

Restriction: None

Register Availability

Status	Availability
Normal Mode on,Idle Mode Off,Sleep Out	Yes
Normal Mode on,Idle Mode On,Sleep Out	Yes
Partial Mode on,Idle Mode Off,Sleep Out	Yes
Partial Mode on,Idle Mode On,Sleep Out	Yes
Sleep In	Yes

Default

Status	Default Value(D7 to D0)
Power On Sequence	8’h08
HW Reset	8’h08

12.1.6. RDD_COLMOD (0Ch)

<table>
<tr><td>0CH</td><td colspan="11">Read Display Pixel Format</td></tr>
<tr><td></td><td>D/CX</td><td>RDX</td><td>WRX</td><td>D7</td><td>D6</td><td>D5</td><td>D4</td><td>D3</td><td>D2</td><td>D1</td><td>D0</td><td>HEX</td></tr>
<tr><td>Command</td><td>L</td><td>R</td><td>H</td><td>0</td><td>0</td><td>0</td><td>0</td><td>1</td><td>1</td><td>0</td><td>0</td><td>0CH</td></tr>
<tr><td>Parameter1</td><td>H</td><td>R</td><td>H</td><td>X</td><td>X</td><td>X</td><td>X</td><td>X</td><td>X</td><td>X</td><td>X</td><td>X</td></tr>
<tr><td>Parameter2</td><td>H</td><td>R</td><td>H</td><td></td><td></td><td></td><td></td><td></td><td colspan="4">IFPF[2:0]</td><td>07H</td></tr>
<tr><td>Description</td><td colspan="12">The 1st Parameter is dummy read.

<table>
<tr><th colspan="2">Bit</th><th>Description</th><th>Value</th></tr>
<tr><td>D2</td><td>IFPF2</td><td rowspan="3">Control Interface Color Format</td><td rowspan="3">“011”=12 bit/pixel
“101”=16 bit/pixel
“111”=8 bit/pixel
The other=not defined</td></tr>
<tr><td>D1</td><td>IFPF1</td></tr>
<tr><td>D0</td><td>IFPF0</td></tr>
</table>
</td></tr>
<tr><td>Restriction</td><td colspan="12">None</td></tr>
<tr><td>Register Availability</td><td colspan="12">
<table>
<tr><th>Status</th><th>Availability</th></tr>
<tr><td>Normal Mode on,Idle Mode Off,Sleep Out</td><td>Yes</td></tr>
<tr><td>Normal Mode on,Idle Mode On,Sleep Out</td><td>Yes</td></tr>
<tr><td>Partial Mode on,Idle Mode Off,Sleep Out</td><td>Yes</td></tr>
<tr><td>Partial Mode on,Idle Mode On,Sleep Out</td><td>Yes</td></tr>
<tr><td>Sleep In</td><td>Yes</td></tr>
</table>
</td></tr>
<tr><td>Default</td><td colspan="12">
<table>
<tr><th>Status</th><th>Default Value(D7 to D0)</th></tr>
<tr><td>Power On Sequence</td><td>0000_0111(8bit/pixel)</td></tr>
<tr><td>HW Reset</td><td>0000_0111(8bit/pixel)</td></tr>
</table>
</td></tr>
</table>

12.1.7. RDDIM (0Dh)

<table>
<tr><td>0DH</td><td colspan="12">Read Display Image Mode</td></tr>
<tr><td></td><td>D/CX</td><td>RDX</td><td>WRX</td><td>D7</td><td>D6</td><td>D5</td><td>D4</td><td>D3</td><td>D2</td><td>D1</td><td>D0</td><td>HEX</td></tr>
<tr><td>Command</td><td>L</td><td>R</td><td>H</td><td>0</td><td>0</td><td>0</td><td>0</td><td>1</td><td>1</td><td>0</td><td>1</td><td>0DH</td></tr>
<tr><td>Parameter1</td><td>H</td><td>R</td><td>H</td><td>X</td><td>X</td><td>X</td><td>X</td><td>X</td><td>X</td><td>X</td><td>X</td><td>X</td></tr>
<tr><td>Parameter2</td><td>H</td><td>R</td><td>H</td><td>SCROLL_ON</td><td>0</td><td>INVON</td><td>0</td><td>0</td><td>0</td><td>0</td><td>0</td><td>00H</td></tr>
<tr><td>Description</td><td colspan="12">The 1st Parameter is dummy read.
<table>
<tr><th>Bit</th><th>Description</th><th>Value</th></tr>
<tr><td>D7</td><td>Scrolling On/Off</td><td>“1”=Scrolling is On
“0”=Scrolling is Off</td></tr>
<tr><td>D5</td><td>Inversion On/Off</td><td>“1”=Inversion is On
“0”=Inversion is Off</td></tr>
</table></td></tr>
<tr><td>Restriction</td><td colspan="12">None</td></tr>
<tr><td>Register Availability</td><td colspan="12">
<table>
<tr><th>Status</th><th>Availability</th></tr>
<tr><td>Normal Mode on,Idle Mode Off,Sleep Out</td><td>Yes</td></tr>
<tr><td>Normal Mode on,Idle Mode On,Sleep Out</td><td>Yes</td></tr>
<tr><td>Partial Mode on,Idle Mode Off,Sleep Out</td><td>Yes</td></tr>
<tr><td>Partial Mode on,Idle Mode On,Sleep Out</td><td>Yes</td></tr>
<tr><td>Sleep In</td><td>Yes</td></tr>
</table></td></tr>
<tr><td>Default</td><td colspan="12">
<table>
<tr><th>Status</th><th>Default Value(D7 to D0)</th></tr>
<tr><td>Power On Sequence</td><td>8’h00</td></tr>
<tr><td>HW Reset</td><td>8’h00</td></tr>
</table></td></tr>
</table>

12.1.8. RDDSM (0Eh)

<table>
<tr><th>0EH</th><th colspan="12">Read Display Signal Mode</th></tr>
<tr><td></td><td>D/CX</td><td>RDX</td><td>WRX</td><td>D7</td><td>D6</td><td>D5</td><td>D4</td><td>D3</td><td>D2</td><td>D1</td><td>D0</td><td>HEX</td></tr>
<tr><td>Command</td><td>L</td><td>R</td><td>H</td><td>0</td><td>0</td><td>0</td><td>0</td><td>1</td><td>1</td><td>1</td><td>0</td><td>0EH</td></tr>
<tr><td>Parameter1</td><td>H</td><td>R</td><td>H</td><td>X</td><td>X</td><td>X</td><td>X</td><td>X</td><td>X</td><td>X</td><td>X</td><td>X</td></tr>
<tr><td>Parameter2</td><td>H</td><td>R</td><td>H</td><td>TE_ON</td><td>TELOM</td><td>0</td><td>0</td><td>0</td><td>0</td><td>0</td><td>0</td><td>00H</td></tr>
<tr><td>Description</td><td colspan="12">The 1st Parameter is dummy read.
<table><tr><th>Bit</th><th>Description</th><th>Value</th></tr><tr><td>D7</td><td>Tearing Effect Line On/Off</td><td>“0”=Off, “1”= On</td></tr><tr><td>D6</td><td>Tearing Effect Line Mode</td><td>“0”=Mode1,“1”= Mode2</td></tr></table></td></tr>
<tr><td>Restriction</td><td colspan="12">None</td></tr>
<tr><td>Register Availability</td><td colspan="12"><table><tr><th>Status</th><th>Availability</th></tr><tr><td>Normal Mode on,Idle Mode Off,Sleep Out</td><td>Yes</td></tr><tr><td>Normal Mode on,Idle Mode On,Sleep Out</td><td>Yes</td></tr><tr><td>Partial Mode on,Idle Mode Off,Sleep Out</td><td>Yes</td></tr><tr><td>Partial Mode on,Idle Mode On,Sleep Out</td><td>Yes</td></tr><tr><td>Sleep In</td><td>Yes</td></tr></table></td></tr>
<tr><td>Default</td><td colspan="12"><table><tr><th>Status</th><th>Default Value(D7 to D0)</th></tr><tr><td>Power On Sequence</td><td>8’h00</td></tr><tr><td>HW Reset</td><td>8’h00</td></tr></table></td></tr>
</table>

12.1.9. RDDSDR (0Fh)

0FH	Read Display Self-Diagnostic Result											
	D/CX	**RDX**	**WRX**	D7	D6	D5	D4	D3	D2	D1	D0	HEX
Command	L	R	H	0	0	0	0	1	1	1	1	0FH
Parameter1	H	R	H	X	X	X	X	X	X	X	X	X
Parameter2	H	R	H	4{SELF_DIAG}				FIFO_INTF2OSC_EMPTY	FIFO_INTF2OSC_FULL	FIFO_OSC2TCON_EMPTY	FIFO_OSC2TCON_FULL	FAH
Description	The 1st Parameter is dummy read. If internal function work correctly, send this command will get an inverted result every time.											
Restriction	None											
Register Availability												

Status	Availability
Normal Mode on,Idle Mode Off,Sleep Out	Yes
Normal Mode on,Idle Mode On,Sleep Out	Yes
Partial Mode on,Idle Mode Off,Sleep Out	Yes
Partial Mode on,Idle Mode On,Sleep Out	Yes
Sleep In	Yes

Default

Status	Default Value(D7 to D0)
Power On Sequence	8'hFA
HW Reset	8'hFA

12.1.10. SLPIN (10h)

<table>
<tr><th>10H</th><th colspan="12">Sleep In</th></tr>
<tr><td></td><td>D/CX</td><td>RDX</td><td>WRX</td><td>D7</td><td>D6</td><td>D5</td><td>D4</td><td>D3</td><td>D2</td><td>D1</td><td>D0</td><td>HEX</td></tr>
<tr><td>Command</td><td>L</td><td>H</td><td>R</td><td>0</td><td>0</td><td>0</td><td>1</td><td>0</td><td>0</td><td>0</td><td>0</td><td>10H</td></tr>
<tr><td>Parameter</td><td colspan="12">No Parameter</td></tr>
<tr><td>Description</td><td colspan="12">This command causes the LCD module to enter the minimum power consumption mode.
In this mode e.g. the DC/DC converter is stopped, Internal oscillator is stopped, and panel scanning is stopped.
MCU interface and memory are still working and the memory keeps it’s contents.</td></tr>
<tr><td>Restriction</td><td colspan="12">This command has no effect when module is already in sleep in mode. Sleep In Mode can only be left by the Sleep Out Command (11h).
It will be necessary to wait 5msec before sending next command; this is to allow time for the supply voltages and clock circuits to stabilize.
It will be necessary to wait 120msec after sending Sleep Out command (when in Sleep In Mode) before Sleep In command can be sent.</td></tr>
<tr><td>Register Availability</td><td colspan="12"><table><tr><th>Status</th><th>Availability</th></tr><tr><td>Normal Mode on,Idle Mode Off,Sleep Out</td><td>Yes</td></tr><tr><td>Normal Mode on,Idle Mode On,Sleep Out</td><td>Yes</td></tr><tr><td>Partial Mode on,Idle Mode Off,Sleep Out</td><td>Yes</td></tr><tr><td>Partial Mode on,Idle Mode On,Sleep Out</td><td>Yes</td></tr><tr><td>Sleep In</td><td>Yes</td></tr></table></td></tr>
<tr><td>Default</td><td colspan="12"><table><tr><th>Status</th><th>Default Value(D7 to D0)</th></tr><tr><td>Power On Sequence</td><td>SLPIN</td></tr><tr><td>HW Reset</td><td>SLPIN</td></tr></table></td></tr>
</table>

12.1.11. SLPOUT(11h)

<table>
<tr><th>11H</th><th colspan="12">Sleep Out</th></tr>
<tr><td></td><td>D/CX</td><td>RDX</td><td>WRX</td><td>D7</td><td>D6</td><td>D5</td><td>D4</td><td>D3</td><td>D2</td><td>D1</td><td>D0</td><td>HEX</td></tr>
<tr><td>Command</td><td>L</td><td>H</td><td>R</td><td>0</td><td>0</td><td>0</td><td>1</td><td>0</td><td>0</td><td>0</td><td>1</td><td>11H</td></tr>
<tr><td>Parameter</td><td colspan="12">No Parameter</td></tr>
<tr><td>Description</td><td colspan="12">This command turns off sleep mode.
In this mode e.g. the DC/DC converter is enabled. Internal oscillator is started, and panel scanning is started.</td></tr>
<tr><td>Restriction</td><td colspan="12">This command has no effect when module is already in sleep out mode. Sleep Our Mode can only be left by the Sleep in Command (10h).
It will be necessary to wait 5msec before sending next command; this is to allow time for the clock circuits to stabilize.
The display module loads all display supplier’s factory default values to the registers during this 120msec and there cannot be any abnormal visual effect on the display image if factory default and register values are same when this load is done and when the display module is already Sleep Out-mode.
The display module is doing self-diagnostic function during this 5msec. It will be necessary to wait 120msec after sending Sleep In command (when in Sleep Out mode) before Sleep Out command can be sent.
This command has no effect when module is already in sleep out mode.
Sleep Out Mode can only be left by HW Reset, Sleep In (10h), or a NMI event trigger.</td></tr>
<tr><td>Register Availability</td><td colspan="12"><table><tr><th>Status</th><th>Availability</th></tr><tr><td>Normal Mode on,Idle Mode Off,Sleep Out</td><td>Yes</td></tr><tr><td>Normal Mode on,Idle Mode On,Sleep Out</td><td>Yes</td></tr><tr><td>Partial Mode on,Idle Mode Off,Sleep Out</td><td>Yes</td></tr><tr><td>Partial Mode on,Idle Mode On,Sleep Out</td><td>Yes</td></tr><tr><td>Sleep In</td><td>Yes</td></tr></table></td></tr>
<tr><td>Default</td><td colspan="12"><table><tr><th>Status</th><th>Default Value(D7 to D0)</th></tr><tr><td>Power On Sequence</td><td>SLPIN</td></tr><tr><td>HW Reset</td><td>SLPIN</td></tr></table></td></tr>
</table>

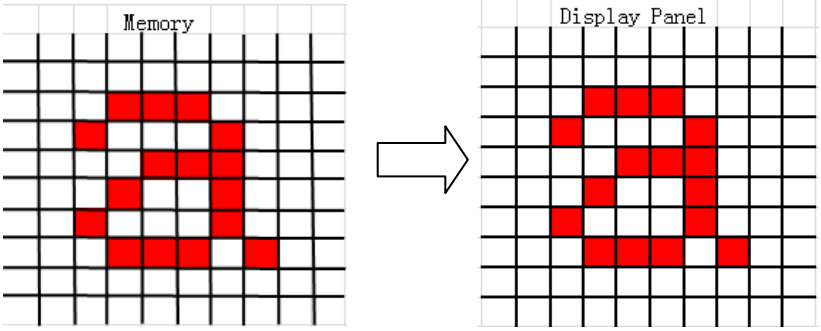
12.1.12. PTLON (12h)

<table>
<tr><th>12H</th><th colspan="12">Partial Display Mode On</th></tr>
<tr><td></td><td>D/CX</td><td>RDX</td><td>WRX</td><td>D7</td><td>D6</td><td>D5</td><td>D4</td><td>D3</td><td>D2</td><td>D1</td><td>D0</td><td>HEX</td></tr>
<tr><td>Command</td><td>L</td><td>H</td><td>R</td><td>0</td><td>0</td><td>0</td><td>1</td><td>0</td><td>0</td><td>1</td><td>0</td><td>12H</td></tr>
<tr><td>Parameter</td><td colspan="12">No Parameter</td></tr>
<tr><td>Description</td><td colspan="12">This command turns on partial mode.The partial mode is described by the Partial Area command (30h).
To leave Partial mode, the Normal Display On command (13h) should be written.
X=Don’t care
Note: If a command is written in a frame cycle, the command becomes effective from the next frame.</td></tr>
<tr><td>Restriction</td><td colspan="12">This command has no effect during Partial mode is active.</td></tr>
<tr><td>Register Availability</td><td colspan="12"><table><tr><th>Status</th><th>Availability</th></tr><tr><td>Normal Mode on,Idle Mode Off,Sleep Out</td><td>Yes</td></tr><tr><td>Normal Mode on,Idle Mode On,Sleep Out</td><td>Yes</td></tr><tr><td>Partial Mode on,Idle Mode Off,Sleep Out</td><td>Yes</td></tr><tr><td>Partial Mode on,Idle Mode On,Sleep Out</td><td>Yes</td></tr><tr><td>Sleep In</td><td>Yes</td></tr></table></td></tr>
<tr><td>Default</td><td colspan="12"><table><tr><th>Status</th><th>Default Value(D7 to D0)</th></tr><tr><td>Power On Sequence</td><td>Normal Display On</td></tr><tr><td>HW Reset</td><td>Normal Display On</td></tr></table></td></tr>
</table>

12.1.13. NORON (13h)

<table>
<tr><th>13H</th><th colspan="12">Normal Display Mode On</th></tr>
<tr><td></td><td>D/CX</td><td>RDX</td><td>WRX</td><td>D7</td><td>D6</td><td>D5</td><td>D4</td><td>D3</td><td>D2</td><td>D1</td><td>D0</td><td>HEX</td></tr>
<tr><td>Command</td><td>L</td><td>H</td><td>R</td><td>0</td><td>0</td><td>0</td><td>1</td><td>0</td><td>0</td><td>1</td><td>1</td><td>13H</td></tr>
<tr><td>Parameter</td><td colspan="12">No Parameter</td></tr>
<tr><td>Description</td><td colspan="12">This command returns the display to normal mode.
Normal display mode on means Partial mode off and Scroll mode off.
Exit from NORON by the Partial mode On command (12h)
X=Don’t care
Note: If a command is written in a frame cycle, the command becomes effective from the next frame.</td></tr>
<tr><td>Restriction</td><td colspan="12">This command has no effect when Normal Display mode is active.</td></tr>
<tr><td>Register Availability</td><td colspan="12"><table><tr><th>Status</th><th>Availability</th></tr><tr><td>Normal Mode on,Idle Mode Off,Sleep Out</td><td>Yes</td></tr><tr><td>Normal Mode on,Idle Mode On,Sleep Out</td><td>Yes</td></tr><tr><td>Partial Mode on,Idle Mode Off,Sleep Out</td><td>Yes</td></tr><tr><td>Partial Mode on,Idle Mode On,Sleep Out</td><td>Yes</td></tr><tr><td>Sleep In</td><td>Yes</td></tr></table></td></tr>
<tr><td>Default</td><td colspan="12"><table><tr><th>Status</th><th>Default Value(D7 to D0)</th></tr><tr><td>Power On Sequence</td><td>Normal Display On</td></tr><tr><td>HW Reset</td><td>Normal Display On</td></tr></table></td></tr>
</table>

12.1.14. INVOFF (20h)

20H	Display Inversion Off											
	D/CX	**RDX**	**WRX**	D7	D6	D5	D4	D3	D2	D1	D0	HEX
Command	L	H	R	0	0	1	0	0	0	0	0	20H
Parameter	No Parameter											
Description	This command is used to recover from display inversion mode. This command makes no change of contents of frame memory. This command does not change any other status. 											
Restriction	This command has no effect when module is already in inversion off mode.											

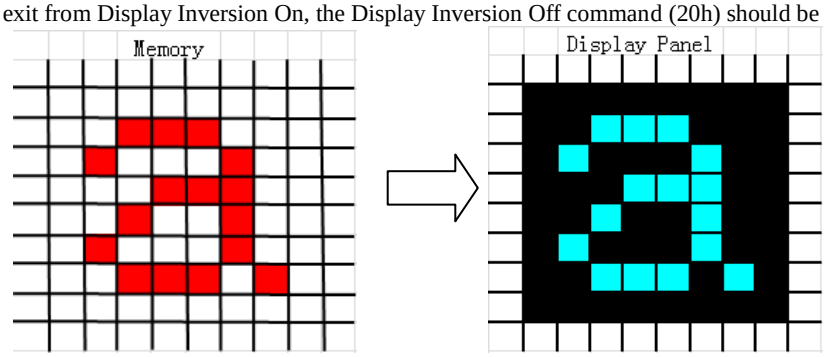
Register Availability

Status	Availability
Normal Mode on,Idle Mode Off,Sleep Out	Yes
Normal Mode on,Idle Mode On,Sleep Out	Yes
Partial Mode on,Idle Mode Off,Sleep Out	Yes
Partial Mode on,Idle Mode On,Sleep Out	Yes
Sleep In	Yes

Default

Status	Default Value(D7 to D0)
Power On Sequence	Normal Display On
HW Reset	Normal Display On

12.1.15. INVON (21h)

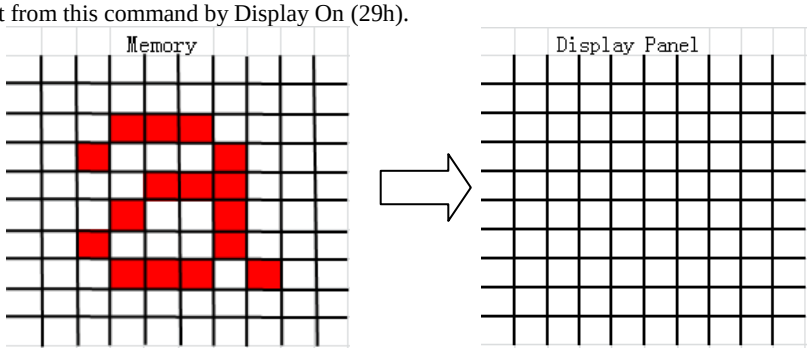
21H	Display Inversion On											
	D/CX	**RDX**	**WRX**	D7	D6	D5	D4	D3	D2	D1	D0	HEX
Command	L	H	R	0	0	1	0	0	0	0	1	21H
Parameter	No Parameter											
Description	This command is used to enter into display inversion mode. This command makes no change of contents of frame memory. Every bit is inverted from the frame memory to the display. This command does not change any other status. To exit from Display Inversion On, the Display Inversion Off command (20h) should be written. 											
Restriction	This command has no effect when module is already in inversion on mode.											
Register Availability												

Status	Availability
Normal Mode on,Idle Mode Off,Sleep Out	Yes
Normal Mode on,Idle Mode On,Sleep Out	Yes
Partial Mode on,Idle Mode Off,Sleep Out	Yes
Partial Mode on,Idle Mode On,Sleep Out	Yes
Sleep In	Yes

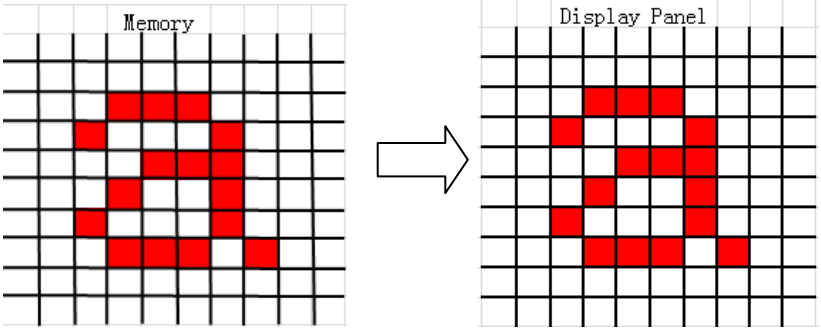
Default

Status	Default Value(D7 to D0)
Power On Sequence	Display Inversion On
HW Reset	Display Inversion On

12.1.16. DISPOFF (28h)

28H	Display Off											
	D/CX	**RDX**	**WRX**	D7	D6	D5	D4	D3	D2	D1	D0	HEX
Command	L	H	R	0	0	1	0	1	0	0	0	28H
Parameter	No Parameter											
Description	This command is used to enter into DISPLAY OFF mode. In this mode, the output from Frame Memory is disabled and blank page inserted. This command makes no change of contents of frame memory. This command does not change any other status. There will be no abnormal visible effect on the display. Exit from this command by Display On (29h). 											
Restriction	This command has no effect when module is already in display off mode.											
Register Availability	**Status** / **Availability** Normal Mode on,Idle Mode Off,Sleep Out — Yes Normal Mode on,Idle Mode On,Sleep Out — Yes Partial Mode on,Idle Mode Off,Sleep Out — Yes Partial Mode on,Idle Mode On,Sleep Out — Yes Sleep In — Yes											
Default	**Status** / **Default Value(D7 to D0)** Power On Sequence — Display Off HW Reset — Display Off											

12.1.17 DISPON (29h)

29H	Display On											
	D/CX	**RDX**	**WRX**	D7	D6	D5	D4	D3	D2	D1	D0	HEX
Command	L	H	R	0	0	1	0	1	0	0	1	29H
Parameter	No Parameter											
Description	This command is used to recover from DISPLAY OFF mode. Output from the Frame Memory is enabled. This command makes no change of contents of frame memory. This command does not change any other status. 											
Restriction	This command has no effect when module is already in display on mode.											
Register Availability												

Status	Availability
Normal Mode on,Idle Mode Off,Sleep Out	Yes
Normal Mode on,Idle Mode On,Sleep Out	Yes
Partial Mode on,Idle Mode Off,Sleep Out	Yes
Partial Mode on,Idle Mode On,Sleep Out	Yes
Sleep In	Yes

Default

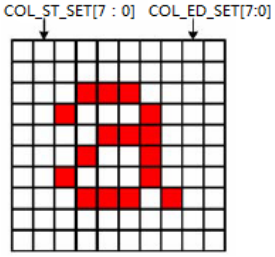
Status	Default Value(D7 to D0)
Power On Sequence	Display On
HW Reset	Display On

12.1.18 CASET (2Ah)

2AH	Column Address Set											
	D/CX	RDX	WRX	D7	D6	D5	D4	D3	D2	D1	D0	HEX
Command	L	H	R	0	0	1	0	1	0	1	0	2AH
Parameter 1	H	H	R	col_st_set_pre[15:8]								00H
Parameter 2	H	H	R	col_st_set_pre[7:0]								18H
Parameter 3	H	H	R	col_ed_set_pre[15:8]								00H
Parameter 4	H	H	R	col_ed_set_pre[7:0]								67H

Description

This command is used to define area of frame memory where MCU can access.
This command makes no change on the other driver status.
The value of col_st_set_pre[7:0] and col_ed_set_pre[7:0] are referred when RAMWR command comes.
Each value represents one column line in the Frame Memory.



Restriction

XS[7:0] always must be equal to or less than XE[7:0].
When XS[7:0] or XE[7:0] is greater than maximum row address like below, data of out of range will be ignored.
80x160 memory base(GM='000', '101')
(Parameter range:26(001AH)≦XS[7:0]≦XE[7:0]≦105(0069H)):MV="0"
(Parameter range:1(0001H)≦XS[7:0]≦XE[7:0]≦160(00A0H)):MV="1"
GM='010','011','001','100','110','111'
(Parameter range:24(0018H)≦XS[7:0]≦XE[7:0]≦103(0067H)):MV="0"
(Parameter range:0(0000H)≦XS[7:0]≦XE[7:0]≦159(009FH)):MV="1"

Register Availability

Status	Availability
Normal Mode on,Idle Mode Off,Sleep Out	Yes
Normal Mode on,Idle Mode On,Sleep Out	Yes
Partial Mode on,Idle Mode Off,Sleep Out	Yes
Partial Mode on,Idle Mode On,Sleep Out	Yes
Sleep In	Yes

Default

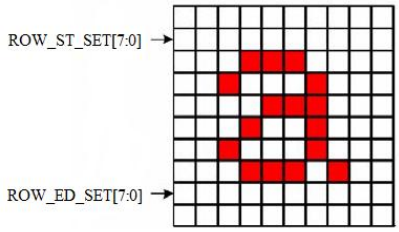
Status	Default Value		
(80x160 GM='000', '101')	XS[7:0]	XE[7:0]	EX[7:0](MV=1)
Power On Sequence	001ah(26)	0069h(105)	
H/W Reset	001ah(26)	0069h(105)	
Status	**Default Value**		
(80x160 GM='010','011','001', '100','110','111')	XS[7:0]	XE[7:0]	EX[7:0](MV=1)
Power On Sequence	0018h(24)	0067h(103)	
H/W Reset	0018h(24)	0067h(103)	

12.1.19 RASET (2Bh)

2BH	Row Address Set											
	D/CX	**RDX**	**WRX**	D7	D6	D5	D4	D3	D2	D1	D0	HEX
Command	L	H	R	0	0	1	0	1	0	1	1	2BH
Parameter 1	H	H	R	row_st_set_pre[15:8]								00H
Parameter 2	H	H	R	row_st_set_pre[7:0]								00H
Parameter 3	H	H	R	row_ed_set_pre[15:8]								00H
Parameter 4	H	H	R	row_ed_set_pre[7:0]								9FH

Description

This command is used to define area of frame memory where MCU can access.
This command makes no change on the other driver status.
The value of row_st_set_pre[7:0] and row_ed_set_pre[7:0] are referred when RAMWR command comes.
Each value represents one Page line in the Frame Memory.



Restriction

YS[7:0] always must be equal to or less than YE[7:0].
When YS[7:0] or YE[7:0] is greater than maximum row address like below,data of out of range will be ignored.
80x160 memory base(GM=’000’, ’101’)
(Parameter range:1(0001H)≦YS[7:0]≦YE[7:0]≦160(00A0H)):MV=”0”
(Parameter range:26(001AH)≦YS[7:0]≦YE[7:0]≦105(0069H)):MV=”1”
GM=’010’,’011’,’001’,’100’,’110’,’111’
(Parameter range:0(0000H)≦YS[7:0]≦YE[7:0]≦159(009FH)):MV=”0”
(Parameter range:24(0018H)≦YS[7:0]≦YE[7:0]≦103(0067H)):MV=”1”

Register Availability

Status	Availability
Normal Mode on,Idle Mode Off,Sleep Out	Yes
Normal Mode on,Idle Mode On,Sleep Out	Yes
Partial Mode on,Idle Mode Off,Sleep Out	Yes
Partial Mode on,Idle Mode On,Sleep Out	Yes
Sleep In	Yes

Default

Status	Default Value		
(80X160 GM=’000’, ’101’)	YS[7:0]	YE[7:0]	YX[7:0](MV=1)
Power On Sequence	0001h(1)	A0h(160)	
H/W Reset	0001h(1)	A0h(160)	
Status	**Default Value**		
(80x160 GM=’010’,’011’,’001’, ’100’,’110’,’111’)	YS[7:0]	YE[7:0]	YX[7:0](MV=1)
Power On Sequence	0000h(0)	9Fh(159)	
H/W Reset	0000h(0)	9Fh(159)	

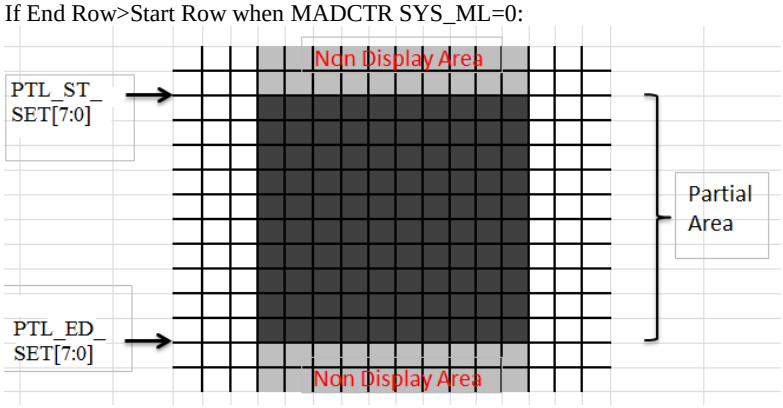
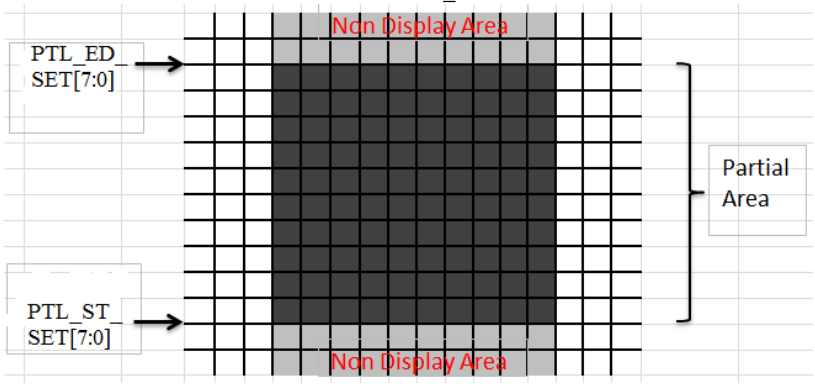
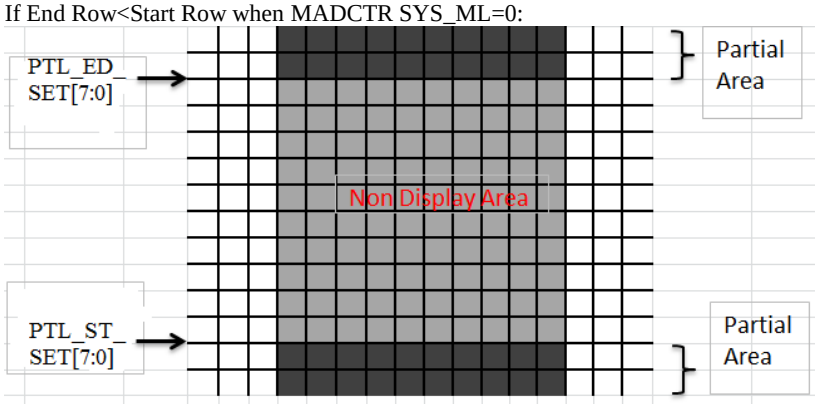
12.1.20 RAMWR (2Ch)

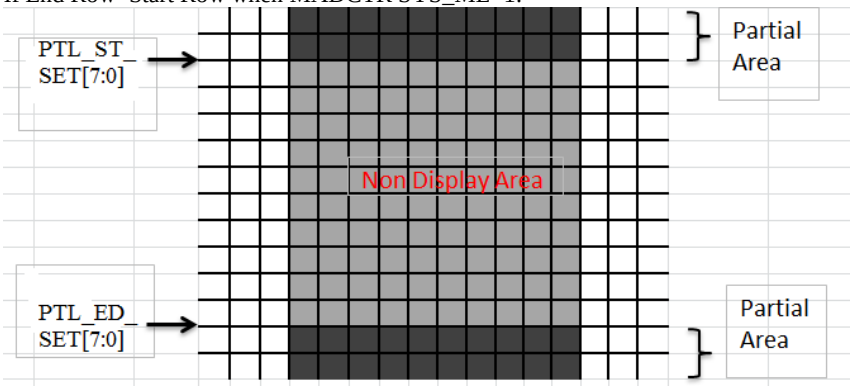
<table>
<tr><th>2CH</th><th colspan="12">Memory Write</th></tr>
<tr><td></td><td>D/CX</td><td>RDX</td><td>WRX</td><td>D7</td><td>D6</td><td>D5</td><td>D4</td><td>D3</td><td>D2</td><td>D1</td><td>D0</td><td>HEX</td></tr>
<tr><td>Command</td><td>L</td><td>H</td><td>R</td><td>0</td><td>0</td><td>1</td><td>0</td><td>1</td><td>1</td><td>0</td><td>0</td><td>2CH</td></tr>
<tr><td>Parameter 1</td><td>H</td><td>H</td><td>R</td><td></td><td></td><td></td><td></td><td></td><td></td><td></td><td></td><td>-</td></tr>
<tr><td>Description</td><td colspan="12">This command is used to transfer data from MCU to frame memory.
This command makes no change to the other driver status.
When this command is accepted, the column register and the page register are reset to the Start Column/Start Page positions.
The Start Column /Start Page positions are different in accordance with MADCTR setting.
Then D[7:0] is stored in frame memory and the column register and the row register incremented.
Sending any other command can stop frame Write.</td></tr>
<tr><td>Restriction</td><td colspan="12">In all color modes, there is no restriction on length of parameters.
80x160 memory base
80x160x16-bit memory can be written by this command.
Memory range(0000h,0000h)->(004Fh,09Fh)</td></tr>
<tr><td>Register Availability</td><td colspan="12"><table><tr><th>Status</th><th>Availability</th></tr><tr><td>Normal Mode on,Idle Mode Off,Sleep Out</td><td>Yes</td></tr><tr><td>Normal Mode on,Idle Mode On,Sleep Out</td><td>Yes</td></tr><tr><td>Partial Mode on,Idle Mode Off,Sleep Out</td><td>Yes</td></tr><tr><td>Partial Mode on,Idle Mode On,Sleep Out</td><td>Yes</td></tr><tr><td>Sleep In</td><td>Yes</td></tr></table></td></tr>
<tr><td>Default</td><td colspan="12"><table><tr><th>Status</th><th>Default Value(D7 to D0)</th></tr><tr><td>Power On Sequence</td><td>Contents of memory is set randomly</td></tr><tr><td>HW Reset</td><td>Contents of memory is not cleared</td></tr></table></td></tr>
</table>

12.1.21 RAMRD (2Eh)

<table>
<tr><th>2EH</th><th colspan="12">Memory Read</th></tr>
<tr><td></td><td>D/CX</td><td>RDX</td><td>WRX</td><td>D7</td><td>D6</td><td>D5</td><td>D4</td><td>D3</td><td>D2</td><td>D1</td><td>D0</td><td>HEX</td></tr>
<tr><td>Command</td><td>L</td><td>R</td><td>R</td><td>0</td><td>0</td><td>1</td><td>0</td><td>1</td><td>1</td><td>1</td><td>0</td><td>2EH</td></tr>
<tr><td>Parameter</td><td>H</td><td>R</td><td>R</td><td></td><td></td><td></td><td></td><td></td><td></td><td></td><td></td><td>–</td></tr>
<tr><td>Description</td><td colspan="12">This command is used to transfer data from frame memory to MCU.
This command makes no change to other driver status.
When this command is accepted, the column register and then row register are reset to the Start Column/Start Row positions.
The Start Column/Start Row positions are different in accordance with MADCTR setting.
Then D[7:0] is read back from the frame memory and the column register and the register incremented.
Frame Read can be stopped by sending any other command.
Display Data Format for color coding (16 bit cases), when there is used 8 data line for image data.</td></tr>
<tr><td>Restriction</td><td colspan="12">In all color modes, the Frame Read is always 16 bit so there is no restriction on length of parameters.
Note: Memory Read is only possible via the Parallel Interface.</td></tr>
<tr><td>Register Availability</td><td colspan="12"><table><tr><th>Status</th><th>Availability</th></tr><tr><td>Normal Mode on,Idle Mode Off,Sleep Out</td><td>Yes</td></tr><tr><td>Normal Mode on,Idle Mode On,Sleep Out</td><td>Yes</td></tr><tr><td>Partial Mode on,Idle Mode Off,Sleep Out</td><td>Yes</td></tr><tr><td>Partial Mode on,Idle Mode On,Sleep Out</td><td>Yes</td></tr><tr><td>Sleep In</td><td>Yes</td></tr></table></td></tr>
<tr><td>Default</td><td colspan="12"><table><tr><th>Status</th><th>Default Value(D7 to D0)</th></tr><tr><td>Power On Sequence</td><td>Contents of memory is set randomly</td></tr><tr><td>HW Reset</td><td>Contents of memory is not cleared</td></tr></table></td></tr>
</table>

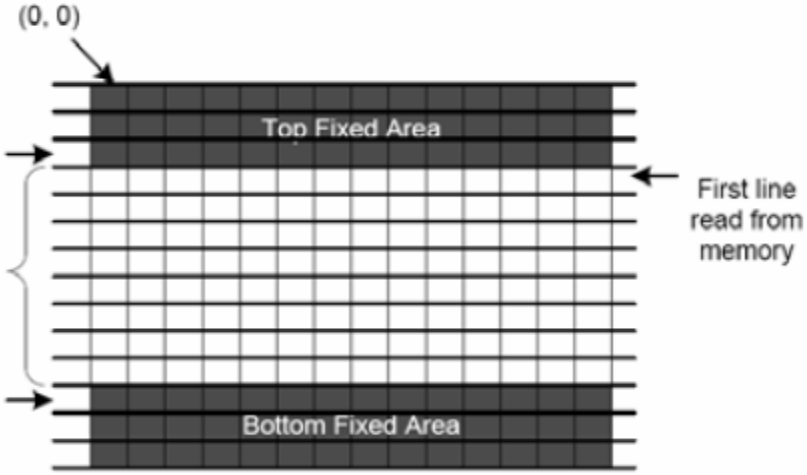
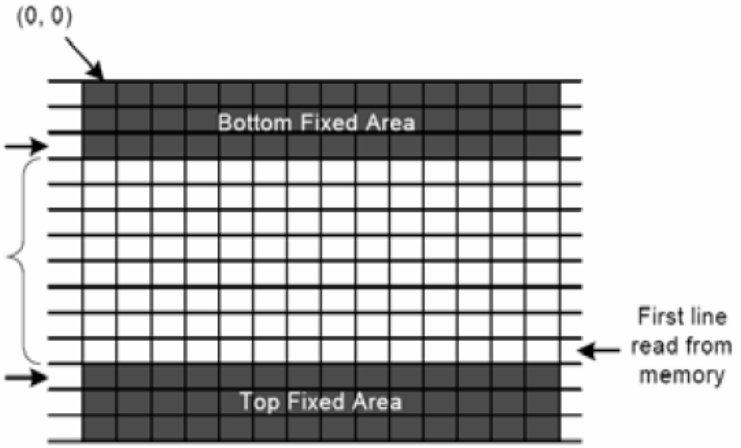
12.1.22 PTLAR(30h)

30H	Partial Area											
	D/CX	RDX	WRX	D7	D6	D5	D4	D3	D2	D1	D0	HEX
Command	L	H	R	0	0	1	1	0	0	0	0	30H
Parameter1	H	H	R	PTL_ST_SET[7:0]								00H
Parameter2	H	H	R	PTL_ED_SET[7:0]								9FH
Description	This command defines the partial mode's display area.There are 2 parameters associated with this command, the first defines the Start Row(PSL) and the second the End Row(PEL),as illustrated in the figure below.PSL and PEL refer to the Frame Memory Line Pointer. If End Row>Start Row when MADCTR SYS_ML=0:  If End Row>Start Row when MADCTR SYS_ML=1:  If End Row<Start Row when MADCTR SYS_ML=0:  											

<table>
<tr><td></td><td>If End Row<Start Row when MADCTR SYS_ML=1:
 

If End Row=Start Row then the Partial Area will be one row deep.</td></tr>
<tr><td>Restriction</td><td>No Restriction</td></tr>
<tr><td>Register Availability</td><td>
<table>
<tr><th>Status</th><th>Availability</th></tr>
<tr><td>Normal Mode on,Idle Mode Off,Sleep Out</td><td>Yes</td></tr>
<tr><td>Normal Mode on,Idle Mode On,Sleep Out</td><td>Yes</td></tr>
<tr><td>Partial Mode on,Idle Mode Off,Sleep Out</td><td>Yes</td></tr>
<tr><td>Partial Mode on,Idle Mode On,Sleep Out</td><td>Yes</td></tr>
<tr><td>Sleep In</td><td>Yes</td></tr>
</table>
</td></tr>
<tr><td>Default</td><td>
<table>
<tr><th rowspan="2">Status</th><th colspan="3">Default Value</th></tr>
<tr><th>PSL[7:0]</th><th colspan="2">PEL[7:0]</th></tr>
<tr><td>GM</td><td>"xxx"</td><td>"000","101"</td><td>"010","011","001","100",
"110","111"</td></tr>
<tr><td>Power On Sequence</td><td>8’h00</td><td>8’hA0</td><td>8’h9F</td></tr>
<tr><td>HW Reset</td><td>8’h00</td><td>8’hA0</td><td>8’h9F</td></tr>
</table>
</td></tr>
</table>

12.1.23 SCRLAR (33h)

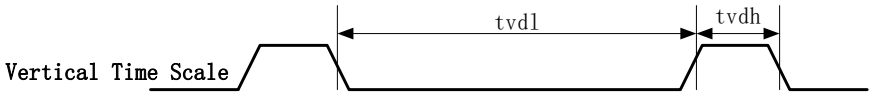
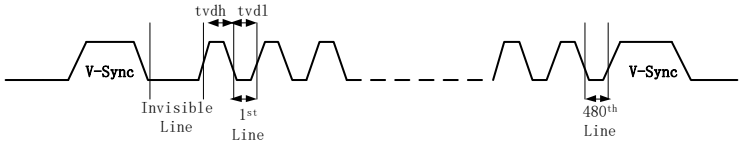
33H	Scroll Area											
	D/CX	**RDX**	**WRX**	D7	D6	D5	D4	D3	D2	D1	D0	HEX
Command	L	H	R	0	0	1	1	0	0	1	1	33H
Parameter1	H	H	R	TFA[7:0]								00H
Parameter2	H	H	R	VSA_SET_PRE[7:0]								00H
Description	This command defines the Vertical Scrolling Area of the display. When MADCTR SYS_ML =0 The 1st parameter TFA[7:0]describes the Top Fixed Area (in No.of lines from TOP of the Frame Memory and Display). The 2nd parameter VSA[7:0]describes the height of the Vertical Scrolling Area(in No.of lines of the Frame Memory[not the display] from the Vertical Scrolling Start Address). The first line read from Frame Memory appears immediately after the bottom most line of the Top Fixed Area. TFA, VSA refer to the Frame Memory Line Point.  When MADCTR SYS_ML=1 The 1st parameter TFA[7:0] describes the Top Fixed Area (in No.of lines from Bottom of the Frame Memory and Display). The 2nd parameter VSA[7:0] describes the height of the Vertical Scrolling Area (in No.of lines of the Frame Memory [not the display] from the Vertical Scrolling Start Address). The first line read from Frame Memory appears immediately after the top most line of the Top Fixed Area.  											

<table>
<tr><td>Restriction</td><td>The condition is (TFA+VSA+BFA)=160 in 80RGBx160
Otherwise Scrolling mode is undefined.
In Vertical Scroll Mode, MADCTR parameter SYS_MV should be set to “0”
this affects the Frame memory Write.</td></tr>
<tr><td>Register Availability</td><td>
<table>
<tr><th>Status</th><th>Availability</th></tr>
<tr><td>Normal Mode on,Idle Mode Off,Sleep Out</td><td>Yes</td></tr>
<tr><td>Normal Mode on,Idle Mode On,Sleep Out</td><td>Yes</td></tr>
<tr><td>Partial Mode on,Idle Mode Off,Sleep Out</td><td>Yes</td></tr>
<tr><td>Partial Mode on,Idle Mode On,Sleep Out</td><td>Yes</td></tr>
<tr><td>Sleep In</td><td>Yes</td></tr>
</table>
</td></tr>
<tr><td>Default</td><td>
<table>
<tr><th rowspan="2">Status</th><th colspan="3">Default Value</th></tr>
<tr><th>PSL[7:0]</th><th colspan="2">PEL[7:0]</th></tr>
<tr><td>GM</td><td>"xxx"</td><td>"000","101"</td><td>"010","011","001","100", "110","111"</td></tr>
<tr><td>Power On Sequence</td><td>8’h00</td><td>8’hA0</td><td>8’h9F</td></tr>
<tr><td>HW Reset</td><td>8’h00</td><td>8’hA0</td><td>8’h9F</td></tr>
</table>
</td></tr>
</table>

12.1.24 TEOFF (34h)

<table>
<tr><th>34H</th><th colspan="12">Tearing Effect Line Off</th></tr>
<tr><td></td><td>D/CX</td><td>RDX</td><td>WRX</td><td>D7</td><td>D6</td><td>D5</td><td>D4</td><td>D3</td><td>D2</td><td>D1</td><td>D0</td><td>HEX</td></tr>
<tr><td>Command</td><td>L</td><td>H</td><td>R</td><td>0</td><td>0</td><td>1</td><td>1</td><td>0</td><td>1</td><td>0</td><td>0</td><td>34H</td></tr>
<tr><td>Parameter</td><td colspan="12">No Parameter</td></tr>
<tr><td>Description</td><td colspan="12">This command is used to turn OFF (Active Low) the Tearing Effect output single from the TE signal line.</td></tr>
<tr><td>Restriction</td><td colspan="12">This command has no effect when Tearing Effect output is already OFF.</td></tr>
<tr><td>Register Availability</td><td colspan="12"><table><tr><th>Status</th><th>Availability</th></tr><tr><td>Normal Mode on,Idle Mode Off,Sleep Out</td><td>Yes</td></tr><tr><td>Normal Mode on,Idle Mode On,Sleep Out</td><td>Yes</td></tr><tr><td>Partial Mode on,Idle Mode Off,Sleep Out</td><td>Yes</td></tr><tr><td>Partial Mode on,Idle Mode On,Sleep Out</td><td>Yes</td></tr><tr><td>Sleep In</td><td>Yes</td></tr></table></td></tr>
<tr><td>Default</td><td colspan="12"><table><tr><th>Status</th><th>Default Value(D7 to D0)</th></tr><tr><td>Power On Sequence</td><td>OFF</td></tr><tr><td>HW Reset</td><td>OFF</td></tr></table></td></tr>
</table>

12.1.25 TEON (35h)

35H	Tearing Effect Line On											
	D/CX	**RDX**	**WRX**	D7	D6	D5	D4	D3	D2	D1	D0	HEX
Command	L	H	R	0	0	1	1	0	1	0	1	35H
Parameter	H	H	R	0	0	0	0	0	0	0	TELOM	00H
Description	This command is used to turn ON the Tearing Effect output signal from the TE signal line.This output is not affected by charging MADCTR bit ML. The Tearing Effect Line On has one parameter which describes the mode of the Tearing Effect Output Line. (X=Don't Care). When TELOM=0: The Tearing Effect Output line consists of V-Blanking information only:  When TELOM =1: The Tearing Effect Output line consists of both V-Blanking and H-Blanking information:  Note: During Sleep In Mode with Tearing Effect Line On, Tearing Effect Output pin will be active LOW. Display Data Format for color coding (8 bit cases), when there is used 8 data line for image data.											
Restriction	This command has no effect when Tearing Effect output is already ON.											

Register Availability

Status	Availability
Normal Mode on,Idle Mode Off,Sleep Out	Yes
Normal Mode on,Idle Mode On,Sleep Out	Yes
Partial Mode on,Idle Mode Off,Sleep Out	Yes
Partial Mode on,Idle Mode On,Sleep Out	Yes
Sleep In	Yes

Default

Status	Default Value(D7 to D0)
Power On Sequence	Tearing effect off & TELOM=0
HW Reset	Tearing effect off & TELOM=0

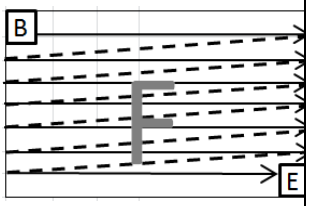
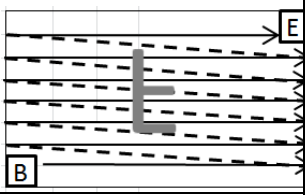
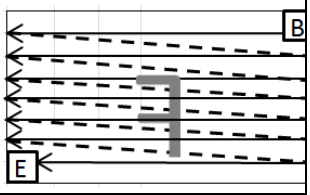
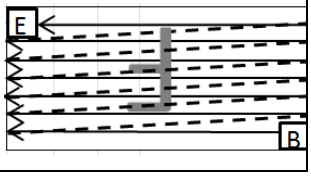
12.1.26 MADCTR (36h)

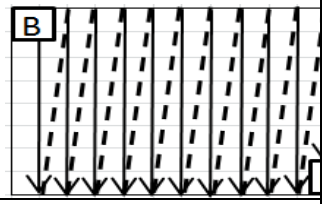
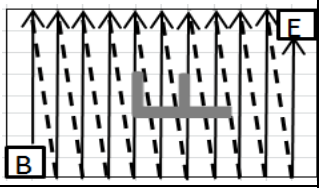
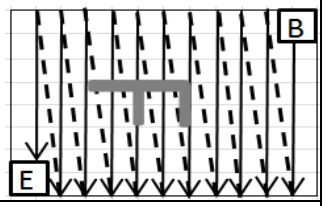
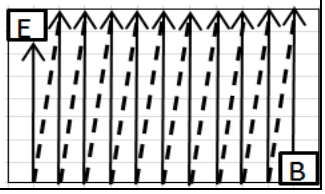
36H	Memory Data Access Control											
	D/CX	**RDX**	**WRX**	D7	D6	D5	D4	D3	D2	D1	D0	HEX
Command	L	H	R	0	0	1	1	0	1	1	0	36H
Parameter	H	H	R	SYS_MY	SYS_MX	SYS_MV	SYS_ML	SYS_RGB				08H

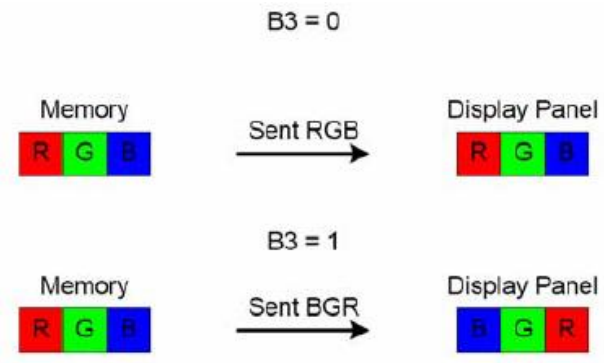
Description

This command defines read/write scanning direction of frame memory.
This command makes no change on the other driver status.
Bit Assignment

Bit	Description	Value
SYS_MY	Row Address Order	These 3 bits controls MPU to memory write/read direction.
SYS_MX	Column Address Order	
SYS_MV	Page/Column Selection	
SYS_ML	Vertical Order	LCD Vertical refresh direction control
SYS_RGB	RGB/BGR Order	Color selector switch control 0=RGB color filter panel 1=BGR color filter panel

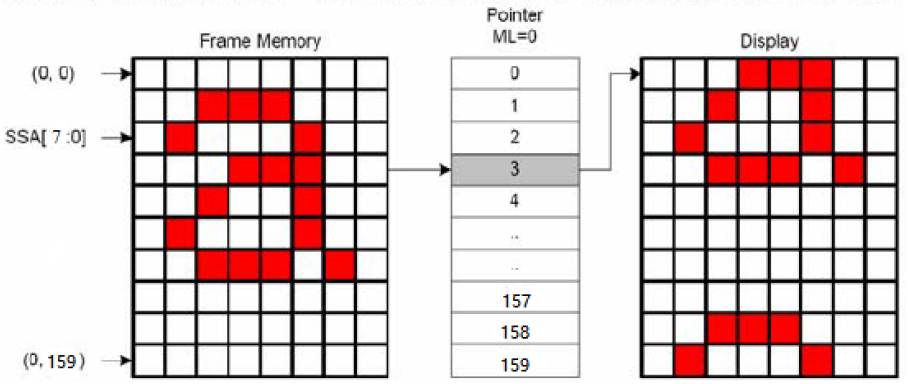
B5	B6	B7	Image in Frame Memory
0	0	0	
0	0	1	
0	1	0	
0	1	1	

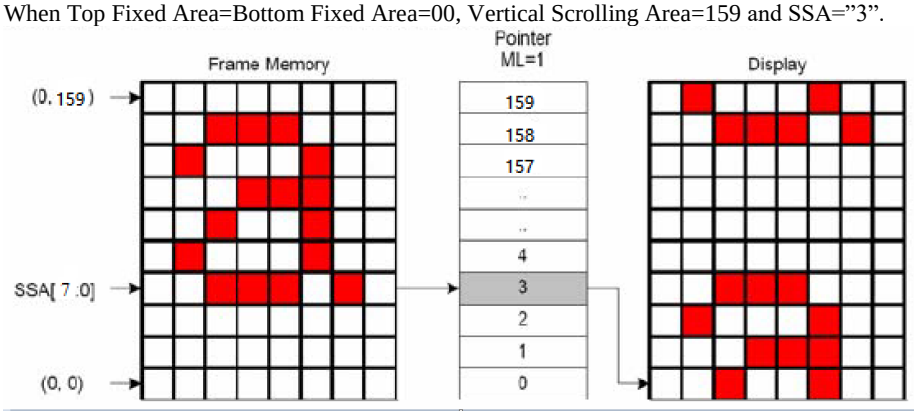
B5	B6	B7	Image in Frame Memory
1	0	0	
1	0	1	
1	1	0	
1	1	1	

<table>
<tr><td></td><td> 
</td></tr>
<tr><td>Restriction</td><td></td></tr>
<tr><td>Register Availability</td><td>
<table>
<tr><th>Status</th><th>Availability</th></tr>
<tr><td>Normal Mode on,Idle Mode Off,Sleep Out</td><td>Yes</td></tr>
<tr><td>Normal Mode on,Idle Mode On,Sleep Out</td><td>Yes</td></tr>
<tr><td>Partial Mode on,Idle Mode Off,Sleep Out</td><td>Yes</td></tr>
<tr><td>Partial Mode on,Idle Mode On,Sleep Out</td><td>Yes</td></tr>
<tr><td>Sleep In</td><td>Yes</td></tr>
</table>
</td></tr>
<tr><td>Default</td><td>
<table>
<tr><th>Status</th><th>Default Value</th></tr>
<tr><td>Power On Sequence</td><td>8’h08</td></tr>
<tr><td>HW Reset</td><td>8’h08</td></tr>
</table>
</td></tr>
</table>

12.1.27 VSCSAD (37h)

<table>
<tr><th>37H</th><th colspan="12">Vertical Scroll Start Address of RAM</th></tr>
<tr><td></td><td>D/CX</td><td>RDX</td><td>WRX</td><td>D7</td><td>D6</td><td>D5</td><td>D4</td><td>D3</td><td>D2</td><td>D1</td><td>D0</td><td>HEX</td></tr>
<tr><td>Command</td><td>L</td><td>H</td><td>R</td><td>0</td><td>0</td><td>1</td><td>1</td><td>0</td><td>1</td><td>1</td><td>1</td><td>37H</td></tr>
<tr><td>Parameter</td><td>H</td><td>H</td><td>R</td><td colspan="8">SSA[7:0]</td><td>00H</td></tr>
<tr><td>Description</td><td colspan="12">This command is used together with Scroll Area(33h).These two command describe the scrolling area and scrolling mode.
The Vertical Scrolling Start Address command has one parameter which describes the address of the line in the Frame Memory that will be written as the first line after the last line of the Top Fixed Area on the display as illustrated below.
This command Start the scrolling.

When MADCTR SYS_ML=0
Example: GM=001,80RGBx160
When Top Fixed Area=Bottom Fixed Area=00, Vertical Scrolling Area=159 and Vertical Scrolling Pointer SSA=”3”.
 

When MADCTR SYS_ML=1
Example:GM=001,80RGBx160
When Top Fixed Area=Bottom Fixed Area=00, Vertical Scrolling Area=159 and SSA=”3”.
 

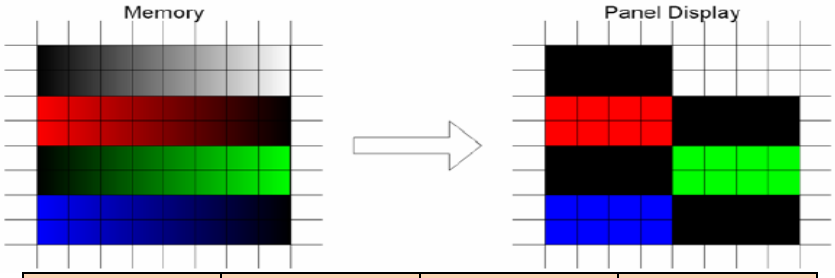
Note:
When new Pointer position and Picture Data are sent, the result on the display will happen at the next Panel Scan to avoid tearing effect. SSA refers to the Frame Memory scan address.</td></tr>
<tr><td>Restriction</td><td colspan="12"></td></tr>
<tr><td>Register Availability</td><td colspan="12"><table><tr><th>Status</th><th>Availability</th></tr><tr><td>Normal Mode on,Idle Mode Off,Sleep Out</td><td>Yes</td></tr><tr><td>Normal Mode on,Idle Mode On,Sleep Out</td><td>Yes</td></tr></table></td></tr>
</table>

	Partial Mode on,Idle Mode Off,Sleep Out: Yes Partial Mode on,Idle Mode On,Sleep Out: Yes Sleep In: Yes
Default	**Status** / **Default Value** Power On Sequence: 8’h00 HW Reset: 8’h00

12.1.28 IDMOFF (38h)

<table>
<tr><th>38H</th><th colspan="11">Idle Mode Off</th></tr>
<tr><td></td><td>D/CX</td><td>RDX</td><td>WRX</td><td>D7</td><td>D6</td><td>D5</td><td>D4</td><td>D3</td><td>D2</td><td>D1</td><td>D0</td><td>HEX</td></tr>
<tr><td>Command</td><td>L</td><td>H</td><td>R</td><td>0</td><td>0</td><td>1</td><td>1</td><td>1</td><td>0</td><td>0</td><td>0</td><td>38H</td></tr>
<tr><td>Parameter</td><td colspan="11">No Parameter</td></tr>
<tr><td>Description</td><td colspan="11">This command is used to recover from Idle mode on.
There will be no abnormal visible effect on the display mode change transition.
In the Idle off mode
1, LCD can display maximum 65,536 colors.
2, Normal frame frequency is applied.</td></tr>
<tr><td>Restriction</td><td colspan="11">This command has no effect when module is already in Idle off mode.</td></tr>
<tr><td>Register Availability</td><td colspan="11"><table><tr><th>Status</th><th>Availability</th></tr><tr><td>Normal Mode on,Idle Mode Off,Sleep Out</td><td>Yes</td></tr><tr><td>Normal Mode on,Idle Mode On,Sleep Out</td><td>Yes</td></tr><tr><td>Partial Mode on,Idle Mode Off,Sleep Out</td><td>Yes</td></tr><tr><td>Partial Mode on,Idle Mode On,Sleep Out</td><td>Yes</td></tr><tr><td>Sleep In</td><td>Yes</td></tr></table></td></tr>
<tr><td>Default</td><td colspan="11"><table><tr><th>Status</th><th>Default Value(D7 to D0)</th></tr><tr><td>Power On Sequence</td><td>Idle Mode Off</td></tr><tr><td>HW Reset</td><td>Idle Mode Off</td></tr></table></td></tr>
</table>

12.1.29 IDMON (39h)

<table>
<tr><th>39H</th><th colspan="12">Idle Mode On</th></tr>
<tr><td></td><td>D/CX</td><td>RDX</td><td>WRX</td><td>D7</td><td>D6</td><td>D5</td><td>D4</td><td>D3</td><td>D2</td><td>D1</td><td>D0</td><td>HEX</td></tr>
<tr><td>Command</td><td>L</td><td>H</td><td>R</td><td>0</td><td>0</td><td>1</td><td>1</td><td>1</td><td>0</td><td>0</td><td>1</td><td>39H</td></tr>
<tr><td>Parameter</td><td colspan="12">No Parameter</td></tr>
<tr><td>Description</td><td colspan="12">This command is used to enter into Idle mode on.
There will be no abnormal visible effect on the display mode change transition.
In the Idle mode.
Color expression is reduced. The primary and the secondary colors using MSB of each R,G and B in the Frame Memory, 8 color depth data is displayed.
8-Color mode frame frequency is applied.
Exit from IDMON by Idle Mode Off (38h) command.
 

<table>
<tr><th>Reduced Color</th><th>R[5:0]</th><th>G[5:0]</th><th>B[5:0]</th></tr>
<tr><td>Black</td><td>0XXXX</td><td>0XXXX</td><td>0XXXX</td></tr>
<tr><td>Blue</td><td>0XXXX</td><td>0XXXX</td><td>1XXXX</td></tr>
<tr><td>Red</td><td>1XXXX</td><td>0XXXX</td><td>0XXXX</td></tr>
<tr><td>Magenta</td><td>1XXXX</td><td>0XXXX</td><td>1XXXX</td></tr>
<tr><td>Green</td><td>0XXXX</td><td>1XXXX</td><td>0XXXX</td></tr>
<tr><td>Cyan</td><td>0XXXX</td><td>1XXXX</td><td>1XXXX</td></tr>
<tr><td>Yellow</td><td>1XXXX</td><td>1XXXX</td><td>0XXXX</td></tr>
<tr><td>White</td><td>1XXXX</td><td>1XXXX</td><td>1XXXX</td></tr>
</table></td></tr>
<tr><td>Restriction</td><td colspan="12">This command has no effect when module is already in idle on mode.</td></tr>
<tr><td>Register Availability</td><td colspan="12"><table>
<tr><th>Status</th><th>Availability</th></tr>
<tr><td>Normal Mode on,Idle Mode Off,Sleep Out</td><td>Yes</td></tr>
<tr><td>Normal Mode on,Idle Mode On,Sleep Out</td><td>Yes</td></tr>
<tr><td>Partial Mode on,Idle Mode Off,Sleep Out</td><td>Yes</td></tr>
<tr><td>Partial Mode on,Idle Mode On,Sleep Out</td><td>Yes</td></tr>
<tr><td>Sleep In</td><td>Yes</td></tr>
</table></td></tr>
<tr><td>Default</td><td colspan="12"><table>
<tr><th>Status</th><th>Default Value(D7 to D0)</th></tr>
<tr><td>Power On Sequence</td><td>Idle Mode Off</td></tr>
<tr><td>HW Reset</td><td>Idle Mode Off</td></tr>
</table></td></tr>
</table>

12.1.30 COLMOD (3Ah)

<table>
<tr><th>3AH</th><th colspan="12">Interface Color Mode Set</th></tr>
<tr><td></td><td>D/CX</td><td>RDX</td><td>WRX</td><td>D7</td><td>D6</td><td>D5</td><td>D4</td><td>D3</td><td>D2</td><td>D1</td><td>D0</td><td>HEX</td></tr>
<tr><td>Command</td><td>L</td><td>H</td><td>R</td><td>0</td><td>0</td><td>1</td><td>1</td><td>1</td><td>0</td><td>1</td><td>0</td><td>3AH</td></tr>
<tr><td>Parameter</td><td>H</td><td>H</td><td>R</td><td></td><td></td><td></td><td></td><td></td><td colspan="3">IFPF[2:0]</td><td>07H</td></tr>
<tr><td>Description</td><td colspan="13">This command is used to define the format of RGB picture data,which is to be transferred via the MCU interface.The formats are shown in the table:
<table>
<tr><th>Bit</th><th>Description</th><th>Value</th></tr>
<tr><td>IFPF2</td><td rowspan="3">Control Interface Color Format</td><td rowspan="3">"011"=12 bit/pixel
"101"=16 bit/pixel
"111"=8 bit/pixel
The others=not defined</td></tr>
<tr><td>IFPF1</td></tr>
<tr><td>IFPF0</td></tr>
</table>
Note:
In 12-bit /Pixel, 16-bit/Pixel, 8-bit/Pixel mode, the LUT is applied to transfer data into the Frame Memory.</td></tr>
<tr><td>Restriction</td><td colspan="13">This command has no effect when module is already in Idle off mode.</td></tr>
<tr><td>Register Availability</td><td colspan="13">
<table>
<tr><th>Status</th><th>Availability</th></tr>
<tr><td>Normal Mode on,Idle Mode Off,Sleep Out</td><td>Yes</td></tr>
<tr><td>Normal Mode on,Idle Mode On,Sleep Out</td><td>Yes</td></tr>
<tr><td>Partial Mode on,Idle Mode Off,Sleep Out</td><td>Yes</td></tr>
<tr><td>Partial Mode on,Idle Mode On,Sleep Out</td><td>Yes</td></tr>
<tr><td>Sleep In</td><td>Yes</td></tr>
</table>
</td></tr>
<tr><td>Default</td><td colspan="13">
<table>
<tr><th>Status</th><th>Default Value(D7 to D0)</th></tr>
<tr><td>Power On Sequence</td><td>8’h07</td></tr>
<tr><td>HW Reset</td><td>8’h07</td></tr>
</table>
</td></tr>
</table>

12.1.31 RDID1 (DAh)

DAH	Read ID1 Value											
	D/CX	RDX	WRX	D7	D6	D5	D4	D3	D2	D1	D0	HEX
Command	L	H	R	1	1	0	1	1	0	1	0	DAH
Parameter	H	R	H	SYS_ID1[7:0]								33H
Description	The return value is 33h											
Restriction	Should set “FF=A5” before configure this registers											
Register Availability												

Status	Availability
Normal Mode on,Idle Mode Off,Sleep Out	Yes
Normal Mode on,Idle Mode On,Sleep Out	Yes
Partial Mode on,Idle Mode Off,Sleep Out	Yes
Partial Mode on,Idle Mode On,Sleep Out	Yes
Sleep In	Yes

Default

Status	Default Value(D7 to D0)
Power On Sequence	8’h33
HW Reset	8’h33

12.1.32 RDID2 (DBh)

<table>
<tr><th>DBH</th><th colspan="12">Read ID2 Value</th></tr>
<tr><td></td><td>D/CX</td><td>RDX</td><td>WRX</td><td>D7</td><td>D6</td><td>D5</td><td>D4</td><td>D3</td><td>D2</td><td>D1</td><td>D0</td><td>HEX</td></tr>
<tr><td>Command</td><td>L</td><td>H</td><td>R</td><td>1</td><td>1</td><td>0</td><td>1</td><td>1</td><td>0</td><td>1</td><td>1</td><td>DBH</td></tr>
<tr><td>Parameter</td><td>H</td><td>R</td><td>H</td><td colspan="8">SYS_ID2[7:0]</td><td>30H</td></tr>
<tr><td>Description</td><td colspan="12">The return value is 30h</td></tr>
<tr><td>Restriction</td><td colspan="12">Should set “FF=A5” before configure this registers</td></tr>
<tr><td>Register Availability</td><td colspan="12"><table><tr><th>Status</th><th>Availability</th></tr><tr><td>Normal Mode on,Idle Mode Off,Sleep Out</td><td>Yes</td></tr><tr><td>Normal Mode on,Idle Mode On,Sleep Out</td><td>Yes</td></tr><tr><td>Partial Mode on,Idle Mode Off,Sleep Out</td><td>Yes</td></tr><tr><td>Partial Mode on,Idle Mode On,Sleep Out</td><td>Yes</td></tr><tr><td>Sleep In</td><td>Yes</td></tr></table></td></tr>
<tr><td>Default</td><td colspan="12"><table><tr><th>Status</th><th>Default Value(D7 to D0)</th></tr><tr><td>Power On Sequence</td><td>8’h30</td></tr><tr><td>HW Reset</td><td>8’h30</td></tr></table></td></tr>
</table>

12.1.33 RDID3 (DCh)

<table>
<tr><th>DCH</th><th colspan="12">Read ID3 Value</th></tr>
<tr><td></td><td>D/CX</td><td>RDX</td><td>WRX</td><td>D7</td><td>D6</td><td>D5</td><td>D4</td><td>D3</td><td>D2</td><td>D1</td><td>D0</td><td>HEX</td></tr>
<tr><td>Command</td><td>L</td><td>H</td><td>R</td><td>1</td><td>1</td><td>0</td><td>1</td><td>1</td><td>1</td><td>0</td><td>0</td><td>DCH</td></tr>
<tr><td>Parameter</td><td>H</td><td>R</td><td>H</td><td colspan="8">SYS_ID3[7:0]</td><td>23H</td></tr>
<tr><td>Description</td><td colspan="12">The return value is 23h</td></tr>
<tr><td>Restriction</td><td colspan="12">Should set “FF=A5” before configure this registers</td></tr>
<tr><td>Register Availability</td><td colspan="12"><table><tr><th>Status</th><th>Availability</th></tr><tr><td>Normal Mode on,Idle Mode Off,Sleep Out</td><td>Yes</td></tr><tr><td>Normal Mode on,Idle Mode On,Sleep Out</td><td>Yes</td></tr><tr><td>Partial Mode on,Idle Mode Off,Sleep Out</td><td>Yes</td></tr><tr><td>Partial Mode on,Idle Mode On,Sleep Out</td><td>Yes</td></tr><tr><td>Sleep In</td><td>Yes</td></tr></table></td></tr>
<tr><td>Default</td><td colspan="12"><table><tr><th>Status</th><th>Default Value(D7 to D0)</th></tr><tr><td>Power On Sequence</td><td>8’h23</td></tr><tr><td>HW Reset</td><td>8’h23</td></tr></table></td></tr>
</table>

12.2 Panel Function Command List and Description

Addr	Name	R/W	D7	D6	D5	D4	D3	D2	D1	D0
0x51	DDVDH_CLAMP	W				ddvdh_clp_en		ddvdh_clamp[2:0]		
0x52	VGH_CLAMP	W				vgh_clp_en			vgh_clamp[1:0]	
0x53	VGL_CLAMP	W				vgl_clp_en			vgl_clamp[1:0]	
0x61	DISP_HBP	W	hbp[7:0]							
0x62	DISP_HFP	W	hfp[7:0]							
0x63	DISP_VBP	W		vbp[6:0]						
0x64	DISP_VFP	W		vfp[6:0]						
0x84	GATE_ST	W	gate_st[7:0]							
0x85	GATE_ED	W	gate_ed[7:0]							
0x87	VRH_TEST	W				vrh_pre[4:0]				
0x88	VCM_TEST	W			vcm[5:0]					
0x89	VDV_TEST	W			vcom_vdv_pre[5:0]					
0x93	SRC_POL	W				ndl			src_en_sel	polar_sel
0x94	SRC_ST	W	src_st[7:0]							
0x95	PCHG_ST	W	pchg_st[7:0]							
0xB1	CHP_TEST1	W					bth[1:0]		btl[1:0]	
0xB2	CHP_TEST2	W	ddvdh_clk_sel[1:0]		vgh_clk_sel[1:0]		vgl_clk_sel[1:0]		vcl_clk_sel[1:0]	
0xB5	GAM_FIX	W			chop_sel[1:0]		gam_flw_fix		gam_bias_fix[1:0]	
0xC3	GATE_REV	W				src_ss				gate_gs
0xC4	REGU_TEST	W					regu_ad[2:0]			
0xC5	REGU_TEST1	W					bgr_ad[3:0]			
0xE9	SPI2SET	W						spi2line_en	spi2line_mode	

12.2.1 DDVDH_CLAMP (51h)

<table>
<tr><td>51H</td><td colspan="12">Power Control 1</td></tr>
<tr><td></td><td>D/CX</td><td>RDX</td><td>WRX</td><td>D7</td><td>D6</td><td>D5</td><td>D4</td><td>D3</td><td>D2</td><td>D1</td><td>D0</td><td>HEX</td></tr>
<tr><td>Command</td><td>L</td><td>H</td><td>R</td><td>0</td><td>1</td><td>0</td><td>1</td><td>0</td><td>0</td><td>0</td><td>1</td><td>51H</td></tr>
<tr><td>Parameter</td><td>H</td><td>H</td><td>R</td><td></td><td></td><td></td><td>DDVDH_CLP_EN</td><td></td><td colspan="3">DDVDH_CLAMP[2:0]</td><td>15H</td></tr>
<tr><td>Description</td><td colspan="12">DDVDH_CLP_EN: This command is used to define DDVDHAC clamp enable.
<table><tr><th>DDVDH_CLP_EN</th><th>Description</th></tr><tr><td>0</td><td>Disable the clamp</td></tr><tr><td>1</td><td>Enable the clamp</td></tr></table>
DDVDH_CLAMP[2:0]: This command is used to define DDVDHAC clamp value
<table><tr><th>DDVDH_CLAMP[2:0]</th><th>DDVDHAC (V)</th></tr><tr><td>000</td><td>4.5</td></tr><tr><td>001</td><td>4.6</td></tr><tr><td>010</td><td>4.7</td></tr><tr><td>011</td><td>4.8</td></tr><tr><td>100</td><td>4.9</td></tr><tr><td>101</td><td>5</td></tr><tr><td>110</td><td>5.1</td></tr><tr><td>111</td><td>5.2</td></tr></table></td></tr>
<tr><td>Restriction</td><td colspan="12">Should set “FF=A5” before configure this registers</td></tr>
<tr><td>Register Availability</td><td colspan="12"><table><tr><th>Status</th><th>Availability</th></tr><tr><td>Normal Mode on,Idle Mode Off,Sleep Out</td><td>Yes</td></tr><tr><td>Normal Mode on,Idle Mode On,Sleep Out</td><td>Yes</td></tr><tr><td>Partial Mode on,Idle Mode Off,Sleep Out</td><td>Yes</td></tr><tr><td>Partial Mode on,Idle Mode On,Sleep Out</td><td>Yes</td></tr><tr><td>Sleep In</td><td>Yes</td></tr></table></td></tr>
<tr><td>Default</td><td colspan="12"><table><tr><th>Status</th><th>Default Value(D7 to D0)</th></tr><tr><td>Power On Sequence</td><td>8’h15</td></tr><tr><td>HW Reset</td><td>8’h15</td></tr></table></td></tr>
</table>

12.2.2 VGH_CLAMP (52h)

<table>
<tr><td>52H</td><td colspan="12">VGH_CLAMP</td></tr>
<tr><td></td><td>D/CX</td><td>RDX</td><td>WRX</td><td>D7</td><td>D6</td><td>D5</td><td>D4</td><td>D3</td><td>D2</td><td>D1</td><td>D0</td><td>HEX</td></tr>
<tr><td>Command</td><td>L</td><td>H</td><td>R</td><td>0</td><td>1</td><td>0</td><td>1</td><td>0</td><td>0</td><td>1</td><td>0</td><td>52H</td></tr>
<tr><td>Parameter</td><td>H</td><td>H</td><td>R</td><td></td><td></td><td></td><td>VGH_CLP_EN</td><td></td><td></td><td colspan="2">VGH_CLAMP[1:0]</td><td>13H</td></tr>
<tr><td>Description</td><td colspan="12">VGH_CLP_EN: This command is used to define VGH clamp enable.
<table>
<tr><th>VGH_CLP_EN</th><th>Description</th></tr>
<tr><td>0</td><td>Disable the clamp</td></tr>
<tr><td>1</td><td>Enable the clamp</td></tr>
</table>
VGH_CLAMP[1:0]: This command is used to define VGH clamp value.
<table>
<tr><th>VGH_CLAMP[1:0]</th><th>VGH (unit:V)</th></tr>
<tr><td>00</td><td>10</td></tr>
<tr><td>01</td><td>12</td></tr>
<tr><td>10</td><td>13.5</td></tr>
<tr><td>11</td><td>15</td></tr>
</table></td></tr>
<tr><td>Restriction</td><td colspan="12">Should set “FF=A5” before configure this registers</td></tr>
<tr><td>Register Availability</td><td colspan="12"><table>
<tr><th>Status</th><th>Availability</th></tr>
<tr><td>Normal Mode on,Idle Mode Off,Sleep Out</td><td>Yes</td></tr>
<tr><td>Normal Mode on,Idle Mode On,Sleep Out</td><td>Yes</td></tr>
<tr><td>Partial Mode on,Idle Mode Off,Sleep Out</td><td>Yes</td></tr>
<tr><td>Partial Mode on,Idle Mode On,Sleep Out</td><td>Yes</td></tr>
<tr><td>Sleep In</td><td>Yes</td></tr>
</table></td></tr>
<tr><td>Default</td><td colspan="12"><table>
<tr><th>Status</th><th>Default Value(D7 to D0)</th></tr>
<tr><td>Power On Sequence</td><td>8’h13</td></tr>
<tr><td>HW Reset</td><td>8’h13</td></tr>
</table></td></tr>
</table>

12.2.3 VGL_CLAMP (53h)

<table>
<tr><td>53H</td><td colspan="12">VGL_CLAM</td></tr>
<tr><td></td><td>D/CX</td><td>RDX</td><td>WRX</td><td>D7</td><td>D6</td><td>D5</td><td>D4</td><td>D3</td><td>D2</td><td>D1</td><td>D0</td><td>HEX</td></tr>
<tr><td>Command</td><td>L</td><td>H</td><td>R</td><td>0</td><td>1</td><td>0</td><td>1</td><td>0</td><td>0</td><td>1</td><td>1</td><td>53H</td></tr>
<tr><td>Parameter</td><td>H</td><td>H</td><td>R</td><td>0</td><td>0</td><td>0</td><td>VGL_CLP_EN</td><td>0</td><td>0</td><td colspan="2">VGL_CLAMP[1:0]</td><td>11H</td></tr>
<tr><td>Description</td><td colspan="12">This command is used to set the VGL clamp voltage.
<table>
<tr><th>VGL_CLAMP[1:0]</th><th>VGL (unit:V)</th></tr>
<tr><td>00</td><td>-7.5</td></tr>
<tr><td>01</td><td>-10</td></tr>
<tr><td>10</td><td>-12.5</td></tr>
<tr><td>11</td><td>-13</td></tr>
</table></td></tr>
<tr><td>Restriction</td><td colspan="12">Should set “FF=A5” before configure this registers</td></tr>
<tr><td>Register Availability</td><td colspan="12">
<table>
<tr><th>Status</th><th>Availability</th></tr>
<tr><td>Normal Mode on,Idle Mode Off,Sleep Out</td><td>Yes</td></tr>
<tr><td>Normal Mode on,Idle Mode On,Sleep Out</td><td>Yes</td></tr>
<tr><td>Partial Mode on,Idle Mode Off,Sleep Out</td><td>Yes</td></tr>
<tr><td>Partial Mode on,Idle Mode On,Sleep Out</td><td>Yes</td></tr>
<tr><td>Sleep In</td><td>Yes</td></tr>
</table></td></tr>
<tr><td>Default</td><td colspan="12">
<table>
<tr><th>Status</th><th>Default Value(D7 to D0)</th></tr>
<tr><td>Power On Sequence</td><td>8’h11</td></tr>
<tr><td>HW Reset</td><td>8’h11</td></tr>
</table></td></tr>
</table>

12.2.4 DISP_HBP (61h)

<table>
<tr><td>61H</td><td colspan="11">Frame Rate Control 1</td></tr>
<tr><td></td><td>D/CX</td><td>RDX</td><td>WRX</td><td>D7</td><td>D6</td><td>D5</td><td>D4</td><td>D3</td><td>D2</td><td>D1</td><td>D0</td><td>HEX</td></tr>
<tr><td>Command</td><td>L</td><td>H</td><td>R</td><td>0</td><td>1</td><td>1</td><td>0</td><td>0</td><td>0</td><td>0</td><td>1</td><td>61H</td></tr>
<tr><td>Parameter</td><td>H</td><td>H</td><td>R</td><td colspan="8">HBP[7:0]</td><td>05H</td></tr>
<tr><td>Description</td><td colspan="12">Internal frame rate control register, used to enlarge horizontal scan cycle.
Frame rate=fosc/10*(80+HBP+HFP)*(160+VBP+VFP)
Fosc=20MHz</td></tr>
<tr><td>Restriction</td><td colspan="12">Should set “FF=A5” before configure this registers</td></tr>
<tr><td>Register Availability</td><td colspan="12"><table><tr><th>Status</th><th>Availability</th></tr><tr><td>Normal Mode on,Idle Mode Off,Sleep Out</td><td>Yes</td></tr><tr><td>Normal Mode on,Idle Mode On,Sleep Out</td><td>Yes</td></tr><tr><td>Partial Mode on,Idle Mode Off,Sleep Out</td><td>Yes</td></tr><tr><td>Partial Mode on,Idle Mode On,Sleep Out</td><td>Yes</td></tr><tr><td>Sleep In</td><td>Yes</td></tr></table></td></tr>
<tr><td>Default</td><td colspan="12"><table><tr><th>Status</th><th>Default Value(D7 to D0)</th></tr><tr><td>Power On Sequence</td><td>8’h05</td></tr><tr><td>HW Reset</td><td>8’h05</td></tr></table></td></tr>
</table>

12.2.5 DISP_HFP (62h)

<table>
<tr><td>62H</td><td colspan="11">Frame Rate Control 2</td></tr>
<tr><td></td><td>D/CX</td><td>RDX</td><td>WRX</td><td>D7</td><td>D6</td><td>D5</td><td>D4</td><td>D3</td><td>D2</td><td>D1</td><td>D0</td><td>HEX</td></tr>
<tr><td>Command</td><td>L</td><td>H</td><td>R</td><td>0</td><td>1</td><td>1</td><td>0</td><td>0</td><td>0</td><td>1</td><td>0</td><td>62H</td></tr>
<tr><td>Parameter</td><td>H</td><td>H</td><td>R</td><td colspan="8">HFP[7:0]</td><td>10H</td></tr>
<tr><td>Description</td><td colspan="12">Internal frame rate control register, used to enlarge horizontal scan cycle.
Frame rate=fosc/10*(80+HBP+HFP)*(160+VBP+VFP)
Fosc=20MHz</td></tr>
<tr><td>Restriction</td><td colspan="12">Should set “FF=A5” before configure this registers</td></tr>
<tr><td>Register Availability</td><td colspan="12"><table><tr><th>Status</th><th>Availability</th></tr><tr><td>Normal Mode on,Idle Mode Off,Sleep Out</td><td>Yes</td></tr><tr><td>Normal Mode on,Idle Mode On,Sleep Out</td><td>Yes</td></tr><tr><td>Partial Mode on,Idle Mode Off,Sleep Out</td><td>Yes</td></tr><tr><td>Partial Mode on,Idle Mode On,Sleep Out</td><td>Yes</td></tr><tr><td>Sleep In</td><td>Yes</td></tr></table></td></tr>
<tr><td>Default</td><td colspan="12"><table><tr><th>Status</th><th>Default Value(D7 to D0)</th></tr><tr><td>Power On Sequence</td><td>8’h10</td></tr><tr><td>HW Reset</td><td>8’h10</td></tr></table></td></tr>
</table>

12.2.6 DISP_VBP (63h)

63H	Frame Rate Control 3											
	D/CX	**RDX**	**WRX**	D7	D6	D5	D4	D3	D2	D1	D0	HEX
Command	L	H	R	0	1	1	0	0	0	1	1	63H
Parameter	H	H	R		VBP[6:0]							01H
Description	Internal frame rate control register, used to enlarge vertical scan cycle. Frame rate=fosc/10*(80+HBP+HFP)*(160+VBP+VFP) Fosc=20MHz											
Restriction	Should set “FF=A5” before configure this registers											

Register Availability

Status	Availability
Normal Mode on,Idle Mode Off,Sleep Out	Yes
Normal Mode on,Idle Mode On,Sleep Out	Yes
Partial Mode on,Idle Mode Off,Sleep Out	Yes
Partial Mode on,Idle Mode On,Sleep Out	Yes
Sleep In	Yes

Default

Status	Default Value(D7 to D0)
Power On Sequence	8’h01
HW Reset	8’h01

12.2.7 DISP_VFP (64h)

<table>
<tr><td>64H</td><td colspan="11">Frame Rate Control 4</td></tr>
<tr><td></td><td>D/CX</td><td>RDX</td><td>WRX</td><td>D7</td><td>D6</td><td>D5</td><td>D4</td><td>D3</td><td>D2</td><td>D1</td><td>D0</td><td>HEX</td></tr>
<tr><td>Command</td><td>L</td><td>H</td><td>R</td><td>0</td><td>1</td><td>1</td><td>0</td><td>0</td><td>1</td><td>0</td><td>0</td><td>64H</td></tr>
<tr><td>Parameter</td><td>H</td><td>H</td><td>R</td><td></td><td colspan="6">VFP[6:0]</td><td>01H</td></tr>
<tr><td>Description</td><td colspan="12">Internal frame rate control register, used to enlarge vertical scan cycle.
Frame rate=fosc/10*(80+HBP+HFP)*(160+VBP+VFP)
Fosc=20MHz</td></tr>
<tr><td>Restriction</td><td colspan="12">Should set “FF=A5” before configure this registers</td></tr>
<tr><td>Register Availability</td><td colspan="12"><table><tr><th>Status</th><th>Availability</th></tr><tr><td>Normal Mode on,Idle Mode Off,Sleep Out</td><td>Yes</td></tr><tr><td>Normal Mode on,Idle Mode On,Sleep Out</td><td>Yes</td></tr><tr><td>Partial Mode on,Idle Mode Off,Sleep Out</td><td>Yes</td></tr><tr><td>Partial Mode on,Idle Mode On,Sleep Out</td><td>Yes</td></tr><tr><td>Sleep In</td><td>Yes</td></tr></table></td></tr>
<tr><td>Default</td><td colspan="12"><table><tr><th>Status</th><th>Default Value(D7 to D0)</th></tr><tr><td>Power On Sequence</td><td>8’h01</td></tr><tr><td>HW Reset</td><td>8’h01</td></tr></table></td></tr>
</table>

12.2.8 GATEST (84h)

84H	GATE START SETTING											
	D/CX	**RDX**	**WRX**	D7	D6	D5	D4	D3	D2	D1	D0	HEX
Command	L	H	R	1	0	0	0	0	1	0	0	84H
Parameter	H	H	R	GATE_ST[7:0]								0DH
Description	Gate enables start position setting.											
Restriction	Should set “FF=A5” before configure this registers											
Register Availability												

Status	Availability
Normal Mode on,Idle Mode Off,Sleep Out	Yes
Normal Mode on,Idle Mode On,Sleep Out	Yes
Partial Mode on,Idle Mode Off,Sleep Out	Yes
Partial Mode on,Idle Mode On,Sleep Out	Yes
Sleep In	Yes

Default

Status	Default Value(D7 to D0)
Power On Sequence	8’h0D
HW Reset	8’h0D

12.2.9 GATE_ED (85h)

<table>
<tr><td>85H</td><td colspan="11">GATE END SETTING</td></tr>
<tr><td></td><td>D/CX</td><td>RDX</td><td>WRX</td><td>D7</td><td>D6</td><td>D5</td><td>D4</td><td>D3</td><td>D2</td><td>D1</td><td>D0</td><td>HEX</td></tr>
<tr><td>Command</td><td>L</td><td>H</td><td>R</td><td>1</td><td>0</td><td>0</td><td>0</td><td>0</td><td>1</td><td>0</td><td>1</td><td>85H</td></tr>
<tr><td>Parameter</td><td>H</td><td>H</td><td>R</td><td colspan="8">GATE_ED[7:0]</td><td>50H</td></tr>
<tr><td>Description</td><td colspan="12">Gate enables end position setting.</td></tr>
<tr><td>Restriction</td><td colspan="12">Should set “FF=A5” before configure this registers</td></tr>
<tr><td>Register Availability</td><td colspan="12"><table><tr><th>Status</th><th>Availability</th></tr><tr><td>Normal Mode on,Idle Mode Off,Sleep Out</td><td>Yes</td></tr><tr><td>Normal Mode on,Idle Mode On,Sleep Out</td><td>Yes</td></tr><tr><td>Partial Mode on,Idle Mode Off,Sleep Out</td><td>Yes</td></tr><tr><td>Partial Mode on,Idle Mode On,Sleep Out</td><td>Yes</td></tr><tr><td>Sleep In</td><td>Yes</td></tr></table></td></tr>
<tr><td>Default</td><td colspan="12"><table><tr><th>Status</th><th>Default Value(D7 to D0)</th></tr><tr><td>Power On Sequence</td><td>8’h50</td></tr><tr><td>HW Reset</td><td>8’h50</td></tr></table></td></tr>
</table>

12.2.10 VRH_TEST (87h)

<table>
<tr><td>87H</td><td colspan="11">Power Control 4</td></tr>
<tr><td></td><td>D/CX</td><td>RDX</td><td>WRX</td><td>D7</td><td>D6</td><td>D5</td><td>D4</td><td>D3</td><td>D2</td><td>D1</td><td>D0</td><td>HEX</td></tr>
<tr><td>Command</td><td>L</td><td>H</td><td>R</td><td>1</td><td>0</td><td>0</td><td>0</td><td>0</td><td>1</td><td>1</td><td>1</td><td>87H</td></tr>
<tr><td>Parameter</td><td>H</td><td>H</td><td>R</td><td></td><td></td><td></td><td colspan="4">VRH_PRE[4:0]</td><td>17H</td></tr>
<tr><td>Description</td><td colspan="11">vrh_pre[4:0]: This command is used to define GVDD trimming value.
<table>
<tr><th>VRH_PRE[4:0]</th><th>GVDD</th><th>VRH_PRE[4:0]</th><th>GVDD</th></tr>
<tr><td>00000</td><td></td><td>10000</td><td>4.15</td></tr>
<tr><td>00001</td><td></td><td>10001</td><td>4.20</td></tr>
<tr><td>00010</td><td></td><td>10010</td><td>4.25</td></tr>
<tr><td>00011</td><td>3.50</td><td>10011</td><td>4.30</td></tr>
<tr><td>00100</td><td>3.55</td><td>10100</td><td>4.35</td></tr>
<tr><td>00101</td><td>3.60</td><td>10101</td><td>4.40</td></tr>
<tr><td>00110</td><td>3.65</td><td>10110</td><td>4.45</td></tr>
<tr><td>00111</td><td>3.70</td><td>10111</td><td>4.50</td></tr>
<tr><td>01000</td><td>3.75</td><td>11000</td><td>4.55</td></tr>
<tr><td>01001</td><td>3.80</td><td>11001</td><td>4.60</td></tr>
<tr><td>01010</td><td>3.85</td><td>11010</td><td>4.65</td></tr>
<tr><td>01011</td><td>3.90</td><td>11011</td><td>4.70</td></tr>
<tr><td>01100</td><td>3.95</td><td>11100</td><td>4.75</td></tr>
<tr><td>01101</td><td>4.00</td><td>11101</td><td></td></tr>
<tr><td>01110</td><td>4.05</td><td>11110</td><td></td></tr>
<tr><td>01111</td><td>4.10</td><td>11111</td><td></td></tr>
</table></td></tr>
<tr><td>Restriction</td><td colspan="11">Should set “FF=A5” before configure this registers</td></tr>
<tr><td>Register Availability</td><td colspan="11">
<table>
<tr><th>Status</th><th>Availability</th></tr>
<tr><td>Normal Mode on,Idle Mode Off,Sleep Out</td><td>Yes</td></tr>
<tr><td>Normal Mode on,Idle Mode On,Sleep Out</td><td>Yes</td></tr>
<tr><td>Partial Mode on,Idle Mode Off,Sleep Out</td><td>Yes</td></tr>
<tr><td>Partial Mode on,Idle Mode On,Sleep Out</td><td>Yes</td></tr>
<tr><td>Sleep In</td><td>Yes</td></tr>
</table></td></tr>
<tr><td>Default</td><td colspan="11">
<table>
<tr><th>Status</th><th>Default Value(D7 to D0)</th></tr>
<tr><td>Power On Sequence</td><td>8’h17</td></tr>
<tr><td>HW Reset</td><td>8’h17</td></tr>
</table></td></tr>
</table>

12.2.11 VCM_TEST (88h)

88H	VCOM Control 1											
	D/CX	**RDX**	**WRX**	D7	D6	D5	D4	D3	D2	D1	D0	HEX
Command	L	H	R	1	0	0	0	1	0	0	0	88H
Parameter	H	H	R			VCM[5:0]						17H

Description

VCM[5:0]: This command is used to define VCOMH trimming value.

VCM[5:0]	VCOMH	VCM[5:0]	VCOMH
000000		100000	GVDD*338/400
000001		100001	GVDD*340/400
000010		100010	GVDD*342/400
000011		100011	GVDD*344/400
000100		000100	GVDD*346/400
000101		000101	GVDD*348/400
000110		100110	GVDD*350/400
000111	GVDD*288/400	100111	GVDD*352/400
001000	GVDD*290/400	101000	GVDD*354/400
001001	GVDD*292/400	101001	GVDD*356/400
001010	GVDD*294/400	101010	GVDD*358/400
001011	GVDD*296/400	101011	GVDD*360/400
001100	GVDD*298/400	101100	GVDD*362/400
001101	GVDD*300/400	101101	GVDD*364/400
001110	GVDD*302/400	101110	GVDD*366/400
001111	GVDD*304/400	101111	GVDD*368/400
010000	GVDD*306/400	110000	GVDD*370/400
010001	GVDD*308/400	110001	GVDD*372/400
010010	GVDD*310/400	110010	GVDD*374/400
010011	GVDD*312/400	110011	GVDD*376/400
010100	GVDD*314/400	110100	GVDD*378/400
010101	GVDD*316/400	110101	GVDD*380/400
010110	GVDD*318/400	110110	GVDD*382/400
010111	**GVDD*320/400**	110111	GVDD*384/400
011000	GVDD*322/400	111000	GVDD*386/400
011001	GVDD*324/400	111001	
011010	GVDD*326/400	111010	
011011	GVDD*328/400	111011	
011100	GVDD*330/400	111100	
011101	GVDD*332/400	111101	
011110	GVDD*334/400	111110	
011111	GVDD*336/400	111111	

Note: This value need GVDD set in 4.5V.

Restriction	Should set “FF=A5” before configure this registers

<table>
<tr><td>Register Availability</td><td>
<table>
<tr><th>Status</th><th>Availability</th></tr>
<tr><td>Normal Mode on,Idle Mode Off,Sleep Out</td><td>Yes</td></tr>
<tr><td>Normal Mode on,Idle Mode On,Sleep Out</td><td>Yes</td></tr>
<tr><td>Partial Mode on,Idle Mode Off,Sleep Out</td><td>Yes</td></tr>
<tr><td>Partial Mode on,Idle Mode On,Sleep Out</td><td>Yes</td></tr>
<tr><td>Sleep In</td><td>Yes</td></tr>
</table>
</td></tr>
<tr><td>Default</td><td>
<table>
<tr><th>Status</th><th>Default Value(D7 to D0)</th></tr>
<tr><td>Power On Sequence</td><td>8’h17</td></tr>
<tr><td>HW Reset</td><td>8’h17</td></tr>
</table>
</td></tr>
</table>

12.2.12 VDV_TEST (89h)

89H	VCOM Control 2											
	D/CX	**RDX**	**WRX**	D7	D6	D5	D4	D3	D2	D1	D0	HEX
Command	L	H	R	1	0	0	0	1	0	0	1	89H
Parameter	H	H	R			VCOM_VDV_PRE[5:0]						1CH

Description

VCOM_VDV[5:0]: This command is used to define VCOML trimming value.

VCOM_VDV_PRE [5:0]	VCOML	VCOM_VDV_PRE [5:0]	VCOML
000000		100000	VCOMH-(GVDD*205)/200
000001		100001	VCOMH-(GVDD*206)/200
000010		100010	VCOMH-(GVDD*207)/200
000011		100011	VCOMH-(GVDD*208)/200
000100		100100	VCOMH-(GVDD*209)/200
000101		100101	VCOMH-(GVDD*210)/200
000110		100110	VCOMH-(GVDD*211)/200
000111	VCOMH-(GVDD*178)/200	100111	VCOMH-(GVDD*212)/200
001000	VCOMH-(GVDD*180)/200	101000	VCOMH-(GVDD*213)/200
001001	VCOMH-(GVDD*182)/200	101001	VCOMH-(GVDD*214)/200
001010	VCOMH-(GVDD*183)/200	101010	VCOMH-(GVDD*215)/200
001011	VCOMH-(GVDD*184)/200	101011	VCOMH-(GVDD*216)/200
001100	VCOMH-(GVDD*185)/200	101100	VCOMH-(GVDD*217)/200
001101	VCOMH-(GVDD*186)/200	101101	VCOMH-(GVDD*218)/200
001110	VCOMH-(GVDD*187)/200	101110	VCOMH-(GVDD*219)/200
001111	VCOMH-(GVDD*188)/200	101111	VCOMH-(GVDD*220)/200
010000	VCOMH-(GVDD*189)/200	110000	VCOMH-(GVDD*221)/200
010001	VCOMH-(GVDD*190)/200	110001	VCOMH-(GVDD*222)/200
010010	VCOMH-(GVDD*191)/200	110010	VCOMH-(GVDD*223)/200
010011	VCOMH-(GVDD*192)/200	110011	VCOMH-(GVDD*224)/200
010100	VCOMH-(GVDD*193)/200	110100	VCOMH-(GVDD*225)/200
010101	VCOMH-(GVDD*194)/200	110101	VCOMH-(GVDD*226)/200
010110	VCOMH-(GVDD*195)/200	110110	VCOMH-(GVDD*227)/200
010111	VCOMH-(GVDD*196)/200	110111	VCOMH-(GVDD*229)/200
011000	VCOMH-(GVDD*197)/200	111000	VCOMH-(GVDD*231)/200

	011001	VCOMH-(GVDD*198)/200	111001	
	011010	VCOMH-(GVDD*199)/200	111010	
	011011	VCOMH-(GVDD*200)/200	111011	
	011100	**VCOMH-(GVDD*201)/200**	111100	
	011101	VCOMH-(GVDD*202)/200	111101	
	011110	VCOMH-(GVDD*203)/200	111110	
	011111	VCOMH-(GVDD*204)/200	111111	
	Note: This value need VCOMH set in 3.6V, GVDD set in 4.5V.			
Restriction	Should set “FF=A5” before configure this registers			

Register Availability

Status	Availability
Normal Mode on,Idle Mode Off,Sleep Out	Yes
Normal Mode on,Idle Mode On,Sleep Out	Yes
Partial Mode on,Idle Mode Off,Sleep Out	Yes
Partial Mode on,Idle Mode On,Sleep Out	Yes
Sleep In	Yes

Default

Status	Default Value(D7 to D0)
Power On Sequence	8’h1C
HW Reset	8’h1C

12.2.13 SRC_POL (93h)

93H	SOURCE SET											
	D/CX	**RDX**	**WRX**	D7	D6	D5	D4	D3	D2	D1	D0	HEX
Command	L	H	R	1	0	0	1	0	0	1	1	93H
Parameter	H	H	R				NDL			SRC_EN_SEL	POLAR_SEL	10H
Description	NDL: Select max or min voltage to clear screen. SRC_EN_SEL: Decide if the source enable is open during v-porch area. “1” is OPEN, “0” is CLOSE; POLAR_SEL: “0” is frame inverse; “1” is line inverse											
Restriction	Should set “FF=A5” before configure this registers											
Register Availability												

Status	Availability
Normal Mode on,Idle Mode Off,Sleep Out	Yes
Normal Mode on,Idle Mode On,Sleep Out	Yes
Partial Mode on,Idle Mode Off,Sleep Out	Yes
Partial Mode on,Idle Mode On,Sleep Out	Yes
Sleep In	Yes

Default

Status	Default Value(D7 to D0)
Power On Sequence	8’h10
HW Reset	8’h10

12.2.14 SRC_ST (94h)

<table>
<tr><td>94H</td><td colspan="11">SOURCE START SETTING</td></tr>
<tr><td></td><td>D/CX</td><td>RDX</td><td>WRX</td><td>D7</td><td>D6</td><td>D5</td><td>D4</td><td>D3</td><td>D2</td><td>D1</td><td>D0</td><td>HEX</td></tr>
<tr><td>Command</td><td>L</td><td>H</td><td>R</td><td>1</td><td>0</td><td>0</td><td>1</td><td>0</td><td>1</td><td>0</td><td>0</td><td>94H</td></tr>
<tr><td>Parameter</td><td>H</td><td>H</td><td>R</td><td colspan="8">SRC_ST[7:0]</td><td>0AH</td></tr>
<tr><td>Description</td><td colspan="12">src_out_ctrl start position setting.</td></tr>
<tr><td>Restriction</td><td colspan="12">Should set “FF=A5” before configure this registers</td></tr>
<tr><td>Register Availability</td><td colspan="12"><table><tr><th>Status</th><th>Availability</th></tr><tr><td>Normal Mode on,Idle Mode Off,Sleep Out</td><td>Yes</td></tr><tr><td>Normal Mode on,Idle Mode On,Sleep Out</td><td>Yes</td></tr><tr><td>Partial Mode on,Idle Mode Off,Sleep Out</td><td>Yes</td></tr><tr><td>Partial Mode on,Idle Mode On,Sleep Out</td><td>Yes</td></tr><tr><td>Sleep In</td><td>Yes</td></tr></table></td></tr>
<tr><td>Default</td><td colspan="12"><table><tr><th>Status</th><th>Default Value(D7 to D0)</th></tr><tr><td>Power On Sequence</td><td>8’h0A</td></tr><tr><td>HW Reset</td><td>8’h0A</td></tr></table></td></tr>
</table>

12.2.15 PCHGST (95h)

95H	SOURCE PRECHARGE START SETTING											
	D/CX	**RDX**	**WRX**	D7	D6	D5	D4	D3	D2	D1	D0	HEX
Command	L	H	R	1	0	0	1	0	1	0	1	95H
Parameter	H	H	R	PCHG_ST[7:0]								03H
Description	Source pre-charge start position setting.											
Restriction	Should set “FF=A5” before configure this registers											
Register Availability												

Status	Availability
Normal Mode on,Idle Mode Off,Sleep Out	Yes
Normal Mode on,Idle Mode On,Sleep Out	Yes
Partial Mode on,Idle Mode Off,Sleep Out	Yes
Partial Mode on,Idle Mode On,Sleep Out	Yes
Sleep In	Yes

Default

Status	Default Value(D7 to D0)
Power On Sequence	8’h03
HW Reset	8’h03

12.2.16 CHP_TEST1 (B1h)

<table>
<tr><td>B1H</td><td colspan="11">Power Control 5</td></tr>
<tr><td></td><td>D/CX</td><td>RDX</td><td>WRX</td><td>D7</td><td>D6</td><td>D5</td><td>D4</td><td>D3</td><td>D2</td><td>D1</td><td>D0</td><td>HEX</td></tr>
<tr><td>Command</td><td>L</td><td>H</td><td>R</td><td>1</td><td>0</td><td>1</td><td>1</td><td>0</td><td>0</td><td>0</td><td>1</td><td>B1H</td></tr>
<tr><td>Parameter</td><td>H</td><td>H</td><td>R</td><td></td><td></td><td></td><td></td><td colspan="2">BTH[1:0]</td><td colspan="2">BTL[1:0]</td><td>0FH</td></tr>
<tr><td>Description</td><td colspan="12">BTH[1:0]: is used to define the ratio in VGH pump.
<table>
<tr><th>BTH[1:0]</th><th>VGH</th></tr>
<tr><td>00</td><td>4*VCI</td></tr>
<tr><td>01</td><td>5*VCI</td></tr>
<tr><td>10</td><td>6*VCI</td></tr>
<tr><td>11</td><td>6*VCI</td></tr>
</table>
BTL[1:0]: is used to define the ratio in VGL pump.
<table>
<tr><th>BTL[1:0]</th><th>VGL</th></tr>
<tr><td>00</td><td>-3*VCI</td></tr>
<tr><td>01</td><td>-4*VCI</td></tr>
<tr><td>10</td><td>-5*VCI</td></tr>
<tr><td>11</td><td>-5*VCI</td></tr>
</table></td></tr>
<tr><td>Restriction</td><td colspan="12">Should set “FF=A5” before configure this registers</td></tr>
<tr><td>Register Availability</td><td colspan="12"><table>
<tr><th>Status</th><th>Availability</th></tr>
<tr><td>Normal Mode on,Idle Mode Off,Sleep Out</td><td>Yes</td></tr>
<tr><td>Normal Mode on,Idle Mode On,Sleep Out</td><td>Yes</td></tr>
<tr><td>Partial Mode on,Idle Mode Off,Sleep Out</td><td>Yes</td></tr>
<tr><td>Partial Mode on,Idle Mode On,Sleep Out</td><td>Yes</td></tr>
<tr><td>Sleep In</td><td>Yes</td></tr>
</table></td></tr>
<tr><td>Default</td><td colspan="12"><table>
<tr><th>Status</th><th>Default Value(D7 to D0)</th></tr>
<tr><td>Power On Sequence</td><td>8’h0F</td></tr>
<tr><td>HW Reset</td><td>8’h0F</td></tr>
</table></td></tr>
</table>

12.2.17 CHP_TEST2 (B2h)

B2H	Power Control 6											
	D/CX	**RDX**	**WRX**	D7	D6	D5	D4	D3	D2	D1	D0	HEX
Command	L	H	R	1	0	1	1	0	0	1	0	B2H
Parameter	H	H	R	DDVDH_CLK_SEL [1:0]		VGH_CLK_ SEL[1:0]		VGL_CLK_SEL [1:0]		VCL_CLK_SEL [1:0]		A4H

Description

DDVDH_CLK_SEL[1:0] is used to set the operation frequency of DDVDH pump.

DDVDH_CLK_SEL[1:0]	Frequency
00	1/6 * OSC
01	1/4 * OSC
10	**1/8* OSC**
11	1/16 * OSC

VGH_CLK_SEL[1:0] is used to set the operation frequency of VGH pump.

VGH_CLK_SEL[1:0]	Frequency
00	1/6 * OSC
01	1/4 * OSC
10	**1/8 * OSC**
10	1/16 * OSC

VGL_CLK_SEL[1:0] is used to set the operation frequency of VGL pump.

VGL_CLK_SEL[1:0]	Frequency
00	1/16 * OSC
01	**1/8 * OSC**
10	1/4 * OSC
11	1/6 * OSC

VCL_CLK_SEL[1:0] is used to set the operation frequency of VCL pump.

VCL_CLK_SEL[1:0]	Frequency
00	**1/16 * OSC**
01	1/8* OSC
10	1/4 * OSC
11	1/6 * OSC

Restriction: Should set “FF=A5” before configure this registers

Register Availability

Status	Availability
Normal Mode on,Idle Mode Off,Sleep Out	Yes
Normal Mode on,Idle Mode On,Sleep Out	Yes
Partial Mode on,Idle Mode Off,Sleep Out	Yes
Partial Mode on,Idle Mode On,Sleep Out	Yes
Sleep In	Yes

Default	

Status	Default Value(D7 to D0)
Power On Sequence	8'hA4
HW Reset	8'hA4

12.2.18 GAM_FIX (B5h)

<table>
<tr><td>B5H</td><td colspan="11">GAM_FIX</td></tr>
<tr><td></td><td>D/CX</td><td>RDX</td><td>WRX</td><td>D7</td><td>D6</td><td>D5</td><td>D4</td><td>D3</td><td>D2</td><td>D1</td><td>D0</td><td>HEX</td></tr>
<tr><td>Command</td><td>L</td><td>H</td><td>R</td><td>1</td><td>0</td><td>1</td><td>1</td><td>0</td><td>1</td><td>0</td><td>1</td><td>B5H</td></tr>
<tr><td>Parameter</td><td>H</td><td>H</td><td>R</td><td></td><td></td><td colspan="2">CHOP_SEL[1:0]</td><td>GAM_FLW_FIX</td><td></td><td colspan="2">GAM_BIAS_FIX[1:0]</td><td>30H</td></tr>
<tr><td>Description</td><td colspan="12">CHOPPER_SEL[1:0]: Gamma chopper function option.
<table>
<tr><th>CHOP_SEL[1:0]</th><th>Description</th></tr>
<tr><td>00</td><td>2 frame chopper</td></tr>
<tr><td>01</td><td>32 line chopper (1 frame chopper polarity change)</td></tr>
<tr><td>10</td><td>1 line chopper (2 frame chopper polarity change)</td></tr>
<tr><td>11</td><td>NONE</td></tr>
</table>
GAM_BIAS_FIX[1:0] is used to adjust gamma op’s bias current, default =00.
GAM_FLW_FIX is used to change gamma op’s driver capacity, default =0.</td></tr>
<tr><td>Restriction</td><td colspan="12">Should set “FF=A5” before configure this registers</td></tr>
<tr><td>Register Availability</td><td colspan="12">
<table>
<tr><th>Status</th><th>Availability</th></tr>
<tr><td>Normal Mode on,Idle Mode Off,Sleep Out</td><td>Yes</td></tr>
<tr><td>Normal Mode on,Idle Mode On,Sleep Out</td><td>Yes</td></tr>
<tr><td>Partial Mode on,Idle Mode Off,Sleep Out</td><td>Yes</td></tr>
<tr><td>Partial Mode on,Idle Mode On,Sleep Out</td><td>Yes</td></tr>
<tr><td>Sleep In</td><td>Yes</td></tr>
</table>
</td></tr>
<tr><td>Default</td><td colspan="12">
<table>
<tr><th>Status</th><th>Default Value(D7 to D0)</th></tr>
<tr><td>Power On Sequence</td><td>8’h30</td></tr>
<tr><td>HW Reset</td><td>8’h30</td></tr>
</table>
</td></tr>
</table>

12.2.19 GATE_REV (C3h)

<table>
<tr><th>C3H</th><th colspan="12">GATE_REV</th></tr>
<tr><td></td><td>D/CX</td><td>RDX</td><td>WRX</td><td>D7</td><td>D6</td><td>D5</td><td>D4</td><td>D3</td><td>D2</td><td>D1</td><td>D0</td><td>HEX</td></tr>
<tr><td>Command</td><td>L</td><td>H</td><td>R</td><td>1</td><td>1</td><td>0</td><td>0</td><td>0</td><td>0</td><td>1</td><td>1</td><td>C3H</td></tr>
<tr><td>Parameter</td><td>H</td><td>H</td><td>R</td><td></td><td></td><td></td><td>SRC_SS</td><td></td><td></td><td></td><td>GATE_GS</td><td>10H</td></tr>
<tr><td>Description</td><td colspan="12">SRC_SS: source reverse scan control. “1” is reverse, “0” is normal.
GATE_GS: gate reverse scan control. “1” is reverse, “0” is normal.</td></tr>
<tr><td>Restriction</td><td colspan="12">Should set “FF=A5” before configure this registers</td></tr>
<tr><td>Register Availability</td><td colspan="12"><table><tr><th>Status</th><th>Availability</th></tr><tr><td>Normal Mode on,Idle Mode Off,Sleep Out</td><td>Yes</td></tr><tr><td>Normal Mode on,Idle Mode On,Sleep Out</td><td>Yes</td></tr><tr><td>Partial Mode on,Idle Mode Off,Sleep Out</td><td>Yes</td></tr><tr><td>Partial Mode on,Idle Mode On,Sleep Out</td><td>Yes</td></tr><tr><td>Sleep In</td><td>Yes</td></tr></table></td></tr>
<tr><td>Default</td><td colspan="12"><table><tr><th>Status</th><th>Default Value(D7 to D0)</th></tr><tr><td>Power On Sequence</td><td>8’h10</td></tr><tr><td>HW Reset</td><td>8’h10</td></tr></table></td></tr>
</table>

12.2.20 REGU_TEST (C4h)

<table>
<tr><td>C4H</td><td colspan="11">REGU_TEST</td></tr>
<tr><td></td><td>D/CX</td><td>RDX</td><td>WRX</td><td>D7</td><td>D6</td><td>D5</td><td>D4</td><td>D3</td><td>D2</td><td>D1</td><td>D0</td><td>HEX</td></tr>
<tr><td>Command</td><td>L</td><td>H</td><td>R</td><td>1</td><td>1</td><td>0</td><td>0</td><td>0</td><td>1</td><td>0</td><td>0</td><td>C4H</td></tr>
<tr><td>Parameter</td><td>H</td><td>H</td><td>R</td><td></td><td></td><td></td><td></td><td colspan="3">REGU_AD[2:0]</td><td></td><td>00H</td></tr>
<tr><td>Description</td><td colspan="12">REGU_AD[2:0]:vdd18 trimming signal.
<table><tr><th>REGU_AD[2:0]</th><th>VDD18</th></tr><tr><td>000</td><td>1.85</td></tr><tr><td>001</td><td>1.90</td></tr><tr><td>010</td><td>1.96</td></tr><tr><td>011</td><td>2.11</td></tr><tr><td>100</td><td>1.60</td></tr><tr><td>101</td><td>1.67</td></tr><tr><td>110</td><td>1.74</td></tr><tr><td>111</td><td>1.80</td></tr></table></td></tr>
<tr><td>Restriction</td><td colspan="12">Should set “FF=A5” before configure this registers</td></tr>
<tr><td>Register Availability</td><td colspan="12"><table><tr><th>Status</th><th>Availability</th></tr><tr><td>Normal Mode on,Idle Mode Off,Sleep Out</td><td>Yes</td></tr><tr><td>Normal Mode on,Idle Mode On,Sleep Out</td><td>Yes</td></tr><tr><td>Partial Mode on,Idle Mode Off,Sleep Out</td><td>Yes</td></tr><tr><td>Partial Mode on,Idle Mode On,Sleep Out</td><td>Yes</td></tr><tr><td>Sleep In</td><td>Yes</td></tr></table></td></tr>
<tr><td>Default</td><td colspan="12"><table><tr><th>Status</th><th>Default Value(D7 to D0)</th></tr><tr><td>Power On Sequence</td><td>8’h00</td></tr><tr><td>HW Reset</td><td>8’h00</td></tr></table></td></tr>
</table>

12.2.21 REGU_TEST1 (C5h)

<table>
<tr><td>C5H</td><td colspan="11">REGU_TEST1</td></tr>
<tr><td></td><td>D/CX</td><td>RDX</td><td>WRX</td><td>D7</td><td>D6</td><td>D5</td><td>D4</td><td>D3</td><td>D2</td><td>D1</td><td>D0</td><td>HEX</td></tr>
<tr><td>Command</td><td>L</td><td>H</td><td>R</td><td>1</td><td>1</td><td>0</td><td>0</td><td>0</td><td>1</td><td>0</td><td>0</td><td>C5H</td></tr>
<tr><td>Parameter</td><td>H</td><td>H</td><td>R</td><td></td><td></td><td></td><td></td><td colspan="4">VBG_AD[3:0]</td><td>08H</td></tr>
<tr><td>Description</td><td colspan="11">VBG_AD[3:0]:vdd18 trimming singal.
<table>
<tr><th>VBG_AD[3:0]</th><th>VBG</th><th>VBG_AD[3:0]</th><th>VBG</th></tr>
<tr><td>0000</td><td>1.03</td><td>1000</td><td>1.2014</td></tr>
<tr><td>0001</td><td>1.05</td><td>1001</td><td>1.2229</td></tr>
<tr><td>0010</td><td>1.07</td><td>1010</td><td>1.24</td></tr>
<tr><td>0011</td><td>1.09</td><td>1011</td><td>1.27</td></tr>
<tr><td>0100</td><td>1.12</td><td>1100</td><td>1.29</td></tr>
<tr><td>0101</td><td>1.14</td><td>1101</td><td>1.31</td></tr>
<tr><td>0110</td><td>1.16</td><td>1110</td><td>1.33</td></tr>
<tr><td>0111</td><td>1.1799</td><td>1111</td><td>1.35</td></tr>
</table></td></tr>
<tr><td>Restriction</td><td colspan="11">Should set “FF=A5” before configure this registers</td></tr>
<tr><td>Register Availability</td><td colspan="11">
<table>
<tr><th>Status</th><th>Availability</th></tr>
<tr><td>Normal Mode on,Idle Mode Off,Sleep Out</td><td>Yes</td></tr>
<tr><td>Normal Mode on,Idle Mode On,Sleep Out</td><td>Yes</td></tr>
<tr><td>Partial Mode on,Idle Mode Off,Sleep Out</td><td>Yes</td></tr>
<tr><td>Partial Mode on,Idle Mode On,Sleep Out</td><td>Yes</td></tr>
<tr><td>Sleep In</td><td>Yes</td></tr>
</table></td></tr>
<tr><td>Default</td><td colspan="11">
<table>
<tr><th>Status</th><th>Default Value(D7 to D0)</th></tr>
<tr><td>Power On Sequence</td><td>8’h08</td></tr>
<tr><td>HW Reset</td><td>8’h08</td></tr>
</table></td></tr>
</table>

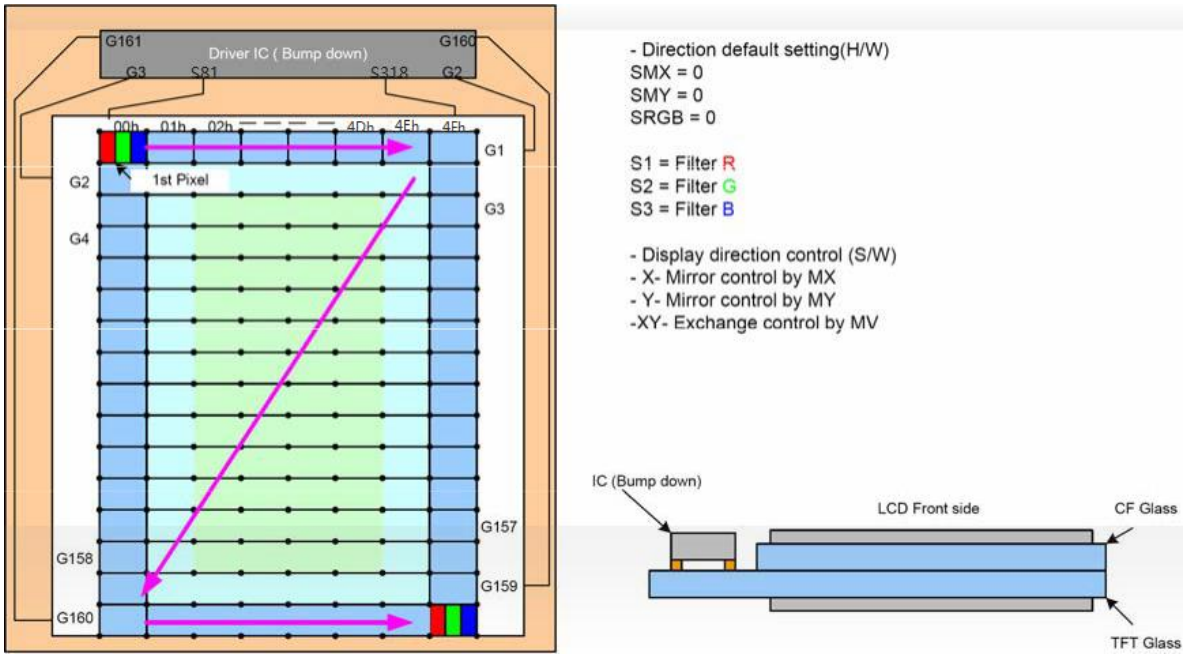
12.2.22 SPI2SET (E9h)

<table>
<tr><td>C5H</td><td colspan="11">REGU_TEST1</td></tr>
<tr><td></td><td>D/CX</td><td>RDX</td><td>WRX</td><td>D7</td><td>D6</td><td>D5</td><td>D4</td><td>D3</td><td>D2</td><td>D1</td><td>D0</td><td>HEX</td></tr>
<tr><td>Command</td><td>L</td><td>H</td><td>R</td><td>1</td><td>1</td><td>1</td><td>0</td><td>1</td><td>0</td><td>0</td><td>1</td><td>E9H</td></tr>
<tr><td>Parameter</td><td>H</td><td>H</td><td>R</td><td></td><td></td><td></td><td></td><td></td><td>SPI2LINE_EN</td><td colspan="2">SPI2LINE_MODE</td><td>00H</td></tr>
<tr><td>Description</td><td colspan="11">SPI2LINE_EN: Enable SPI two data line function. “0” is ON,”1” is OFF;
SPI2LINE_MODE[1:0]: Select different data combination on data lines.
<table>
<tr><th>SPI2LINE_MODE</th><th>Pixel format</th></tr>
<tr><td>2’b01/2’b10</td><td>4-4-4</td></tr>
<tr><td>2’b00/2’b11</td><td>5-6-5</td></tr>
</table></td></tr>
<tr><td>Restriction</td><td colspan="11">Should set “FF=A5” before configure this registers</td></tr>
<tr><td>Register Availability</td><td colspan="11">
<table>
<tr><th>Status</th><th>Availability</th></tr>
<tr><td>Normal Mode on,Idle Mode Off,Sleep Out</td><td>Yes</td></tr>
<tr><td>Normal Mode on,Idle Mode On,Sleep Out</td><td>Yes</td></tr>
<tr><td>Partial Mode on,Idle Mode Off,Sleep Out</td><td>Yes</td></tr>
<tr><td>Partial Mode on,Idle Mode On,Sleep Out</td><td>Yes</td></tr>
<tr><td>Sleep In</td><td>Yes</td></tr>
</table></td></tr>
<tr><td>Default</td><td colspan="11">
<table>
<tr><th>Status</th><th>Default Value(D7 to D0)</th></tr>
<tr><td>Power On Sequence</td><td>8’h00</td></tr>
<tr><td>HW Reset</td><td>8’h00</td></tr>
</table></td></tr>
</table>

13. Example Panel Connection (GM=”000”)

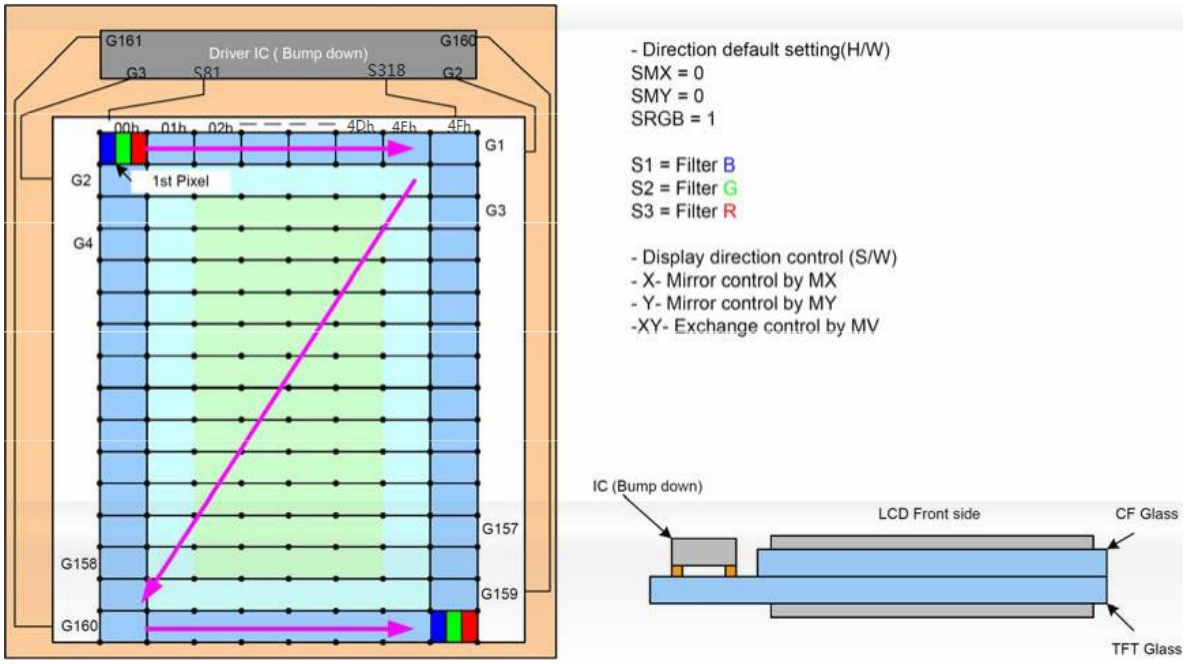
CASE 1: First pixel is at left_top of the panel

RGB order is “R.G.B”



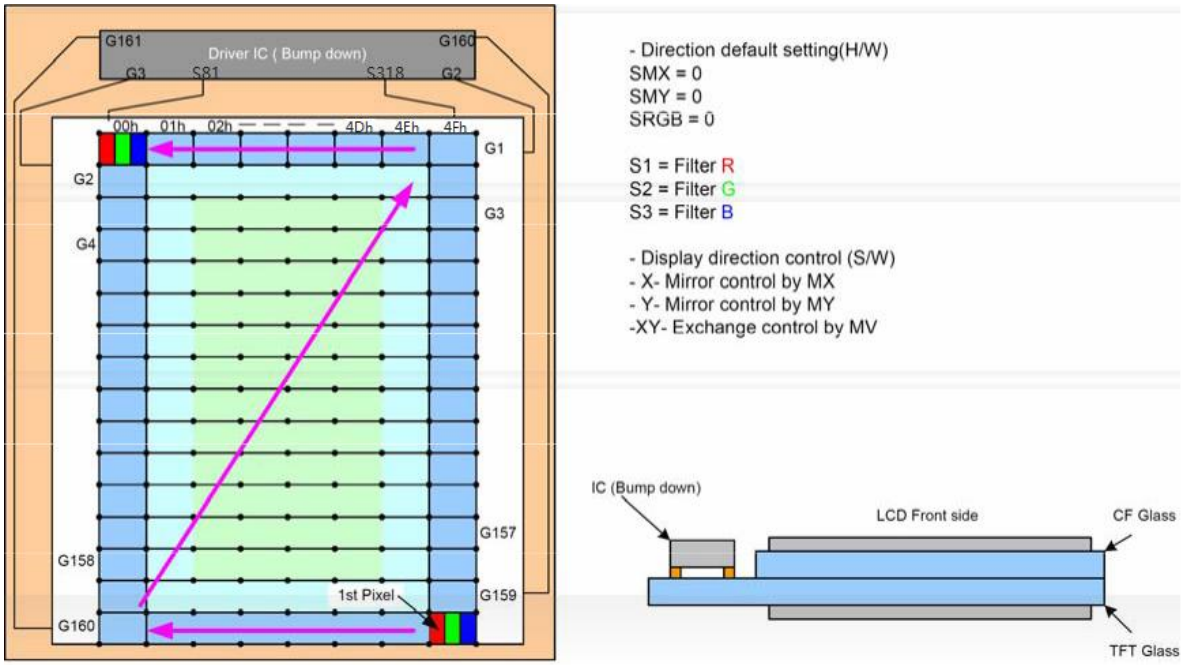
CASE 2: First pixel is at left_top of the panel

RGB order is “B.G.R”



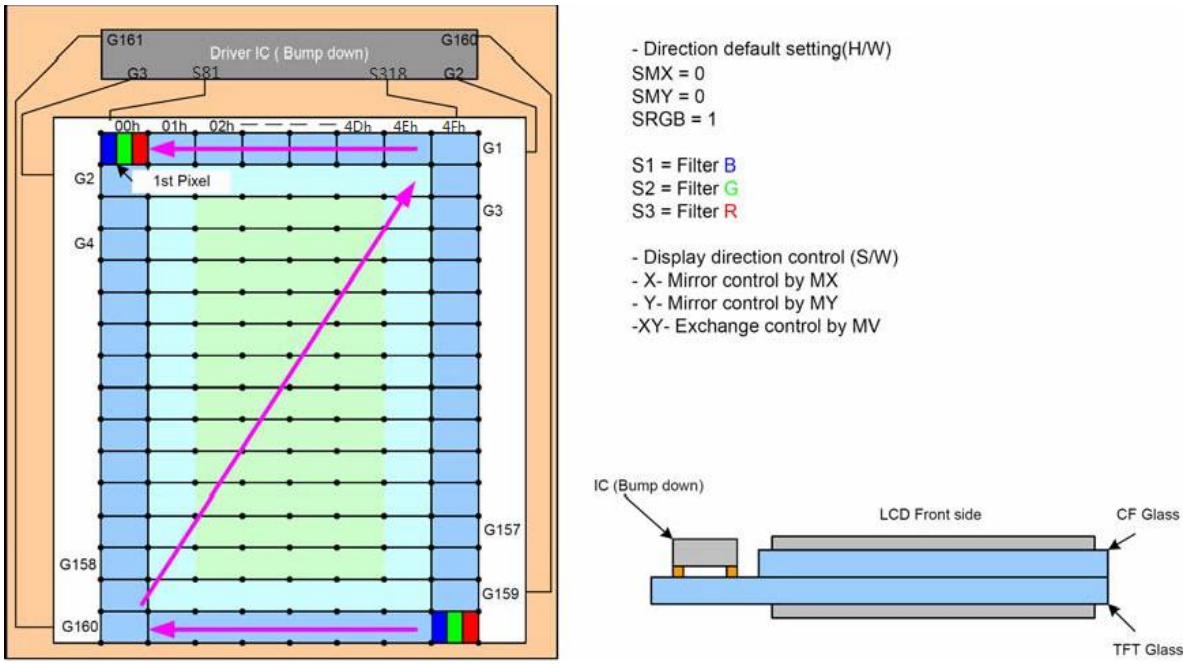
CASE 3: First pixel is at Right_bottom of the panel
RGB order is “R.G.B”

RGB order is “R.G.B”



CASE 4: First pixel is at Right_bottom of the panel

RGB order is “B.G.R”



14. Electrical Characteristics

14.1. Absolute Maximum Ratings

The absolute maximum rating is listed on following table. When NV3022B3 is used out of the absolute maximum ratings, the NV3022B3 may be permanently damaged. So working within the following electrical characteristics limit is strongly recommended during normal operation.

Item	Symbol	Unit	Value Note
Supply voltage	VCI	V	-0.3~+4.8
Supply voltage(Logic)	IOVCC	V	-0.3~+4.6
Supply voltage(Digital)	VCC	V	-0.3~+1.95
Driver supply voltage	VGH-VGL	V	-0.3~+30.0
Logic input voltage range	VIN	V	-0.3~IOVCC+0.3
Logic output voltage range	VO	V	-0.3~IOVCC+0.3
Operation temperature	Topr	℃	-35~+85
Storage temperature	Tstg	℃	-40~+125

Note: If one of the above items is exceeded its maximum limitation momentarily, the quality of the product may be degraded.Absolute maximum limitation, therefore, specify the values exceeding which the product may be physically damaged.Be sure to use the product within the recommend range .

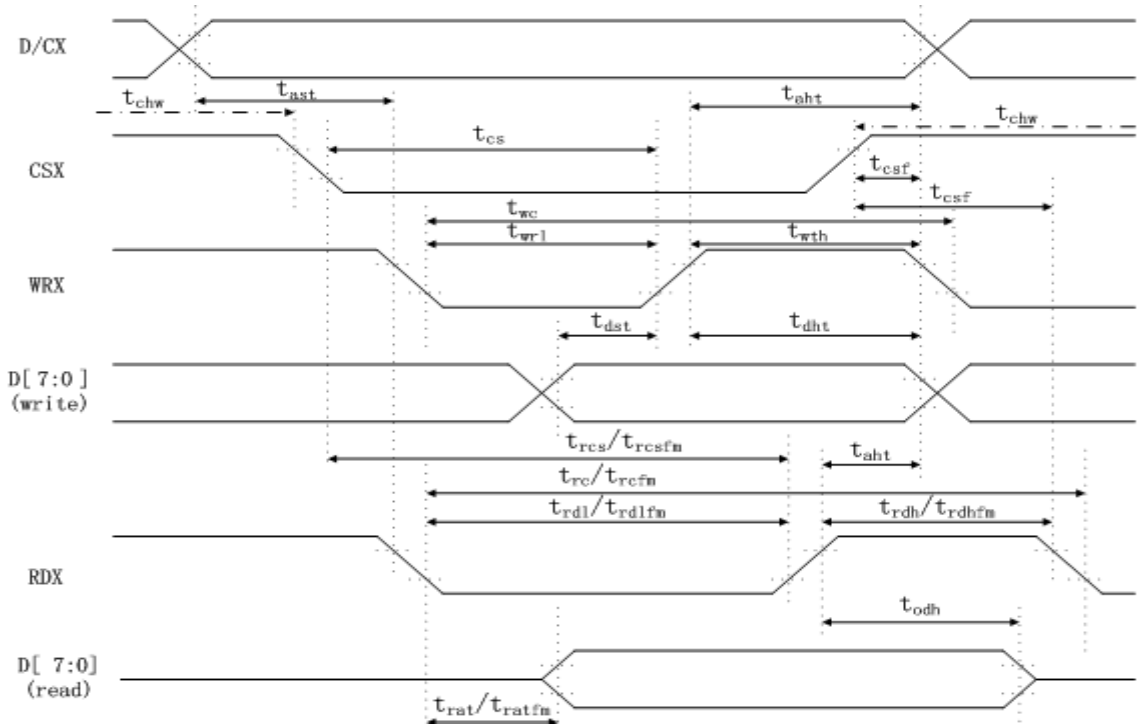
14.2. DC Characteristics

Item	Symbol	Unit	Condition	Min.	Typ.	Max.	Note
Power & Operation Voltage							
Analog Operation Voltage	VCI	V	Operation Voltage	2.5	2.75	3.6	Note2
Logic Operation Voltage	IOVCC	V	I/O Supply Voltage	1.65	1.8	3.6	Note2
Digital Operation Voltage	VCC	V	Digital Supply Voltage		1.8		Note2
Gate Driver High Voltage	VGH	V		10		15	Note3
Gate Driver Low Voltage	VGL	V		-13		-7.5	Note3
Driver Supply Voltage		V	|VGH-VGL|	17.5		28	Note3
Input/Output							
Logic High Level Input Voltage	VIH	V		0.7IOVCC		IOVCC	Note1,2,3
Logic Low Level Input Voltage	VIL	V		VSS		0.3IOVCC	Note1,2,3
Logic High Level Output Voltage	VOH	V	IOH=-1.0mA	0.8IOVCC		IOVCC	Note1,2,3
Logic Low Level Output Voltage	VOL	V	IOL=1.0mA	VSS		0.2IOVCC	Note1,2,3
Logic Input Leakage Current	IIL	uA	VIN=IOVCC or VSS	-0.1		0.1	Note1,2,3
VCOM Operation							
VCOM High Voltage	VCOMH	V	Ccom=12nF	3.24	3.51	4.3425	Note3
VCOM Low Voltage	VCOML	V	Ccom=12nF	-1.5975	-1.0125	-0.4050	Note3
VCOM Amplitude Voltage	VOMA	V	|VCOMH-VCOML|	4		5.5	Note3
Source Driver							
Source Output Range	Vsout	V		0.5		AVDD-0.1	Note4
Gama Reference Voltage	GVDD	V		3.5		4.75	Note3

Note 1: IOVCC=1.65 to 3.6V, VCI=2.5 to 3.6V, AGND=GND=0V, Ta=-30 to 70 (to +85 no damage)
Note2: Please supply digital IOVCC voltage equal or less than analog VCI voltage. (IOVCC≦VCI)
Note2, 3, 4: When the measurements are performed with LCD module. Measurement Points are like below.
Note3: CSX, RDX, WRX, D[23:0], D/CX, RESX, TE, PCLK, VS, HS, DE, SDA, SCL, GM2, GM1, GM0, RCM1, RCM0, P68, IM2, IM1, IM0, SRGB, REV, SMX, SMY, RL, TB, IDM, SHUT, PREG, GS and Test pins.
Note5: Source channel loading = 10pF/channel, Gate channel loading = 50pF/channel

14.3. AC Characteristics

14.3.1 Parallel MCU 8-bit BUS



Note: Logic high and low levels are specified as 30% and 70% of IOVCC for Input signals.

Table 15.3.1 AC characteristics of parallel MCU I/F in asynchronous mode

Signal	Symbol	Parameter	MIN	MAX	UNIT	Description
D/CX	TAST	Address Setup Time	0		ns	
	THAT	Address Hold Time (W/R)	10		ns	
CSX	TCHW	“S” “H” Pulse Width	0		ns	
	TCS	Chip Select Setup Time(W)	10		ns	
	TRCS	Chip Select Setup Time (Read ID)	45		ns	
	TRCSFM	Chip Select Setup Time (Read FM)	355		ns	
	TCSF	Chip Select Wait Time (W/R)	10		ns	
WRX	TWC	Write Cycle	80		ns	
	TWRH	Control Pulse H Duration	40		ns	
	TWRL	Control Pulse L Duration	40		ns	
RDX	TRC	Read Cycle(ID)	160		ns	When Read ID
	TRDH	Control Pulse H Duration(ID)	90		ns	
	TRDL	Control Pulse L Duration(ID)	45		ns	

RDX	TRCFM	Read Cycle(FM)	450		ns	When Read From Frame Memory
	TRDHFM	Control Pulse H Duration(FM)	90		ns	
	TRDLFM	Control Pulse L Duration(FM)	355		ns	
D[7:0]	TDST	Data Setup Time	10		ns	CLmax=30pF Clmin=8pF
	TDHT	Data Hold Time	10		ns	
	TRAT	Read Access Time(ID)		40	ns	
	TRATFM	Read Access Time(FM)		340	ns	
	TODH	Output Disable Time	20	80	ns	

Note 1: IOVCC 1.65 to 3.6V, VCI=2.5 to 3.6V, VSSA=VSS=0V, Ta=-30 to 70 ℃ (to +85 ℃ no damage)
Note 2: This input signal rise time and fall time (tr, tf) is specified at 15 ns or less. Logic high and low levels are specified as 30% and 70% of IOVCC for input signals

14.3.2 Display Serial Interface(SPI)

14.3.2.1 SPI 3-Wire Interface

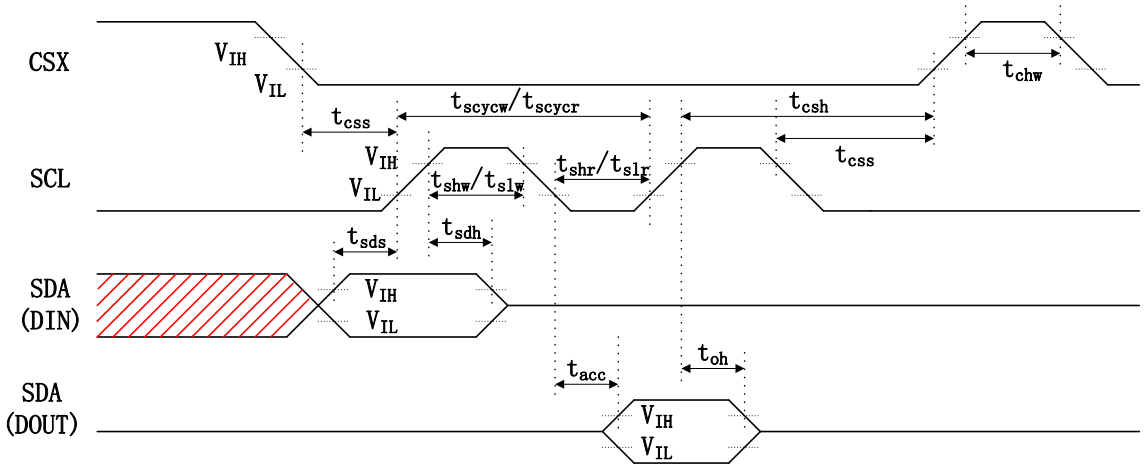


Table 15.3.2.1: 3-pin Serial Interface Characteristics

Signal	Symbol	Parameter	MIN	MAX	UNIT	Description
CSX	TCSS	Chip Select Setup Time	10		ns	
	TCSH	Chip Select Hold Time	30		ns	
	TCHW	Chip Select “H” Pulse Width	30		ns	
SCL	TSCYCW	Serial Clock Cycle(Write)	16		ns	
	TSHW	S”L””H” Pulse Width(Write)	8		ns	
	TSLW	S”L””L” Pulse Width(Write)	8		ns	
	TSCYCR	Serial Clock	150		ns	

Signal	Symbol	Parameter	MIN	MAX	UNIT	Description
		Cycle(Read)				
	TSHR	S”L””H” Pulse Width(Read)	60		ns	
	TSLR	S”L””L” Pulse Width(Read)	60		ns	
SDA(DIN) /(DOUT)	TSDS	Data Setup Time	15		ns	
	TSDH	Data Hold Time	5		ns	
	TACC	Access Time	5	50	ns	CLmax=30pF CLmin=8pF
	TOH	Output Disable Time	10		ns	

Note 1: IOVCC=1.65 to 3.6V, VCI=2.5 to 3.6V, AGND=GND=0V. Ta=-30 to 70℃ (to +85℃ no damage)
Note 2: The input signal rise time and fall time(tr, tf) is specified at 15 ns or less. Logic high and low levels are specified as 10% and 90% of IOVCC for Input signals.

14.3.2.2 SPI 4-Wire Interface

Table 15.3.2.2: 4 pin Serial Interface Characteristics

Signal	Symbol	Parameter	MIN	MAX	UNIT	Description
CSX	TCSS	Chip Select Setup Time	10		ns	
	TCSH	Chip Select Hold Time	30		ns	
	TCHW	Chip Select “H” Pulse Width	30		ns	
SCL	TSCYCW	Serial Clock Cycle(Write)	16		ns	
	TSHW	S”L””H” Pulse Width(Write)	8		ns	
	TSLW	S”L””L” Pulse Width(Write)	8		ns	
	TSCYCR	Serial Clock Cycle(Read)	150		ns	
	TSHR	S”L””H” Pulse Width(Read)	60		ns	
	TSLR	S”L””L” Pulse Width(Read)	60		ns	
D/CX	TDCS	D/CX Setup Time	5		ns	
	TDCH	D/CX Hold Time	5		ns	
SDA(DIN) (DOUT)	TSDS	Data Setup Time	15		ns	
	TSDH	Data Hold Time	5		ns	
	TACC	Access Time	5	50	ns	CLmax=30pF CLmin=8pF
	TOH	Output Disable Time	10		ns	

Note 1: IOVCC=1.65 to 3.6V, VCI=2.5 to 3.6V, AGND=GND=0V. Ta=-30 to 70℃ (to +85℃ no damage)
Note 2: The input signal rise time and fall time (tr, tf) is specified at 15 ns or less. Logic high and low levels are specified as 10% and 90% of IOVCC for Input signals.

15. Revision History

Date	Revision	Page	Description
2020/09/22	V0.1	All	First Release by lvjj
2021/09/15	V0.2	All	Update by lvjj
2022/04/25	V0.3	All	Update by shin
2022/12/26	V0.4	5~7	1. Update the GM mapping 2. Update the Bump Arrangement and Pin Connection
2023/02/03	V0.5	68、69	Update the register 2Ah and 2Bh
2023/05/10	V0.6		Update the register section